Sign compression for Muon: SignMuon, MuonSign, and the Limits of Error Feedback

Anonymous Author(s)
Affiliation
Address
email

Abstract

SignMuon compresses the Muon update to one bit per parameter by taking the elementwise sign of the Muon Linear Minimization Oracle (LMO) direction: the most direct way to give a matrix-aware optimizer a one-bit budget. It outperforms SignSGD empirically, yet it can ascend even on a linear function. Signing the gradient *before* the oracle rather than after does not repair this: we exhibit a small explicit instance on which sign-before (MuonUSign) and sign-on-both-sides (MuonSign) ascend as well, so no placement of the sign around the oracle descends in general. Error feedback does not repair sign-after either: applied to the *output* of the oracle it fails for every smoothness constant, step size and momentum. Applied to the *gradient* it does work, and EF21-MuonUSign and EF21-MuonSign attain the standard $\mathcal{O}(T^{-1/2})$ rate for the squared gradient norm on smooth nonconvex problems, the latter at one bit in each direction. Our experiments then reverse this order: across CIFAR-10, its federated split and the nanoGPT speedrun the strongest compressed method is sign-*after*, precisely the placement we prove divergent, with the provably convergent variants behind it.

1 Introduction

In distributed training over a limited link, communication rather than computation is the binding cost. Compression is the standard remedy Bernstein et al. [2018, 2019], Beznosikov et al. [2023]. SignSGD is its extreme point: one bit per coordinate, at a small cost in accuracy, but it discards matrix structure. Muon exploits that structure, orthonormalizing the momentum matrix before stepping Jordan et al. [2024b], Bernstein and Newhouse [2024], Shah et al. [2025]: its step is the Linear Minimization Oracle (LMO) of the spectral norm, and it is transmitted as a *dense* matrix at full precision. We compress that step with a sign, placed *after* the oracle, *before* it, or on both sides: **SignMuon**, $\mathrm{sign}(\mathrm{LMO}(\cdot))$; **MuonUSign**, $\mathrm{LMO}(\mathrm{sign}(\cdot))$; and **MuonSign**, $\mathrm{sign}(\mathrm{LMO}(\mathrm{sign}(\cdot)))$. SignMuon and MuonSign emit a ± 1-valued step, so uplink and downlink alike cost one bit.

SignMuon, MuonUSign and MuonSign do not descend in general: each can *ascend* on a linear objective at every step size and every momentum, SignMuon on a 4×4 gradient (Theorem 1), MuonUSign and MuonSign on a 5×5 gradient they share (Theorems 2–3). Error feedback is the standard repair for a biased compressor. For sign-after it does not help: applied to the *output* of the oracle it fails for every triple (L, η, μ) (Theorem 4). No step-size rule built from the smoothness and momentum constants can make it convergent. Applied to the *gradient*, however, it restores convergence: adapting EF21-Muon [Gruntkowska et al., 2025] gives **EF21-MuonUSign** and **EF21-MuonSign**, which attain $\min_{t\le T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$, the standard rate for smooth nonconvex problems, at a one- and a two-sided one-bit budget.

The guarantees do not predict which method performs best in practice. On centralized CIFAR-10 with a ResNet-18, on federated CIFAR-10 (Appendix C), and on the nanoGPT speedrun, the

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

strongest compressed method signs *after* the oracle, the placement error feedback cannot repair, and EF21-MuonUSign and EF21-MuonSign consistently trail it. The counterexamples are worst-case constructions; random instances do not reproduce them (Section 3).

Related work. We build on the interpretation of Muon as an LMO method [Bernstein and Newhouse, 2024, Pethick et al., 2025a, Riabinin et al., 2025] and on the error-feedback framework of Gruntkowska et al. [2025] for bidirectionally compressed Muon. The sign-before and both-sides placements are new, as are the two EF21 methods; SignMuon was proposed concurrently by Mishra et al. [2026], whose guarantee, we show in Appendix H.5, applies to a different method. Appendix A gives the full account.

2 Problem Statement

We consider $\min_{\mathbf{X}\in\mathcal{X}}\{f(\mathbf{X}) := \mathbb{E}_{\xi\sim\mathcal{D}}[f(\mathbf{X};\xi)]\}$ with $f(\cdot\,;\xi)$ continuously differentiable and possibly non-convex; the federated form, over N clients, is (7) in Appendix C. The model is a tuple of layers $\mathbf{X} = [\mathbf{X}_1, \ldots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i\times n_i}$, with ∇_i the gradient block of layer i. This is the Gluon parameter space [Riabinin et al., 2025] under a unified geometry: each layer is equipped with the spectral norm $\|\cdot\|_{2\to2}$ with nuclear dual $\|\cdot\|_*$, and $\|\nabla f(\mathbf{X})\|_*^2 := \sum_i \|\nabla_i f(\mathbf{X})\|_*^2$. On a block of width one the oracle only normalizes, so all three methods coincide with SignSGD; the results below need $\min(m_i, n_i) \geq 2$ (Appendix G).

Assumption 1 (Lower boundedness) *$f(\mathbf{X}) \geq f^*$ for all $\mathbf{X}$; where explicitly invoked, each local $f_j \geq f_j^*$ as well.*

Assumption 2 (Layer-wise smoothness) *For every layer i and all $\mathbf{X}, \mathbf{Y}$, $\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \leq L_i^g \|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to2}$ for $g = f$ (constant L_i) and $g = f_j$ (constant $L_{i,j}$). Appendix F gives a weaker (L^0, L^1) form, used only in Corollary 2, and records the strength of the quantifier over arbitrary $\mathbf{X}, \mathbf{Y}$.*

An LMO minimizes the first-order model over the unit ball of a norm, returning the steepest-descent direction in it. For $\mathbf{M} \in \mathbb{R}^{m\times n}$ with rank-r SVD $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ the LMO direction is $A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top$; we write $\mathrm{polar}(\mathbf{M}) := -A(\mathbf{M})$, an orthogonalization, unscaled. The reference implementation and the RMS→RMS interpretation rescale it by positive per-layer constants, invisible to $\mathrm{sign}(\cdot)$; we absorb them into the step size (Appendices F and P).

3 Theory

Three placements of the sign. All three build the same momentum estimate $\tilde{\mathbf{M}}_t$ from the stochastic gradients: an exponential moving average at coefficient $\mu \in [0, 1)$, or its Nesterov variant (Appendix B). All three then step $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t\, \mathbf{s}_t$ at learning rate η_t. They differ only in where the sign acts on $\tilde{\mathbf{M}}_t$:

$$\mathbf{s}_t = \begin{cases} \mathrm{sign}\big(\mathrm{polar}(\tilde{\mathbf{M}}_t)\big) & \text{(SignMuon: after)},\\ \mathrm{polar}\big(\mathrm{sign}(\tilde{\mathbf{M}}_t)\big) & \text{(MuonUSign: before)},\\ \mathrm{sign}\big(\mathrm{polar}(\mathrm{sign}(\tilde{\mathbf{M}}_t))\big) & \text{(MuonSign: both)}. \end{cases} \tag{1}$$

Throughout, $\mathrm{sign}(0)$ is resolved to an independent random ±1, keeping the transmitted alphabet binary. Every method transmits one bit per matrix parameter on the uplink. On the downlink SignMuon and MuonSign distribute a ±1-valued matrix and cost one bit again, whereas MuonUSign applies $\mathrm{polar}(\cdot)$ on the server, which is dense and goes at full precision. Appendix Q gives the algorithms and the federated protocol.

3.1 Divergence of the Three Sign Placements

The sign step is itself an LMO, steepest descent in ℓ_∞ [Bernstein and Newhouse, 2024]. Signing a Muon direction therefore composes two oracles, each sound alone. The composition is an LMO for *no* norm. An oracle for a norm returns $\mathbf{D} = -A(\mathbf{G})$ with $\langle \mathbf{G}, \mathbf{D}\rangle = \|\mathbf{G}\|_{\mathrm{dual}} \geq 0$; Theorem 1 provides a gradient on which the composition drives that inner product strictly negative.

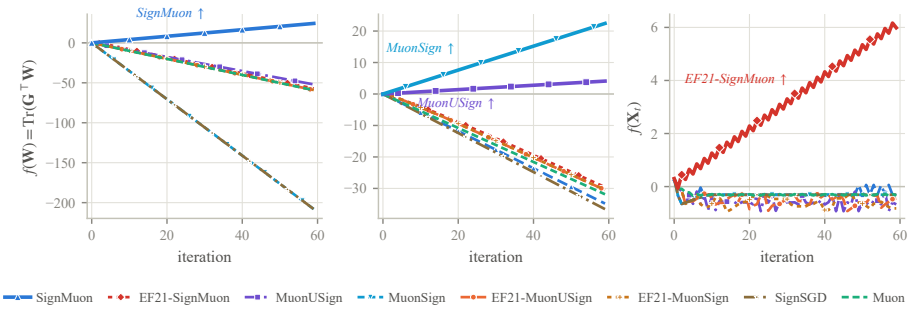


Figure 1: All eight methods on the three counterexample instances. *Left:* SignMuon, 4×4 (Theorem 1). *Centre:* MuonUSign and MuonSign on their shared 5×5 instance (Theorems 2–3); both panels plot (2). *Right:* EF21-SignMuon, 2×2 (Theorem 4), normalized units at $\eta = 1$. Trajectories are momentum-free without loss (Proposition 1; Lemma 4 for EF21-SignMuon).

We refute the descent property on *linear* objectives:

$$f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X} \rangle, \qquad \nabla f(\mathbf{X}) \equiv \mathbf{G}. \tag{2}$$

Gradient descent and full-precision Muon drive $f \to -\infty$ here, so a rule that drives $f \to +\infty$ is unambiguously ascending. (Remark 1 restores Assumption 1 without moving any trajectory.) Whatever the momentum, the step is the constant matrix $\mathbf{s}(\mathbf{G})$, so momentum affects neither convergence nor divergence (Proposition 1). The increment is

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \left\langle \mathbf{G},\ \mathbf{s}(\mathbf{G}) \right\rangle. \tag{3}$$

A single $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ therefore makes f increase at every step. Such a $\mathbf{G}$ exists for each placement at small size.

Theorems 1–3 (summary). *There exists $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ for SignMuon, and $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ for MuonUSign and for MuonSign simultaneously. On the corresponding objective (2) each method ascends at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$ and either momentum rule, and $f(\mathbf{X}_t) \to +\infty$ whenever $\sum_t \eta_t = \infty$.*

Both instances are explicit, with exact algebraic pairings. Appendices H.2–H.4 state the theorems and prove them; Figure 1 (left, centre) shows the ascents. Negative results of this kind exist for uncompressed Muon as well [Parshakova et al., 2026]. Ours are due to the compressor rather than the geometry.

On a deterministic convex quadratic the alignment $\rho_t = \langle \nabla f, \mathbf{D}_t \rangle / (\|\nabla f\|_F \|\mathbf{D}_t\|_F)$ between the gradient and the update direction actually taken can be measured directly. Theorems 1–3 give instances with $\rho_t < 0$; on random instances all three placements keep ρ_t away from zero (Appendix K). The constructions therefore describe a worst case rather than a typical one. The experiments below run contrary to them.

3.2 Error Feedback on the Polar Factor Fails

The classical remedy for a biased compressor is *error feedback*: keep the discarded residual and fold it into the next message [Seide et al., 2014, Karimireddy et al., 2019]. We work in its EF21 form [Richtárik et al., 2021], which stores an estimator and compresses the *difference* to it. No bounded-gradient assumption is then needed. The most direct use keeps the geometry, signing *after* the LMO, and applies error feedback to the polar factor. **EF21-SignMuon** (Algorithm 3) tracks an estimate $\mathbf{d}_t^{\mathrm{est}}$ of the polar factor $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$ by a scaled sign of the residual,

$$\mathbf{d}_t^{\mathrm{est}} = \mathbf{d}_{t-1}^{\mathrm{est}} + \alpha_t \,\mathrm{sign}\big(\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big), \qquad \alpha_t = \mathrm{mean}\big|\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big|, \tag{4}$$

then steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \, \mathbf{d}_t^{\mathrm{est}}$. This does not repair the placement.

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0, 1)$, and either momentum variant, there is an L-smooth (Assumption 2), bounded-below*

(Assumption 1) $f : \mathbb{R}^{2\times 2} \to \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit $c = c(f) > 0$, $f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = c\,L\eta^2 > 0$ for $t \geq 3$, so $f(\mathbf{X}_t) \to +\infty$. No step-size rule $\eta = \eta(L, \mu)$ using only the smoothness and momentum constants can make the method convergent.

The construction (Appendix H.7) turns the shared magnitude against the method. A 2×2 target whose large off-diagonal reverses sign every step holds α_t at $\Theta(1)$; the small diagonal constant is then overshot at every step, its estimate settling into a period-two cycle whose average has the *wrong* sign. Neither dimension nor momentum is implicated (Appendix H.7): 2×2 is where the worst-case contraction $1/d$ of the scaled sign is most favourable.

3.3 Error Feedback on the Gradient Works

The defect is the target, not the shared magnitude. Error feedback needs a target that varies with the step size; $\mathrm{polar}(\tilde{\mathbf{M}}_t)$ is not Lipschitz in its argument, so it can move by $\Theta(1)$ between consecutive rounds however small η is. Compressing the *gradient* in its place restores that dependence. The resulting method is EF21-Muon [Gruntkowska et al., 2025] with the scaled sign $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}\,|\mathbf{Y}|\,\mathrm{sign}(\mathbf{Y})$ on every compressed link: one bit per parameter, plus one scalar per layer. **EF21-MuonUSign** takes the pair $(\mathcal{C}, I)$: a compressed uplink, a full-precision model back. It maintains a gradient estimator $\mathbf{g}_t^{\mathrm{est}}$ from the residual $\tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\mathrm{est}}$ exactly as in (4), then steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ with $\mathbf{D}_t = -A(\mathbf{g}_t^{\mathrm{est}})$, the polar factor of the estimator rather than of ∇f. **EF21-MuonSign** takes $(\mathcal{C}, \mathcal{C})$, one bit each way. Its downlink is a second error-feedback loop, on the *model* rather than the gradient, splitting the method into a server model $\mathbf{X}_t$, which the guarantee bounds, and a broadcast model $\mathbf{W}_t$, where every gradient is evaluated (Appendix C).

Under Assumptions 1–3, with the horizon-tuned sharp steps of Corollary 1 or Theorem 6, both methods with EMA or Nesterov reach $\frac{1}{T}\sum_{t<T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$. Corollary 3 gives explicit momentum and query budgets with a $1/N$ leading variance term. For fixed momentum and constant/cosine learning rates, Corollary 4 bounds the weighted mean gradient norm by an $\mathcal{O}(T^{-1})$ transient plus explicit stepsize and noise residuals, under ordinary smoothness and exact polar updates. The scaled-sign downlink adds rank dependence (Remark 5).

4 Experiments

We compare all six sign-based methods against Muon, SignSGD, SGD and Adam. Only the base rate η_0 is tuned: layer ℓ, of shape $m \times n$, gets $\eta_\ell = \eta_0 \lambda_\ell$ with

$$\lambda_\ell = \sqrt{m}\,/\,\|\mathbf{P}_\ell\|_F \quad \Longrightarrow \quad \lambda_\ell^{\mathrm{LMO}} = \sqrt{\max(1, m/n)}, \qquad \lambda_\ell^{\mathrm{SIGN}} = 1/\sqrt{n}, \tag{5}$$

where $\mathbf{P}_\ell$ is the idealized family step (Appendix P). The criterion is that every layer receive the same root-mean-square input–output gain [Yang et al., 2023, Large et al., 2024]. Its LMO multiplier reproduces the aspect-ratio factor the reference Muon implementation applies [Jordan et al., 2024b]; we apply the same criterion to the SIGN family, for which no convention exists (Appendix P).

4.1 Image Classification

We train a ResNet-18 adapted to low-resolution images on CIFAR-10 Krizhevsky [2009] for 75 epochs, η_0 cosine-annealed to zero and selected per method on a held-out split. Numbers average three seeds; differences below the seed spread are not claimed (Appendix J).

SignMuon ranks first, by less than one seed spread over Muon (Figure 2). The separation that resolves is the 1.2 points over SignSGD. SignSGD spends the same budget without the LMO, so the geometry accounts for the gap, not the compressor. Error feedback is not free here: EF21-SignMuon lies 0.29 points below SignMuon, against standard deviations of 0.11 and 0.15. Both families order the placements after, before, both sides.

4.2 Language Modelling

We test where the matrices are large enough for the layer-rank dependence of Corollary 1 to take effect: the modded-nanoGPT speedrun (record #40) on $8\times$H100 [Jordan et al., 2024a], whose hidden matrices are 768×3072. Each method replaces Muon on the hidden matrices and gates; embeddings,

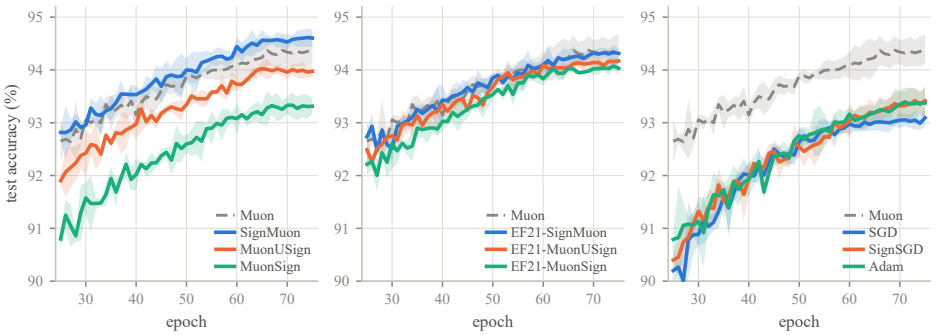


Figure 2: Centralized ResNet-18 on CIFAR-10: test accuracy at the η_0 selected for each method, from epoch 25 (Table 2). Bands show ± 1 s.d. over three seeds.

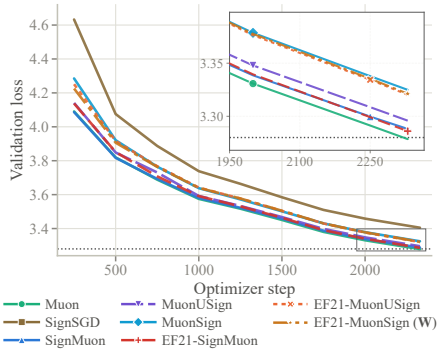


Figure 3: NanoGPT speedrun, 8×H100: validation loss against optimizer step, one run per method. EF21-MuonSign is drawn at its broadcast model $\mathbf{W}$.

scalar parameters, and the output head retain distributed Adam. Rates are fixed a priori by (5), one η_0 per *family*, so every contrast is matched-hyperparameter (Appendix O).

All six sign–LMO compositions improve on SignSGD in validation loss (Figure 3, Table 3). The best compressed methods, EF21-SignMuon and SignMuon, achieve validation losses close to full-precision Muon at similar wall-clock times. The target they compress is an orthogonal factor whose entries stay evenly spread.

The two models of EF21-MuonSign separate at this width. Gradients are evaluated at the broadcast model $\mathbf{W}$, whose validation loss is close to EF21-MuonUSign's. The server model $\mathbf{X}$, which the guarantee bounds, has 2.2 nats higher validation loss. Appendix O reports the largest relative model gap on the zero-initialized MLP output projections.

5 Conclusion

Every placement of the sign around the Muon oracle can turn the update into an ascent direction on a linear objective, at every step size and momentum: SignMuon after the oracle, MuonUSign before it, MuonSign on both sides. Whether error feedback repairs this depends on where it is applied. Applied to the *gradient*, it attains the standard nonconvex rate; applied to the *polar factor*, it fails for every (L, η, μ). In all three settings the strongest compressed method signs *after* the oracle, where no guarantee is available, and the two provably convergent methods trail it. The counterexamples rule out a guarantee for every problem in the class. Whether a guarantee holds under further conditions that ordinary problems satisfy is the open question.

References

Kwangjun Ahn, Byron Xu, Natalie Abreu, Ying Fan, Gagik Magakyan, Pratyusha Sharma, Zheng Zhan, and John Langford. Dion: Distributed orthonormalized updates, 2025. URL https://arxiv.org/abs/2504.05295.

Dan Alistarh, Demjan Grubic, Jerry Li, Ryota Tomioka, and Milan Vojnovic. QSGD: Communication-efficient SGD via gradient quantization and encoding. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 1709–1720, 2017.

Jeremy Bernstein. Deriving Muon. https://jeremybernste.in/writing/deriving-muon, 2025. Blog post.

Jeremy Bernstein and Laker Newhouse. Old optimizer, new norm: An anthology, 2024. URL https://arxiv.org/abs/2409.20325.

Jeremy Bernstein, Yu-Xiang Wang, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD: Compressed optimisation for non-convex problems. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 560–569. PMLR, 2018. URL https://proceedings.mlr.press/v80/bernstein18a.html.

Jeremy Bernstein, Jiawei Zhao, Kamyar Azizzadenesheli, and Anima Anandkumar. signSGD with majority vote is communication efficient and fault tolerant. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://openreview.net/forum?id=BJxhijAcY7. arXiv:1810.05291.

Aleksandr Beznosikov, Samuel Horváth, Peter Richtárik, and Mher Safaryan. On biased compression for distributed learning. *Journal of Machine Learning Research*, 24(276):1–50, 2023. URL https://www.jmlr.org/papers/v24/21-1548.html.

Arman Bolatov, Artem Riabinin, Nikita Kornilov, Andrey Veprikov, Samuel Horváth, Martin Takáč, and Aleksandr Beznosikov. LionMuon: Alternating spectral and sign descent for efficient training, 2026. URL https://arxiv.org/abs/2605.19811.

Franz Louis Cesista. Steepest descent under Schatten-p norms. https://leloykun.github.io/ponder/steepest-descent-schatten-p/, 2025. Blog post.

Xiangyi Chen, Tiancong Chen, Haoran Sun, Zhiwei Steven Wu, and Mingyi Hong. Distributed training with heterogeneous data: Bridging median- and mean-based algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. arXiv:1906.01736.

Ashok Cutkosky and Harsh Mehta. Momentum improves normalized SGD. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 2260–2268. PMLR, 2020. URL https://proceedings.mlr.press/v119/cutkosky20b.html.

Arthur Douillard, Qixuan Feng, Andrei A. Rusu, Rachita Chhaparia, Yani Donchev, Adhiguna Kuncoro, Marc'Aurelio Ranzato, Arthur Szlam, and Jiajun Shen. DiLoCo: Distributed low-communication training of language models, 2023. URL https://arxiv.org/abs/2311.08105.

Kaja Gruntkowska, Alexander Gaponov, Zhirayr Tovmasyan, and Peter Richtárik. Error feedback for Muon and friends, 2025. URL https://arxiv.org/abs/2510.00643.

Aman Gupta, Rafael Celente, Abhishek Shivanna, D. T. Braithwaite, Gregory Dexter, Shao Tang, Hiroto Udagawa, Daniel Silva, Rohan Ramanath, and S. Sathiya Keerthi. Effective quantization of Muon optimizer states, 2025. URL https://arxiv.org/abs/2509.23106.

Samuel Horváth, Dmitry Kovalev, Konstantin Mishchenko, Peter Richtárik, and Sebastian Stich. Stochastic distributed learning with gradient quantization and double-variance reduction. *Optimization Methods and Software*, 38(1):91–106, 2023. doi: 10.1080/10556788.2022.2117355.

Richeng Jin, Yuding Liu, Yufan Huang, Xiaofan He, Tianfu Wu, and Huaiyu Dai. Sign-based gradient descent with heterogeneous data: Convergence and Byzantine resilience. *IEEE Transactions on Neural Networks and Learning Systems*, 36(2):3834–3846, 2025. doi: 10.1109/TNNLS.2023.3345367.

Keller Jordan, Jeremy Bernstein, Brendan Rappazzo, @fernbear.bsky.social, Vlado Boza, Jiacheng You, Franz Cesista, Braden Koszarsky, and @Grad62304977. modded-nanogpt: Speedrunning the NanoGPT baseline. https://github.com/KellerJordan/modded-nanogpt, 2024a. Code repository.

Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks. https://kellerjordan.github.io/posts/muon/, 2024b. Blog post.

Sai Praneeth Karimireddy, Quentin Rebjock, Sebastian Stich, and Martin Jaggi. Error feedback fixes SignSGD and other gradient compression schemes. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3252–3261. PMLR, 2019. URL https://proceedings.mlr.press/v97/karimireddy19a.html.

Dmitry Kovalev. Understanding gradient orthogonalization for deep learning via non-Euclidean trust-region optimization, 2025. URL https://arxiv.org/abs/2503.12645.

Alexey Kravatskiy, Ivan Kozyrev, Nikolai Kozlov, Alexander Vinogradov, Daniil Merkulov, and Ivan Oseledets. Ky Fan norms and beyond: Dual norms and combinations for matrix optimization, 2025. URL https://arxiv.org/abs/2512.09678.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

Tim Large, Yang Liu, Minyoung Huh, Hyojin Bahng, Phillip Isola, and Jeremy Bernstein. Scalable optimization in the modular norm. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.14813.

Jiaxiang Li and Mingyi Hong. A note on the convergence of Muon, 2025. URL https://arxiv.org/abs/2502.02900.

Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is scalable for LLM training, 2025. URL https://arxiv.org/abs/2502.16982.

Neel Mishra, Kushagara Trivedi, and Pawan Kumar. SignMuon: Communication-efficient distributed Muon optimization, 2026. URL https://arxiv.org/abs/2605.16311.

Tetiana Parshakova, Ahmed Khaled, Michael Crawshaw, Guillaume Garrigos, and Robert M. Gower. Muon does not converge on convex Lipschitz functions, 2026. URL https://arxiv.org/abs/2605.08980.

Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin Raffel, Leandro Von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. arXiv:2406.17557.

Thomas Pethick, Wanyun Xie, Kimon Antonakopoulos, Zhenyu Zhu, Antonio Silveti-Falls, and Volkan Cevher. Training deep learning models with norm-constrained LMOs. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 49069–49104. PMLR, 2025a. URL https://proceedings.mlr.press/v267/pethick25a.html.

Thomas Pethick, Wanyun Xie, Mete Erdogan, Kimon Antonakopoulos, Antonio Silveti-Falls, and Volkan Cevher. Generalized gradient norm clipping & non-Euclidean (L_0, L_1)-smoothness. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025b. arXiv:2506.01913.

Xun Qian, Alexander Gaponov, Grigory Malinovsky, and Peter Richtárik. Communication-efficient Gluon in federated learning, 2026. URL https://arxiv.org/abs/2604.10689.

Artem Riabinin, Egor Shulgin, Kaja Gruntkowska, and Peter Richtárik. Gluon: Making Muon & Scion great again! (Bridging theory and practice of LMO-based optimizers for LLMs), 2025. URL https://arxiv.org/abs/2505.13416.

Peter Richtárik, Igor Sokolov, and Ilyas Fatkhullin. EF21: A new, simpler, theoretically better, and practically faster error feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 4384–4396, 2021.

Mher Safaryan and Peter Richtárik. Stochastic sign descent methods: New algorithms and better theory. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9224–9234. PMLR, 2021. URL https://proceedings.mlr.press/v139/safaryan21a.html.

Frank Seide, Hao Fu, Jasha Droppo, Gang Li, and Dong Yu. 1-bit stochastic gradient descent and its application to data-parallel distributed training of speech DNNs. In *Interspeech 2014*, pages 1058–1062, 2014. doi: 10.21437/Interspeech.2014-274.

Ishaan Shah, Anthony M. Polloreno, Karl Stratos, Philip Monk, Adarsh Chaluvaraju, Andrew Hojel, Andrew Ma, Anil Thomas, Ashish Tanwer, Darsh J. Shah, Khoi Nguyen, Kurt Smith, Michael Callahan, Michael Pust, Mohit Parmar, Peter Rushton, Platon Mazarakis, Ritvik Kapila, Saurabh Srivastava, Somanshu Singla, Tim Romanski, Yash Vanjani, and Ashish Vaswani. Practical efficiency of Muon for pretraining, 2025. URL https://arxiv.org/abs/2505.02222.

Tao Sun, Qingsong Wang, Dongsheng Li, and Bao Wang. Momentum ensures convergence of SIGNSGD under weaker assumptions. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 33077–33099. PMLR, 2023. URL https://proceedings.mlr.press/v202/sun23l.html.

Yuki Takezawa, Anastasia Koloskova, Xiaowen Jiang, and Sebastian U. Stich. FedMuon: Federated learning with bias-corrected LMO-based optimization. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://openreview.net/forum?id=9k7bvBVenZ. arXiv:2509.26337.

Hanlin Tang, Chen Yu, Xiangru Lian, Tong Zhang, and Ji Liu. DoubleSqueeze: Parallel stochastic gradient descent with double-pass error-compensated compression. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 6155–6165. PMLR, 2019. URL https://proceedings.mlr.press/v97/tang19d.html.

Benjamin Thérien, Xiaolong Huang, Aaron Defazio, Irina Rish, and Eugene Belilovsky. MuLoCo: Muon is a practical inner optimizer for DiLoCo, 2025. URL https://arxiv.org/abs/2505.23725.

Greg Yang, Edward J. Hu, Igor Babuschkin, Szymon Sidor, Xiaodong Liu, David Farhi, Nick Ryder, Jakub Pachocki, Weizhu Chen, and Jianfeng Gao. Tuning large neural networks via zero-shot hyperparameter transfer. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 17084–17097, 2021. arXiv:2203.03466.

Greg Yang, James B. Simon, and Jeremy Bernstein. A spectral condition for feature learning, 2023. URL https://arxiv.org/abs/2310.17813.

Xinwen Zhang and Hongchang Gao. On provable benefits of Muon in federated learning, 2025. URL https://arxiv.org/abs/2510.03866.

A Related Work

Sign compression. The sign is the most widely used compressor in this literature [Alistarh et al., 2017, Horváth et al., 2023, Beznosikov et al., 2023]: one bit per coordinate, no index set or scale sent beside it, and a majority vote of signs that is again a sign, so both directions stay at one bit [Bernstein et al., 2019]. Its theory is correspondingly well developed. SignSGD [Bernstein et al., 2018] needs growing batches to converge; its bias is otherwise repaired by error feedback [Seide et al., 2014, Karimireddy et al., 2019], sharpened into EF21 [Richtárik et al., 2021], by momentum [Cutkosky and Mehta, 2020, Sun et al., 2023], or by randomizing the sign operator [Chen et al., 2020, Safaryan and Richtárik, 2021, Jin et al., 2025].

LMO optimizers. Muon [Jordan et al., 2024b] is the spectral-norm instance of a norm-constrained LMO step [Bernstein and Newhouse, 2024, Pethick et al., 2025a], analysed by Li and Hong [2025] and Kovalev [2025], then generalized to arbitrary layer norms by Gluon [Riabinin et al., 2025].

Compressed and federated Muon. Gruntkowska et al. [2025] give the error-feedback framework for bidirectionally compressed Muon/Gluon that EF21-MuonUSign and EF21-MuonSign instantiate. Around it: Qian et al. [2026] compress Gluon with SARAH-type variance reduction; Takezawa et al. [2026] and Zhang and Gao [2025] study federated LMO steps without compression; Ahn et al. [2025] make the orthonormalization low-rank with error feedback, Gupta et al. [2025] quantize Muon's optimizer states, Thérien et al. [2025] quantize the delta of DiLoCo's Muon inner loop [Douillard et al., 2023] to two bits at next to no loss. Concurrently, Mishra et al. [2026] proposed plain SignMuon with an $\mathcal{O}(1/\sqrt{T})$ guarantee, which we show in Appendix H.5 to be a guarantee for a different method: the rate is proved only for the gradient-sign oracle, which is SignSGD's update and invokes no polar factor, while the one bound covering the polar-sign update leaves an unestimated residual that exceeds the bounded quantity exactly where the expected step ascends, as it does throughout the instance of Theorem 1. Two further methods relate sign descent to Muon differently. Bolatov et al. [2026] *alternate* spectral and sign steps rather than composing them; Kravatskiy et al. [2025] mix the geometries *inside* the oracle, so S-Muon stays an LMO for an explicit norm, with the attendant theory. Our placements put the sign *around* the oracle, where it is a compressor and not a geometry, so the counterexamples bear on neither (Appendix H.6).

B The Sign A Family

The three placements of (1) are instances of one iteration, stated here for any optimizer A built on LMO directions. A Sign A method initializes $\mathbf{M}_0 = \mathbf{0}$ and at each iteration $t \geq 1$ performs

$$\begin{aligned} &\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\,\mathbf{G}_t, \qquad \tilde{\mathbf{M}}_t = \mathbf{M}_t \;\text{ or }\; (1-\mu)\mathbf{G}_t + \mu\mathbf{M}_t \;\text{ (Nesterov)},\\ &\mathbf{D}_t = -A(\tilde{\mathbf{M}}_t), \qquad \mathbf{s}_t = \operatorname{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m\times n}, \qquad \mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t, \end{aligned} \tag{6}$$

with $\mathbf{G}_t = \nabla f(\mathbf{X}_{t-1}; \xi_t)$ the stochastic gradient, $\mu \in [0, 1)$ the momentum coefficient and $\operatorname{sign}(0)$ resolved to an independent random ± 1. Momentum is in exponential-moving-average form throughout; the heavy-ball form differs by the positive factor $1/(1-\mu)$, which $\operatorname{sign}(\cdot)$ and the LMO both absorb.

C Federated Learning

In the federated setting the data are split across clients. The training problem becomes

$$\min_{\mathbf{X}\in\mathcal{X}} \Big\{ f(\mathbf{X}) := \tfrac{1}{N}\textstyle\sum_{j=1}^{N} f_j(\mathbf{X}) \Big\}, \tag{7}$$

where $N \geq 1$ is the number of clients and $f_j(\mathbf{X}) = \mathbb{E}_{\xi_j\sim\mathcal{D}_j}[f_j(\mathbf{X}; \xi_j)]$ is the loss on the data $\mathcal{D}_j$ held by client $j \in [N] := \{1, \ldots, N\}$.

Where the oracle runs. The placement decides this. With the sign *after* it, each client orthogonalizes its own momentum and uploads the sign of the result; the server aggregates by majority vote, as SignSGD does [Bernstein et al., 2019]. With the sign *before*, clients upload $\operatorname{sign}(\tilde{\mathbf{M}}_t^{(j)})$; the server

Optimizer	η_0	Up	Down	Rds. to 80%	Test acc (%)
Muon	0.1	32	32	260	85.98 ± 0.26
SignMuon	0.1	**1.09**	**1.09**	540	85.72 ± 0.24
EF21-SignMuon	0.02	**1.09**	32	640	84.71 ± 0.15
MuonUSign	0.05	**1.09**	32	500	84.56 ± 0.35
EF21-MuonSign	0.05	**1.09**	**1.09**	980	83.99 ± 0.28
EF21-MuonUSign	0.01	**1.09**	32	900	83.56 ± 0.41
MuonSign	0.02	**1.09**	**1.09**	860	82.94 ± 0.19
SGD	0.05	32	32	1240	81.69 ± 0.37
SignSGD	0.01	**1.09**	**1.09**	1140	81.44 ± 0.15
Adam	0.001	32	32	—	77.12 ± 1.04

Table 1: CIFAR-10 federated learning on CNN2: $N = 11$ clients, 2000 rounds. Test accuracy is averaged over the final five evaluations and reported as mean $\pm$ s.d. over five seeds; Up and Down are bits per parameter per round in each direction (Appendix M).

votes, then applies a single LMO. Error feedback changes the uplink message rather than where the oracle runs. Its estimator is dense, so the vote argument lapses: the model must be broadcast at full precision unless a second loop compresses it, as EF21-MuonSign does. SignSGD meets the same obstruction: error feedback is analysed there for a single worker only [Karimireddy et al., 2019], and its distributed forms compress the return path with a second loop of their own [Tang et al., 2019]. Appendix Q gives all of this in full, with the two templates and the reason each aggregation is forced; Algorithms 8–9 are the procedures.

The two models of EF21-MuonSign. Its downlink loop splits the method into two iterates,

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t && \text{(server model)}, \\ \mathbf{W}_t &= \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \operatorname{sign}(\mathbf{X}_t - \mathbf{W}_{t-1}) && \text{(broadcast model)}, \end{aligned} \tag{8}$$

with $\mathbf{D}_t = -A(\mathbf{g}_t^{\text{est}})$ and $\alpha_t^{\downarrow} = \operatorname{mean} |\mathbf{X}_t - \mathbf{W}_{t-1}|$. The server model takes the exact LMO step and never leaves the server. The downlink transmits the one-bit update of $\mathbf{W}$, the only model the rest of the method observes: gradients, momentum and the uplink residual are all computed there, whereas the guarantee bounds the gradient at $\mathbf{X}$. The two models separate in Section 4.2.

Memory is the other cost of error feedback, one model-sized buffer per compressed channel. EF21-MuonUSign holds the gradient estimator on each client. EF21-MuonSign holds that estimator and, on the server, the broadcast model $\mathbf{W}$. Only the client-side buffer is a practical constraint: the server is the better-provisioned side.

Results. We compare the ten methods of Figure 2 on CNN2 (two convolutional blocks with BatchNorm and an MLP classifier), with CIFAR-10 split homogeneously across $N = 11$ clients and no local steps; N is odd, so the majority vote cannot tie. Rates are tuned per method at the reporting horizon on a split held out *before* the client partition; two schemes cover all six, worker- and server-side LMO (Algorithms 8–9, Table 14).

Federation separates the methods far more than centralized training does. At one bit per parameter in *both* directions SignMuon reaches 85.72% against Muon's 85.98%, about one seed standard deviation apart. It exceeds SignSGD by 4.3 points, far more than centrally. Five seeds resolve the three placements: they span 2.8 points in the order after, before, both sides, all three clearing SignSGD, the last by 1.5 points where centrally it was level. The second error-feedback loop is not the source of the cost: EF21-MuonSign, compressed in both directions, stands marginally ahead of the uplink-only EF21-MuonUSign.

Less favourably for the theory, EF21-MuonUSign and EF21-MuonSign trail SignMuon by several seed spreads, as they do centrally. The guarantee's cost here is charged to the placement. EF21-MuonSign is scored at its server model $\mathbf{X}_t$, not the model its clients hold (8); the two track each other here, and Section 4.2 shows that they need not (Remark 4). Appendix N adds the learning curves and the two orderings the table does not show.

D Centralized Learning: the Numbers

Method	η_0	Test acc (%)	Ep. to 90%
SignMuon	0.02	94.60 ± 0.15	7.7
Muon	0.1	94.35 ± 0.27	7.7
EF21-SignMuon	0.02	94.31 ± 0.11	9.0
EF21-MuonUSign	0.05	94.14 ± 0.07	10.3
EF21-MuonSign	0.005	94.04 ± 0.10	11.0
MuonUSign	0.02	93.98 ± 0.12	10.3
Adam	0.001	93.37 ± 0.27	20.0
SignSGD	0.002	93.37 ± 0.26	19.0
MuonSign	0.1	93.31 ± 0.21	17.7
SGD	0.02	93.04 ± 0.14	21.3

Table 2: Centralized training of ResNet-18 on CIFAR-10, 75 epochs. Test accuracy is the mean $\pm$ s.d. over three seeds of the last five epochs; "Ep. to 90%" is the mean epoch at which it first reaches 90%.

Table 2 gives the numbers behind Figure 2, including the epochs each method takes to reach 90%, which the curves do not show cleanly. Appendix L adds the training loss, the full 75-epoch curves and the learning-rate sensitivity.

E Language Modelling: the Numbers

Table 3 gives the numbers behind Figure 3, where the differences are too small to read off the curves. Appendix O gives the grouping, the compressor diagnostics and the layer-type localization of the $\mathbf{X}/\mathbf{W}$ gap.

F Preliminaries: LMO and Muon

This appendix states in full the definitions that Section 2 uses in abbreviated form.

Equip $\mathbb{R}^{m\times n}$ with the inner product $\langle \mathbf{M}, \mathbf{D}\rangle = \sum_{i,j} M_{ij} D_{ij}$, let $\|\cdot\|$ be a norm on it, let $\mathcal{B} = \{\mathbf{D} \in \mathbb{R}^{m\times n} : \|\mathbf{D}\| \le 1\}$ be its unit ball, and let $\|\mathbf{M}\|_{\text{dual}} = \max_{\mathbf{D}\in\mathcal{B}} \langle \mathbf{M}, \mathbf{D}\rangle$ be the dual norm. The *Linear Minimization Oracle* (LMO) of $\|\cdot\|$ minimizes a linear form over the unit ball: it outputs

$$A(\mathbf{M}) \in \arg\min_{\mathbf{D}\in\mathcal{B}} \langle \mathbf{M}, \mathbf{D}\rangle = -\big\{\mathbf{D} \in \mathcal{B} : \langle \mathbf{M}, \mathbf{D}\rangle = \|\mathbf{M}\|_{\text{dual}}\big\}, \tag{9}$$

Hence $\langle \mathbf{M}, A(\mathbf{M})\rangle = -\|\mathbf{M}\|_{\text{dual}} \le 0$: the oracle returns a steepest-descent direction with respect to $\|\cdot\|$, normalized to the unit ball. The minimizer need not be unique, whence the inclusion; at rank-deficient $\mathbf{M}$ we use the rank-r selection fixed in the next paragraph. An LMO method minimizes a differentiable objective F by stepping along this direction: at the iterate $\mathbf{X}$ it forms an *effective update direction* $\mathbf{M}$, a stochastic gradient or a momentum estimate in every method below, and

Method	Step	η_0	Val. loss	Steps to 3.35
Muon (record #40)	LMO	0.06	**3.2785**	1.00×
EF21-SignMuon	LMO	0.06	3.2860	1.02×
SignMuon	SIGN	0.03	3.2881	1.02×
MuonUSign	LMO	0.06	3.2959	1.05×
EF21-MuonUSign	LMO	0.06	3.3203	1.13×
EF21-MuonSign ($\mathbf{W}$)	LMO	0.06	3.3213	1.14×
MuonSign	SIGN	0.03	3.3249	1.14×
SignSGD	SIGN	0.03	3.4049	–
EF21-MuonSign ($\mathbf{X}$)	LMO	0.06	5.5198	–

Table 3: NanoGPT after 2330 steps (611M tokens), one run per method. "Steps to 3.35" is relative to Muon. The last row is the server model $\mathbf{X}$ of the same run as the $\mathbf{W}$ row.

moves along $A(\mathbf{M})$, the minimizer over $\mathcal{B}$ of the first-order model $\mathbf{D} \mapsto F(\mathbf{X}) + \langle \mathbf{M}, \mathbf{D} \rangle$. Each layer of Section 2 is given its own norm and so its own oracle [Riabinin et al., 2025]; for the role of the construction in optimizer design see [Bernstein and Newhouse, 2024, Bernstein, 2025, Pethick et al., 2025a, Kovalev, 2025, Riabinin et al., 2025, Cesista, 2025, Kravatskiy et al., 2025].

Muon is the instance in which each matrix layer is equipped with the spectral norm $\|\cdot\|_{2\to 2}$, so that the oracle uses the matrix structure of the gradient and the update direction is obtained by orthogonalizing the gradient matrix Bernstein and Newhouse [2024]. Let $\mathbf{M} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the singular value decomposition (SVD) of the matrix $\mathbf{M} \in \mathbb{R}^{m\times n}$, where $\mathbf{U} \in \mathbb{R}^{m\times r}$ and $\mathbf{V} \in \mathbb{R}^{n\times r}$ are the orthonormal matrices of singular vectors, $\mathbf{\Sigma} \in \mathbb{R}^{r\times r}$ is the diagonal matrix of singular values, and $r = \operatorname{rank}(\mathbf{M})$. Then the LMO direction is $A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top$: Muon selects an orthonormal update direction corresponding to the solution of the linear minimization problem induced by the spectral norm geometry. Orthogonalization equalizes the singular values of the update, so that the directions in which the gradient is weak are not crowded out by the dominant ones [Jordan et al., 2024b]; large-scale pretraining studies report a corresponding efficiency gain over AdamW [Liu et al., 2025, Shah et al., 2025]. The reference implementation obtains $\mathbf{U}_t\mathbf{V}_t^\top$ from a fixed number of Newton–Schulz iterations rather than from an SVD [Jordan et al., 2024b]; Algorithm 1 gives the procedure and the coefficients we use.

Choice of layer norm and the shape factor. Muon is also presented in the RMS→RMS operator norm, $\sqrt{n/m}\,\|\cdot\|_{2\to 2}$ on $\mathbb{R}^{m\times n}$ [Bernstein and Newhouse, 2024, Pethick et al., 2025a]. The two presentations give different oracles: the unit ball of $\sqrt{n/m}\,\|\cdot\|_{2\to 2}$ is the spectral ball scaled by $\sqrt{m/n}$, so its LMO is $-\sqrt{m/n}\,\mathbf{U}\mathbf{V}^\top$. The unscaled $-\mathbf{U}\mathbf{V}^\top$ therefore belongs to the spectral norm and to no other, since a norm whose oracle it is has $\|\mathbf{M}\|_{\mathrm{dual}} = \langle \mathbf{M}, \mathbf{U}\mathbf{V}^\top \rangle = \|\mathbf{M}\|_*$ for every $\mathbf{M}$ and hence equals $\|\cdot\|_{2\to 2}$ by biduality. We take the spectral norm, in which Assumption 2 and every statement below are stated, and treat the shape factor as a step-size question, for three reasons. It is a positive per-layer constant, so it changes neither $\operatorname{sign}(\cdot)$ nor the sign of any descent inner product, and no result of this paper is sensitive to it. The RMS→RMS value $\sqrt{m/n}$ is not in fact the factor Muon uses: the reference implementation applies $\sqrt{\max(1, m/n)}$ [Jordan et al., 2024b], which agrees with it only for $m \geq n$ and is the factor our unit-gain rule returns (Appendix P). And that rule must also scale the two sign-terminated placements, for which no norm supplies a scale at all: by Theorems 1 and 3 they are oracles for none.

Layerwise generalized smoothness. Corollary 2 requires Assumption 2 only in its weaker layerwise (L^0, L^1) form, which replaces the constant L_i^g by $L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*$ [Riabinin et al., 2025, Pethick et al., 2025b]: for every layer i and all parameter tuples $\mathbf{X}, \mathbf{Y}$,

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \leq \left(L_i^{0,g} + L_i^{1,g}\|\nabla_i g(\mathbf{X})\|_*\right) \|\mathbf{X}_i - \mathbf{Y}_i\|_{2\to 2}, \tag{10}$$

again with one pair of constants per layer for $g = f$ and per layer and client for $g = f_j$; at $L_i^{1,g} = 0$ the display is Assumption 2. It is Assumptions 8–9 of Gruntkowska et al. [2025], the hypotheses of the theorem Corollary 2 quotes, stated there as here for *arbitrary* pairs: the tuples may differ in every block although only block i enters the bound. That quantifier is strong. Fixing $\mathbf{X}_i = \mathbf{Y}_i$ while varying the remaining blocks forces $\nabla_i g(\mathbf{X}) = \nabla_i g(\mathbf{Y})$, so each $\nabla_i g$ depends on its own block alone, and a function satisfying the bound with finite constants is additively separable across layers. At $p = 1$, the setting of every counterexample and synthetic measurement in this paper, the restriction is empty and the display is ordinary (L^0, L^1)-smoothness. For a multilayer network it is the framework's idealization, and we inherit it unweakened: the descent lemma behind Theorem 5 applies the bound along a step in which every layer moves, which the restriction to pairs differing in one block would not license.

G Width-one blocks: vector parameters

A vector parameter is represented as a $1 \times n$ or $m \times 1$ block. The spectral, nuclear and Euclidean norms coincide, and $\operatorname{polar}(\mathbf{g}) = \mathbf{g}/\|\mathbf{g}\|_2$ for $\mathbf{g} \neq \mathbf{0}$. The norm-equivalence constant in Corollary 1 is $\bar{\rho}_i = 1$.

For a row vector, the sign placements satisfy

$$\operatorname{sign}(\operatorname{polar}(\mathbf{g})) = \operatorname{sign}(\mathbf{g}), \qquad \operatorname{polar}(\operatorname{sign}(\mathbf{g})) = \tfrac{1}{\sqrt{n}}\operatorname{sign}(\mathbf{g}),$$

The two-sided placement also returns $\operatorname{sign}(\mathbf{g})$, so all three reduce to SignSGD up to a positive step-size factor.

Parameter groups in the experiments. CIFAR runs assign biases, normalization parameters and the classifier head to AdamW. NanoGPT assigns scalars, embeddings and the head to distributed Adam, and applies the tested method to hidden matrices and gate weights, including the 1×12 `smear_gate`.

H Divergence results

H.1 Ascent criterion on linear objectives

The criterion quoted in Section 3 collapses each of the three methods to a single scalar inequality and removes momentum from the discussion entirely.

Proposition 1 (Ascent criterion on linear objectives) *Run any of the three methods (1) on the linear objective (2) with $\mathbf{G} \neq \mathbf{0}$, from an arbitrary $\mathbf{X}_0$, with any momentum coefficient $\mu \in [0, 1)$ under either the Standard or the Nesterov rule. Assume that every argument of* sign *in the chosen placement, evaluated at $\mathbf{G}$, has no zero entries. Then $\tilde{\mathbf{M}}_t = \gamma_t \, \mathbf{G}$ with $\gamma_t > 0$, the update direction is the constant matrix $\mathbf{s}(\mathbf{G})$ obtained by substituting $\mathbf{G}$ for $\tilde{\mathbf{M}}_t$ in (1), and*

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \left\langle \mathbf{G},\, \mathbf{s}(\mathbf{G}) \right\rangle.$$

In particular, if $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ then f strictly increases at every iteration for any $\eta_t > 0$, and $f(\mathbf{X}_t) \to +\infty$ whenever $\sum_t \eta_t = \infty$ (for instance, for any constant step size).

Proof of Proposition 1. On the linear objective (2) the gradient is globally constant, $\mathbf{G}_t = \mathbf{G}$, so the momentum buffer of (6) is $\mathbf{M}_t = (1-\mu) \sum_{i=0}^{t-1} \mu^i \, \mathbf{G} = (1 - \mu^t) \, \mathbf{G}$. Both momentum rules then return a positive multiple of $\mathbf{G}$,

$$\tilde{\mathbf{M}}_t = \gamma_t \, \mathbf{G}, \qquad \gamma_t = \begin{cases} 1 - \mu^t, & \text{(Standard)}, \\ 1 - \mu^{t+1}, & \text{(Nesterov)}, \end{cases} \tag{11}$$

both positive for $t \geq 1$ because $\mu \in [0, 1)$; the Nesterov case is $(1-\mu)\mathbf{G} + \mu(1-\mu^t)\mathbf{G}$. (Under the heavy-ball convention every γ_t is multiplied by $1/(1-\mu)$, which changes nothing below.) The elementwise $\operatorname{sign}(\cdot)$ and the Muon LMO $\operatorname{polar}(\cdot)$ are each invariant under multiplication by a positive scalar, so evaluating (1) at $\tilde{\mathbf{M}}_t = \gamma_t \mathbf{G}$ returns the same matrix as evaluating it at $\mathbf{G}$; that is, $\mathbf{s}_t = \mathbf{s}(\mathbf{G})$ for every t, independently of μ and of the momentum variant. Hence $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = \langle \mathbf{G}, \mathbf{X}_t - \mathbf{X}_{t-1} \rangle = -\eta_t \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle$, which is (3). If $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$, then $f(\mathbf{X}_t) = f(\mathbf{X}_0) - \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \sum_{i=1}^{t} \eta_i$ increases strictly at every step and diverges to $+\infty$ whenever $\sum_t \eta_t = \infty$. ■

By Proposition 1, each divergence theorem reduces to exhibiting a single gradient $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$; the three proofs below accomplish exactly this, and momentum requires no further comment.

H.2 Proof of Theorem 1: SignMuon

Theorem 1 (Divergence of SignMuon) *There is a matrix $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\left\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \right\rangle = -\frac{42468}{103} < 0$. Hence SignMuon ascends on (2): f strictly increases at every iteration for all $\eta_t > 0$, all $\mu \in [0, 1)$, and both momentum variants.*

For SignMuon $\mathbf{s}(\mathbf{G}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G}))$, so by (3) it suffices to construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ with $\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \rangle < 0$. Group the SVD $\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ of an invertible $\mathbf{G}$ as the polar decomposition $\mathbf{G} = \mathbf{Q}\mathbf{H}$, with $\mathbf{Q} = \mathbf{U}\mathbf{V}^\top = \operatorname{polar}(\mathbf{G})$ orthogonal and $\mathbf{H} = \mathbf{V}\mathbf{\Sigma}\mathbf{V}^\top \succ 0$ symmetric. Only the symmetric part of $\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q})$ then contributes to the trace against $\mathbf{H}$:

$$\langle \mathbf{G}, \operatorname{sign}(\operatorname{polar}(\mathbf{G})) \rangle = \operatorname{tr}\big(\mathbf{H}\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q})\big) = \big\langle \mathbf{H}, \operatorname{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q})) \big\rangle, \tag{12}$$

so a counterexample with polar factor $\mathbf{Q}$ exists precisely when $\lambda_{\min}\big(\mathrm{sym}(\mathbf{Q}^\top \operatorname{sign}(\mathbf{Q}))\big) < 0$: necessity because $\mathbf{H} \succ 0$, sufficiency by taking $\mathbf{H} = \mathbf{w}\mathbf{w}^\top + \delta\mathbf{I}$ at a minimizing eigenvector $\mathbf{w}$ and $\delta > 0$ small, which keeps $\mathbf{G}$ invertible and its polar factor unique. The question thus concerns orthogonal matrices alone.

We construct $\mathbf{G} \in \mathbb{R}^{4\times 4}$ by defining a specific orthogonal matrix $\mathbf{O}$ and a specific rank-1 principal component $\mathbf{u}_1\mathbf{v}_1^\top$. Let the orthogonal matrix $\mathbf{O}$ be given by the following exact rational numbers:

$$\mathbf{O} = \frac{1}{103}\begin{pmatrix} 101 & 20 & 2 & -2 \\ -20 & 97 & 20 & -20 \\ -2 & 20 & 2 & 101 \\ -2 & 20 & -101 & -2 \end{pmatrix}. \tag{13}$$

Because no entry is zero, its element-wise sign matrix $\mathbf{S} = \operatorname{sign}(\mathbf{O})$ is uniquely defined. Now, let $\mathbf{u}_1$ and $\mathbf{v}_1$ be the following exact unit vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ -3 \\ 10 \\ 10 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{\sqrt{309}}\begin{pmatrix} 10 \\ 3 \\ -10 \\ 10 \end{pmatrix}. \tag{14}$$

A direct check gives $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$: $\mathbf{u}_1$ and $\mathbf{v}_1$ are a left and a right singular vector of this orthogonal block. The crucial feature of this geometry is that the Frobenius inner product between this rank-1 component and the sign matrix $\mathbf{S}$ yields an exact, strictly negative fraction:

$$\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle = \mathbf{u}_1^\top \mathbf{S}\mathbf{v}_1 = -\frac{43}{103}. \tag{15}$$

We construct the gradient matrix $\mathbf{G}$ by assigning a large singular value ($\sigma_1 = 1001$) to this pathological component and a singular value of 1 to the rest of the orthogonal space. Exactly,

$$\mathbf{G} := 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O} = \frac{1}{309}\begin{pmatrix} 100303 & 30060 & -99994 & 99994 \\ -30060 & -8709 & 30060 & -30060 \\ 99994 & 30060 & -99994 & 100303 \\ 99994 & 30060 & -100303 & 99994 \end{pmatrix}, \tag{16}$$

or, to one decimal,

$$\mathbf{G} \approx \begin{pmatrix} 324.6 & 97.3 & -323.6 & 323.6 \\ -97.3 & -28.2 & 97.3 & -97.3 \\ 323.6 & 97.3 & -323.6 & 324.6 \\ 323.6 & 97.3 & -324.6 & 323.6 \end{pmatrix}. \tag{17}$$

Since $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$ and $\mathbf{v}_1^\top\mathbf{v}_1 = 1$,

$$\mathbf{G}\mathbf{v}_1 = 1000\,\mathbf{u}_1(\mathbf{v}_1^\top\mathbf{v}_1) + \mathbf{O}\mathbf{v}_1 = 1001\,\mathbf{u}_1,$$

so $(\mathbf{u}_1, \mathbf{v}_1)$ is a singular pair of $\mathbf{G}$ with $\sigma_1 = 1001$. For any $\mathbf{w} \perp \mathbf{v}_1$ we have $\mathbf{G}\mathbf{w} = \mathbf{O}\mathbf{w}$; as $\mathbf{O}$ is orthogonal and $\mathbf{O}\mathbf{v}_1 = \mathbf{u}_1$, the restriction $\mathbf{O}|_{\mathbf{v}_1^\perp}$ is an isometry onto $\mathbf{u}_1^\perp$, hence $\sigma_2 = \sigma_3 = \sigma_4 = 1$. Moreover, with $\mathbf{u}_k = \mathbf{O}\mathbf{v}_k$ for $k \geq 2$,

$$\textstyle\sum_{k\geq 2} \mathbf{u}_k\mathbf{v}_k^\top = \mathbf{O}\big(\mathbf{I} - \mathbf{v}_1\mathbf{v}_1^\top\big) = \mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top,$$

so $\mathbf{U}_G\mathbf{V}_G^\top = \mathbf{u}_1\mathbf{v}_1^\top + (\mathbf{O} - \mathbf{u}_1\mathbf{v}_1^\top) = \mathbf{O}$. All four singular values are positive, so $\mathbf{G}$ is invertible and $\operatorname{polar}(\mathbf{G}) = \mathbf{O}$ is its unique polar factor. The sign matrix (21) of Theorems 2–3 is full rank as well, so in all three proofs the argument of polar is invertible and its polar factor is unique: no statement depends on the selection rule fixed above for rank-deficient arguments.

Substituting the resulting matrix $\mathbf{G}$ into the descent condition yields:

$$\langle \mathbf{G}, \mathbf{S}\rangle = \langle 1000\,\mathbf{u}_1\mathbf{v}_1^\top + \mathbf{O}, \mathbf{S}\rangle = 1000\langle \mathbf{u}_1\mathbf{v}_1^\top, \mathbf{S}\rangle + \langle \mathbf{O}, \mathbf{S}\rangle. \tag{18}$$

The inner product $\langle \mathbf{O}, \mathbf{S}\rangle$ is equivalent to the L_1 norm (sum of absolute values) of $\mathbf{O}$, which equals exactly $\frac{532}{103}$. Therefore:

$$\langle \mathbf{G}, \mathbf{S}\rangle = 1000\left(-\frac{43}{103}\right) + \frac{532}{103} = -\frac{42468}{103} \approx -412.31. \tag{19}$$

Here $\mathbf{S} = \operatorname{sign}(\mathbf{O}) = \operatorname{sign}(\operatorname{polar}(\mathbf{G})) = \mathbf{s}(\mathbf{G})$, so $\langle \mathbf{G}, \mathbf{s}(\mathbf{G})\rangle = -\frac{42468}{103} < 0$. By Proposition 1, SignMuon strictly ascends, $f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = \frac{42468}{103}\,\eta_t > 0$ at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

H.3 Proof of Theorem 2: MuonUSign

Theorem 2 (Divergence of MuonUSign) *There is a matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *with* $\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\rangle \approx -13.89 < 0$. *Hence MuonUSign ascends on (2) for all* $\eta_t > 0$, *all* $\mu \in [0,1)$, *and both momentum variants.*

MuonUSign (Algorithm 4) applies the sign *before* the LMO, so $\mathbf{s}(\mathbf{G}) = \mathrm{polar}(\mathrm{sign}(\mathbf{G}))$. By (3) it suffices to construct $\mathbf{G} \in \mathbb{R}^{5\times 5}$ with $\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\rangle < 0$.

Here the step depends on $\mathbf{G}$ only through the magnitudes $|G_{ij}|$ and the sign pattern $\mathbf{S} = \mathrm{sign}(\mathbf{G})$; write $\mathbf{D} = \mathrm{polar}(\mathbf{S})$ for the direction it produces. Under the randomized convention $\mathbf{S} \in \{\pm 1\}^{m\times n}$ throughout, and the entrywise identity $G_{ij} = |G_{ij}|\, S_{ij}$ holds without exception, both sides vanishing wherever $G_{ij} = 0$. Consequently

$$\big\langle \mathbf{G}, \mathrm{polar}(\mathrm{sign}(\mathbf{G}))\big\rangle = \textstyle\sum_{i,j} |G_{ij}|\, S_{ij} D_{ij}, \tag{20}$$

and likewise $\langle \mathbf{G}, \mathrm{sign}(\mathbf{D})\rangle = \sum_{i,j} |G_{ij}|\, S_{ij}\, \mathrm{sign}(D_{ij})$ for the MuonSign step of Theorem 3. Every summand in (20) is nonnegative unless some entry is *mismatched*, $S_{ij}D_{ij} < 0$. It therefore suffices to exhibit one sign matrix with a mismatched entry: inflating $|G_{ij}|$ there, with the other magnitudes held fixed, drives the sum below zero.

Fix the full-rank sign matrix $\mathbf{S} \in \{-1,1\}^{5\times 5}$,

$$\mathbf{S} = \begin{pmatrix} -1 & -1 & 1 & 1 & 1 \\ -1 & -1 & 1 & -1 & -1 \\ 1 & -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & -1 & 1 \end{pmatrix}, \tag{21}$$

and let $\mathbf{D} = \mathrm{polar}(\mathbf{S})$ be its (unique, since $\mathbf{S}$ is full rank) polar factor. A direct computation gives $D_{4,2} = -1/\sqrt{17} \approx -0.2425 < 0$ while $\mathrm{sign}(D_{i,j}) = S_{i,j}$ at all 24 other entries; equivalently, $\mathrm{sign}(\mathbf{D})$ and $\mathbf{S}$ disagree at exactly the single entry $(4,2)$, where $S_{4,2} = +1$. We exploit this lone mismatch. Define

$$\mathbf{G} = \epsilon\, \mathbf{S} + (M - \epsilon)\, \mathbf{e}_4 \mathbf{e}_2^\top, \qquad \epsilon > 0,\ M > \epsilon, \tag{22}$$

so that $G_{4,2} = M > 0$ and $G_{i,j} = \epsilon S_{i,j}$ otherwise; hence $\mathrm{sign}(\mathbf{G}) = \mathbf{S}$ and $\mathrm{polar}(\mathrm{sign}(\mathbf{G})) = \mathbf{D}$ for every $M > 0$. The descent inner product splits as

$$\langle \mathbf{G}, \mathbf{D}\rangle = \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j} D_{i,j} \;+\; M\, D_{4,2} = \epsilon\, C + M\, D_{4,2}, \tag{23}$$

where $C := \sum_{(i,j)\neq(4,2)} |D_{i,j}| = 6 + 18/\sqrt{17} \approx 10.366 > 0$ is fixed (every such entry agrees in sign with $\mathbf{S}$), while $M\, D_{4,2} = -M/\sqrt{17} \to -\infty$. Thus $\langle \mathbf{G}, \mathbf{D}\rangle < 0$ for any $M > \sqrt{17}\, C\epsilon \approx 42.7\,\epsilon$; with $\epsilon = 1$, $M = 100$ one obtains $\langle \mathbf{G}, \mathbf{D}\rangle = 6 - 82/\sqrt{17} \approx -13.89$, for the exact polar factor that the theorem is stated over. By Proposition 1, MuonUSign strictly ascends on $f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X}\rangle$ for every $\eta_t > 0$, every $\mu \in [0,1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. ■

H.4 Proof of Theorem 3: MuonSign

Theorem 3 (Divergence of MuonSign) *For the* same *matrix* $\mathbf{G} \in \mathbb{R}^{5\times 5}$ *as in Theorem 2,* $\big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G})))\big\rangle = -76 < 0$. *Hence MuonSign ascends on (2) for all* $\eta_t > 0$, *all* $\mu \in [0,1)$, *and both momentum variants.*

MuonSign (Algorithm 5) signs the polar factor as well, so $\mathbf{s}(\mathbf{G}) = \mathrm{sign}(\mathrm{polar}(\mathrm{sign}(\mathbf{G}))) = \mathrm{sign}(\mathbf{D})$. We reuse the *same* $\mathbf{S}$ and $\mathbf{G}$ of (21)–(22): since $\mathrm{sign}(\mathbf{G}) = \mathbf{S}$, the bidirectional step is the constant matrix $\mathrm{sign}(\mathbf{D})$, which agrees with $\mathbf{S}$ at all 24 entries except $(4,2)$, where $\mathrm{sign}(D_{4,2}) = -1 = -S_{4,2}$. Using $S_{i,j}^2 = 1$ everywhere,

$$\begin{aligned} \langle \mathbf{G}, \mathrm{sign}(\mathbf{D})\rangle &= \epsilon \sum_{(i,j)\neq(4,2)} S_{i,j}\, \mathrm{sign}(D_{i,j}) + M\ \mathrm{sign}(D_{4,2}) \\ &= 24\,\epsilon - M. \end{aligned} \tag{24}$$

This is negative for every $M > 24\epsilon$; with $\epsilon = 1$, $M = 100$ it equals exactly -76. By Proposition 1, MuonSign strictly ascends on $f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X} \rangle$ for every $\eta_t > 0$, every $\mu \in [0, 1)$, and both momentum variants; $f(\mathbf{X}_t) \to +\infty$ under any non-summable step size. In particular, the *same* 5×5 linear instance is an ascent instance for the uplink-only placement (MuonUSign) and for the bidirectional one (MuonSign) alike. ■

H.5 Comparison with the convergence claim for SignMuon

Concurrently, Mishra et al. [2026] introduced the same SignMuon update, with direction $\mathbf{D}_t = \mathrm{sign}(\mathrm{polar}(\mathbf{M}_t))/\sqrt{mn}$. Absorbing $1/\sqrt{mn}$ into the learning rate gives the step of Section 3. Their abstract attributes an $\mathcal{O}(T^{-1/2})$ stationarity rate to SignMuon; the rate theorem is stated for a gradient-sign instantiation.

Sign-error decomposition

For a transmitted sign matrix $\mathbf{S}_t$ and $\mathbf{D}_t = \mathbf{S}_t/\sqrt{mn}$, define

$$\mathcal{G}_T = \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[\frac{\|\nabla f(\mathbf{X}_t)\|_1}{\sqrt{mn}}\right],$$
$$q_{ij,t} = \Pr(\mathbf{S}_{t,ij} \neq \mathrm{sign}([\nabla f(\mathbf{X}_t)]_{ij}) \mid \mathbf{X}_t).$$

The identity

$$\mathbb{E}\big[\langle \nabla f(\mathbf{X}_t), \mathbf{D}_t \rangle \,\big|\, \mathbf{X}_t\big] = \frac{1}{\sqrt{mn}} \sum_{i,j} \big|[\nabla f(\mathbf{X}_t)]_{ij}\big| \big(1 - 2q_{ij,t}\big) \tag{25}$$

and the spectral-smoothness descent lemma give, for any sign oracle,

$$\mathcal{G}_T \;\le\; \frac{f(\mathbf{X}_0) - f^*}{\eta T} + \frac{L_*}{2}\,\eta + \underbrace{\frac{2}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[\frac{1}{\sqrt{mn}} \sum_{i,j} \big|[\nabla f(\mathbf{X}_t)]_{ij}\big|\, q_{ij,t}\right]}_{=:\, R_T}. \tag{26}$$

A convergence rate therefore requires control of R_T.

Gradient-sign convergence rate

The residual estimate uses $\mathbf{S}_t = \mathrm{sign}(\widetilde{\mathbf{G}}_t)$, where $\widetilde{\mathbf{G}}_t$ is an unbiased stochastic gradient with coordinatewise variance at most σ_{ij}^2/n_b at batch size n_b. A Markov–Jensen argument gives

$$\big|[\nabla f(\mathbf{X}_t)]_{ij}\big| q_{ij,t} \le \frac{\sigma_{ij}}{\sqrt{n_b}}, \qquad R_T \le \frac{2\|\sigma\|_1}{\sqrt{mn\, n_b}}, \qquad \|\sigma\|_1 = \sum_{i,j} \sigma_{ij}.$$

With $n_b = T$ and $\eta \asymp T^{-1/2}$, (26) yields $\mathcal{G}_T = \mathcal{O}(T^{-1/2})$. This update is normalized SignSGD without momentum, analysed in single-worker and majority-vote forms under spectral smoothness [Bernstein et al., 2018, Mishra et al., 2026]. For SignMuon, the polar transformation introduces additional sign errors that can persist with exact gradients. Their analysis leaves the corresponding R_T unestimated.

Evaluation on the SignMuon counterexample

For the 4×4 gradient $\mathbf{G}$ of Theorem 1, Proposition 1 gives the constant sign matrix $\mathbf{S} = \mathrm{sign}(\mathrm{polar}(\mathbf{G}))$ for every momentum coefficient and either momentum rule. Thus q_{ij} is the indicator of a sign mismatch, and (25) becomes

$$\sum_{i,j} |\mathbf{G}_{ij}|\,(1 - 2q_{ij}) = \big\langle \mathbf{G}, \mathrm{sign}(\mathrm{polar}(\mathbf{G})) \big\rangle = -\tfrac{42468}{103} \;<\; 0. \tag{27}$$

Since $\sqrt{mn} = 4$,

$$R_T = \mathcal{G}_T + \tfrac{10617}{103} \qquad \text{for every } T. \tag{28}$$

The normalized SignMuon step therefore satisfies

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta\langle \mathbf{G}, \mathbf{D}_t\rangle = \eta\,\tfrac{10617}{103} > 0.$$

The smooth bounded-below modification of Remark 1 preserves this trajectory and all the identities above. On that objective, (26) holds with a finite f^*, yet $R_T > \mathcal{G}_T$ at every horizon. Thus the generic inequality is compatible with ascent, while the proved rate applies to the gradient-sign update.

H.6 Comparison with S-Muon

Taking the norms dual to the Ky Fan k-norms, Kravatskiy et al. [2025] obtain the *Fanion* family, whose updates $\sum_{i\le k} \mathbf{u}_i\mathbf{v}_i^\top$ interpolate between the rank-one step of the nuclear norm and Muon's full-rank $\mathbf{U}\mathbf{V}^\top$ at $k = \min(m,n)$; a conic combination of LMO algorithms is again an LMO algorithm, for the norm dual to the corresponding combination of dual norms. Their S-Muon is one such combination, $\tau\,\mathbf{U}\mathbf{V}^\top + (1-\tau)\,c\,\operatorname{sign}(\mathbf{M}_t)$ with fixed $\tau \in [0,1]$, $c > 0$ (their notation differs; we reserve α for compressor contraction and η for the learning rate). There the sign enters *inside* the oracle, so the step is still an LMO for an explicit norm and inherits the convergence theory of one; our three placements act *around* the oracle. A caution from the same work applies to us directly: they exhibit an LMO method (rank-one Neon) markedly worse than Muon in practice despite sharing its convergence asymptotics in the bounds of Kovalev [2025] and Riabinin et al. [2025], from which our own guarantee descends. A rate of the form $\mathcal{O}(T^{-1/2})$ is not a prediction of the performance a method will attain.

H.7 Proof of Theorem 4: EF21-SignMuon

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0,1)$, and either momentum variant, there is an L-smooth (Assumption 2), bounded-below (Assumption 1) function $f : \mathbb{R}^{2\times 2} \to \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit constant $c = c(f) > 0$,*

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = c\,L\eta^2 > 0 \qquad (t \ge 3), \tag{29}$$

so $f(\mathbf{X}_t) \to +\infty$. In particular, no step-size rule $\eta = \eta(L,\mu)$ using only the smoothness and momentum constants can make the method convergent.

Recall from the main text that EF21-SignMuon (Algorithm 3) does not step along the polar factor $\mathbf{D}_t = \operatorname{polar}(\tilde{\mathbf{M}}_t)$ itself, but along the error-feedback estimate $\mathbf{d}_t^{\mathrm{est}}$ of (4), followed by $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta\,\mathbf{d}_t^{\mathrm{est}}$. A *single* magnitude α_t rescales the signs of *all* entries at once, and it is this coupling that the counterexample exploits. We now prove Theorem 4.

Proof idea

The mechanism. The error-feedback update (4) moves *every* entry of $\mathbf{d}_t^{\mathrm{est}}$ by the same magnitude α_t; only the signs are individual. Suppose then that one entry must track a target alternating between $+1$ and -1 while another must track a small constant $-\varepsilon$. The alternating entry keeps its residual, and with it α_t, at $\Theta(1)$; the constant entry is therefore displaced by $\pm\Theta(1)$ at every step and can only oscillate about its target, never settle on it. Which side of the target the oscillation favours is decided by its *phase*, and a one-bit sign carries no information by which a phase could be corrected. In the unfavourable phase the estimate of the constant entry has a time average of the sign *opposite* to $-\varepsilon$, and the iterate driven by that estimate moves the wrong way forever.

From the sketch to an instance. The sketch is not yet a counterexample: in EF21-SignMuon the quantity tracked is the polar factor $\mathbf{D}_t = \operatorname{polar}(\tilde{\mathbf{M}}_t)$, of unit spectral norm, and the gradients behind it must all come from one smooth function. Both constraints are met at size 2×2 by the reflections $\bar{\mathbf{D}}^{\pm} = \left(\begin{smallmatrix} a & \pm b \\ \pm b & -a \end{smallmatrix}\right)$ with $a^2 + b^2 = 1$: one matrix the oracle can emit carries both roles of the sketch at once, the large alternating entries on the off-diagonal ($b = \frac{24}{25}$) and the small constant ones on the diagonal ($a = \frac{7}{25}$), coupled by the shared α_t. On these targets the estimate enters a period-two cycle in which the time average of the $(2,2)$-entry is positive although every target value is $-\frac{7}{25}$, so $(\mathbf{X}_t)_{22}$ travels to $-\infty$, the direction in which the objective increases (Figure 1, right). Nor does the mechanism rest on degeneracy: the momentum matrices of the divergent tail have condition number $\frac{16}{9}$ throughout.

The role of the preamble. The unfavourable phase must be arranged. From $\mathbf{d}_0^{\mathrm{est}} = \mathbf{0}$, the alternating targets alone lock the estimate into a period-two cycle whose diagonal average has the *correct* sign. The recursion (4) is piecewise affine, the pieces indexed by the sign pattern of the residual, and the harmless cycle and the wrong-sign one lie in different pieces, which the dynamics cannot join. The target sequence of Part 2 therefore opens with two preamble steps, whose sole purpose is to place the estimate in the piece containing the wrong-sign cycle.

Eliminating the parameters. The step size and the smoothness constant only rescale the trajectory, so $\eta = 1$ and one value of L suffice (Part 1). Momentum determines only which gradients produce a given target sequence, not how the recursion (4) responds to it; solving the momentum recursion for the gradients therefore settles every $(\mu, \text{variant})$ at once (Part 3).

Accordingly the proof has three parts, and they are independent. Part 1 removes L and η by rescaling. Part 2 carries the dynamics, and is a finite computation in exact rational arithmetic: on one fixed sequence of LMO targets, the estimate enters a limit cycle whose diagonal has the wrong sign. Part 3 is an existence argument only: it exhibits a smooth function whose gradients generate that target sequence for every μ and either momentum variant. A reader willing to grant that some smooth objective produces the targets can stop after Part 2.

Proof

Part 1: rescaling.

Lemma 1 (Scale reduction) *If $\tilde{f}$ is $\tilde{L}$-smooth with EF21-SignMuon iterates $\tilde{\mathbf{X}}_t$ at step size 1, then $f(\mathbf{X}) := \frac{L\eta^2}{\tilde{L}}\,\tilde{f}(\mathbf{X}/\eta)$ is L-smooth and its run at step size η (same μ, same variant, from $\mathbf{0}$) satisfies $\mathbf{X}_t = \eta\,\tilde{\mathbf{X}}_t$ and $f(\mathbf{X}_t) = \frac{L\eta^2}{\tilde{L}}\,\tilde{f}(\tilde{\mathbf{X}}_t)$.*

Proof. $\nabla f(\mathbf{X}) = \frac{L\eta}{\tilde{L}}\nabla\tilde{f}(\mathbf{X}/\eta)$, so f is L-smooth. Suppose $\mathbf{X}_s = \eta\tilde{\mathbf{X}}_s$ for $s < t$: the two gradients then differ by the positive factor $\frac{L\eta}{\tilde{L}}$, which $\mathbf{M}_t$ and $\tilde{\mathbf{M}}_t$ (positive combinations of past gradients) inherit and polar discards. Hence $\mathbf{D}_t$ and $\mathbf{d}_t^{\mathrm{est}}$ match the normalized run, and $\mathbf{X}_t = \eta\tilde{\mathbf{X}}_t$. ■

It therefore suffices to exhibit, for each $(\mu, \text{variant})$, a C^∞ $\tilde{L}$-smooth $\tilde{f}$ on which the *normalized* run ($\eta = 1$) obeys (29); that $\tilde{f}$ may be taken bounded below (Assumption 1) is shown afterwards (Remark 1). We fix $\eta = 1$ from now on.

Part 2: the limit cycle of the estimate. The dynamical core of the proof is the behavior of the recursion (4) on the fixed target sequence

$$\mathbf{D}_1 = \mathbf{S}_1, \quad \mathbf{D}_2 = \mathbf{S}_2, \quad \mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}\ (t \geq 3), \tag{30}$$

$$\bar{\mathbf{D}}^{\pm} = \begin{pmatrix} 7/25 & \pm 24/25 \\ \pm 24/25 & -7/25 \end{pmatrix},$$
$$\mathbf{S}_1 = \begin{pmatrix} -4/5 & 3/5 \\ -3/5 & -4/5 \end{pmatrix}, \quad \mathbf{S}_2 = \begin{pmatrix} 3/5 & -4/5 \\ 0 & 0 \end{pmatrix}. \tag{31}$$

The divergence originates in the tail ($t \geq 3$): the reflections $\bar{\mathbf{D}}^{\pm}$ share the diagonal $(\frac{7}{25}, -\frac{7}{25})$, while their off-diagonal $\pm\frac{24}{25}$ reverses sign at every step. The rotation $\mathbf{S}_1$ and the rank-one $\mathbf{S}_2$ form a two-step preamble. Only $\mathbf{S}_2$ requires comment: at a rank-deficient argument the spectral-ball LMO is not unique, and Section 2 resolves it through the thin SVD that retains only the nonzero singular directions, $\mathrm{polar}(\mathbf{M}) = \mathbf{U}\mathbf{V}^\top$ with one column of $\mathbf{U}, \mathbf{V}$ per nonzero singular value of $\mathbf{M}$; under that convention a rank-one matrix of unit spectral norm, such as $\mathbf{S}_2$, is its own polar factor, and this is also what the implementation computes. The next lemma says what the preamble is for, and that something like it is unavoidable.

Lemma 2 (The cycle reached without the preamble) *Write $\bar{\mathbf{D}}^{\pm} = \left(\begin{smallmatrix} a & \pm b \\ \pm b & -a \end{smallmatrix}\right)$ with $a^2 + b^2 = 1$ and $0 < a < b$. On the purely alternating targets $\mathbf{D}_t = \bar{\mathbf{D}}^{(-1)^t}$ ($t \geq 1$), the recursion (4) started from $\mathbf{d}_0^{\mathrm{est}} = \mathbf{0}$ enters a period-two cycle immediately, namely $\mathbf{d}_t^{\mathrm{est}} = \frac{a+b}{2}\left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$ for odd t and $\frac{b-a}{2}\left(\begin{smallmatrix} -1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ for even t. Over a period its $(2,2)$-entry averages $-\frac{a}{2}$: the* same *sign as every target value $-a$, at half the magnitude.*

Proof. Put $m = \frac{a+b}{2}$. The residual $\bar{\mathbf{D}}^- - \mathbf{0}$ has signs $\left(\begin{smallmatrix} + & - \\ - & - \end{smallmatrix}\right)$ and mean modulus m, so $\mathbf{d}_1^{\text{est}} = m\left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$. Next, $\bar{\mathbf{D}}^+ - \mathbf{d}_1^{\text{est}} = \left(\begin{smallmatrix} a-m & b+m \\ b+m & m-a \end{smallmatrix}\right)$ has signs $\left(\begin{smallmatrix} - & + \\ + & + \end{smallmatrix}\right)$ (as $a < m$) and mean modulus b, giving $\mathbf{d}_2^{\text{est}} = \frac{b-a}{2}\left(\begin{smallmatrix} -1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. Repeating once returns $\mathbf{d}_1^{\text{est}}$. The average of the two $(2,2)$-entries is $\frac{1}{2}\left(-m + \frac{b-a}{2}\right) = -\frac{a}{2}$. ■

That cycle is harmless: the alternation by itself does not diverge, and no choice of (a, b) makes it do so. The divergence comes from a *second* period-two cycle of the same recursion, whose residuals carry a *uniform* sign pattern ($\left(\begin{smallmatrix} + & + \\ + & + \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} - & - \\ - & - \end{smallmatrix}\right)$) where those of the harmless cycle are mixed. Since (4) is affine on each sign-pattern cell, the dynamics cannot pass from one cycle to the other; the preamble exists solely to place $\mathbf{d}_2^{\text{est}}$ in the cell of the wrong-sign cycle, which is what Lemma 3 verifies.

Lemma 3 (Wrong-sign limit cycle) *On the targets (30), the recursion (4) from $\mathbf{d}_0^{\text{est}} = \mathbf{0}$ enters at $t = 3$ the exact period-two cycle $\mathbf{d}_t^{\text{est}} = \mathbf{d}_B$ (odd t), $\mathbf{d}_A$ (even t), where*

$$\mathbf{d}_B = \tfrac{1}{200}\left(\begin{smallmatrix} -61 & -201 \\ -61 & -61 \end{smallmatrix}\right), \qquad \mathbf{d}_A = \tfrac{1}{200}\left(\begin{smallmatrix} 131 & -9 \\ 131 & 131 \end{smallmatrix}\right). \tag{32}$$

Over a period its $(2,2)$-entry averages $\frac{1}{2}\big((\mathbf{d}_A)_{22} + (\mathbf{d}_B)_{22}\big) = +\frac{7}{40}$, opposite in sign to every target value $(\mathbf{D}_t)_{22} = -\frac{7}{25}$.

Proof. Substituting (30) into (4) gives the values

t	1	2	3	4 (then 2-periodic)
α_t	$\frac{7}{10}$	$\frac{21}{20}$	$\frac{131}{200}$	$\frac{24}{25}$
$\mathbf{d}_t^{\text{est}}$	$\frac{1}{10}\left(\begin{smallmatrix} -7 & 7 \\ -7 & -7 \end{smallmatrix}\right)$	$\frac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right)$	$\mathbf{d}_B$	$\mathbf{d}_A$

(33)

Each residual $\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$ has strictly nonzero entries, so the signs are unambiguous; e.g. at $t = 3$ it is $\frac{1}{100}\left(\begin{smallmatrix} -7 & -61 \\ -131 & -63 \end{smallmatrix}\right)$, all negative. From $t = 3$ the targets are 2-periodic and the pair $(\mathbf{d}_B, \mathbf{d}_A)$ reproduces itself: $\bar{\mathbf{D}}^+ - \mathbf{d}_B$ and $-(\bar{\mathbf{D}}^- - \mathbf{d}_A)$ are both entrywise positive with mean $\frac{24}{25}$, so $\mathbf{d}_B + \frac{24}{25}\mathbf{J} = \mathbf{d}_A$ and $\mathbf{d}_A - \frac{24}{25}\mathbf{J} = \mathbf{d}_B$ ($\mathbf{J}$ all-ones). ■

Since $\mathbf{X}_t = \mathbf{X}_{t-1} - \mathbf{d}_t^{\text{est}}$ (recall $\eta = 1$), over one period the $(2,2)$-coordinate changes by $-(\mathbf{d}_A + \mathbf{d}_B)_{22} = -\frac{7}{20}$. Consequently a term $-\gamma W_{22}$ in $\tilde{f}$, with $\gamma := \frac{7}{12}$, increases by $\gamma \cdot \frac{7}{20} = \frac{49}{240}$ per period. This is the divergence, provided a genuine smooth function produces the targets (30); Part 3 constructs one.

Part 3: realization by a smooth function. Fix (μ, variant) and set $\gamma = \frac{7}{12}$, $\nu = \frac{1}{1+2\mu}$, and

$$\begin{aligned} A &= \frac{1+\mu}{1-\mu} \text{ (standard)}, \\ A &= \frac{1+\mu}{(1-\mu)(1+2\mu)} \text{ (Nesterov)}. \end{aligned} \tag{34}$$

We build $\tilde{f} = g + \sum_{k=1}^{3} b_k$ from a periodic-plus-linear *field* g and three *localized corrections* b_k, all explicit.

The field is

$$g(\mathbf{W}) = -\gamma\, W_{22} + A\,\Phi_1(W_{12}) + A\,\Phi_2(W_{21}), \tag{35}$$

where $\Phi_i(w) = \int_0^w \psi_i$ and $\psi_i \colon \mathbb{R} \to [-1, 1]$ is a fixed C^∞, p_i-periodic function with $\int_0^{p_i} \psi_i = 0$,

$$p_1 = \tfrac{21}{20}, \quad p_2 = \tfrac{7}{20}, \qquad \delta = \tfrac{1}{100},$$

equal to $+1$ on $[\rho_i^+ - \delta, \rho_i^+ + \delta]$ and -1 on $[\rho_i^- - \delta, \rho_i^- + \delta]$ (mod p_i), where

$$\rho_1^+ = \tfrac{131}{200},\ \rho_1^- = \tfrac{140}{200}; \qquad \rho_2^+ = \tfrac{61}{200},\ \rho_2^- = 0.$$

For an explicit choice, let χ be a C^∞ transition with $0 \le \chi \le 1$, $\chi(s) = 0$ for $s \le 0$, and $\chi(s) = 1$ for $s \ge 1$, and put

$$B_p(u) = \chi\left(\frac{(1/50)^2 - \big((p/\pi)\sin(\pi u/p)\big)^2}{(1/50)^2 - \delta^2}\right), \qquad \psi_i(w) = B_{p_i}(w - \rho_i^+) - B_{p_i}(w - \rho_i^+ - 9/200).$$

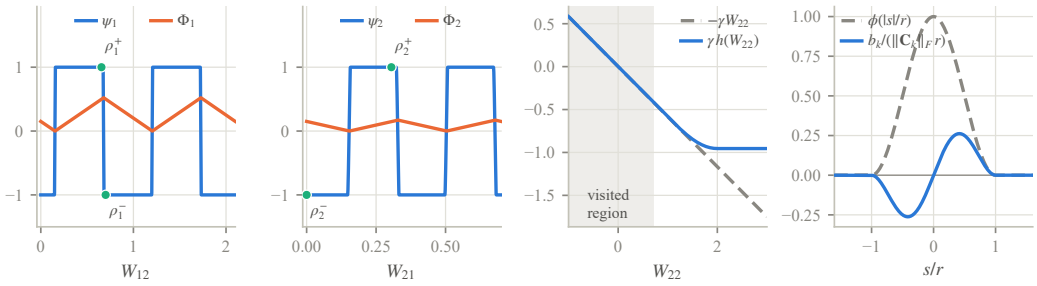


Figure 4: The components of the objective $\tilde{f} = g + \sum_k b_k$ of Part 3, as implemented. The correction b_k is shown along the ray $\mathbf{Z}_k + s\,\mathbf{C}_k/\|\mathbf{C}_k\|_F$, normalized by $\|\mathbf{C}_k\|_F\, r$.

Here $\rho_1^+ + 9/200 = \rho_1^-$ and $\rho_2^+ + 9/200 = p_2 \equiv \rho_2^-$ $(\mathrm{mod}\ p_2)$. The function B_p is smooth, p-periodic, takes values in $[0,1]$, and equals one when $|u| \le \delta$. It is zero when $7/200 \le |u| \le p/2$: Jordan's inequality gives $|(p/\pi)\sin(\pi u/p)| \ge 2|u|/\pi \ge 1/50$ there. Thus ψ_i has the prescribed plateaus and takes values in $[-1,1]$. Its two terms have equal integrals over a period by translation invariance of a periodic integral. Hence $\int_0^{p_i} \psi_i = 0$, making Φ_i periodic and bounded. On each plateau $\Phi_i' = \psi_i = \pm 1$ exactly. Figure 4 draws both ramps, together with the remaining components of $\tilde{f}$.

The corrections pin the first three gradients. Fix once a C^∞ cutoff $\phi\colon \mathbb{R} \to [0,1]$ with $\phi \equiv 1$ on $[0, \frac{1}{2}]$ and $\phi \equiv 0$ on $[1,\infty)$, put $r = \frac{1}{50}$, and set

$$b_k(\mathbf{W}) = \big\langle \mathbf{C}_k,\ \mathbf{W} - \mathbf{Z}_k \big\rangle\, \phi\big(\|\mathbf{W} - \mathbf{Z}_k\|_F / r\big), \tag{36}$$

centered at the first three iterates (from Lemma 3)

$$\mathbf{Z}_1 = \mathbf{0}, \quad \mathbf{Z}_2 = \tfrac{1}{10}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right), \quad \mathbf{Z}_3 = \tfrac{1}{20}\left(\begin{smallmatrix} 7 & -7 \\ 7 & 7 \end{smallmatrix}\right).$$

Each b_k is C^∞, supported in $\{\|\mathbf{W} - \mathbf{Z}_k\|_F \le r\}$; since its linear factor vanishes at $\mathbf{Z}_k$ while $\phi(0) = 1$, $\nabla b_k(\mathbf{Z}_k) = \mathbf{C}_k$. We choose $\mathbf{C}_k := \hat{\mathbf{G}}_k - \nabla g(\mathbf{Z}_k)$ with the explicit $\hat{\mathbf{G}}_k$ of (37) below, so that $\nabla \tilde{f}(\mathbf{Z}_k) = \hat{\mathbf{G}}_k$. Every term of $\tilde{f}$ has a bounded Hessian, so $\tilde{f}$ is C^∞ and $\tilde{L}$-smooth for a finite $\tilde{L}(\mu, \text{variant})$.

Lemma 4 (Realization) *For every $\mu \in [0,1)$ and either variant, EF21-SignMuon on this $\tilde{f}$ ($\eta = 1$, from $\mathbf{0}$) generates exactly the targets (30).*

Proof. *The required gradients.* The buffer recursion $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$ of (6) can be solved for the gradients: prescribing the buffers $(\mathbf{M}_t)_{t\ge 1}$ forces $\mathbf{G}_t = (\mathbf{M}_t - \mu\mathbf{M}_{t-1})/(1-\mu)$, and these are the gradients the function must deliver. Define accordingly the *transient gradients*

$$\hat{\mathbf{G}}_t = \frac{\mathbf{M}_t - \mu\mathbf{M}_{t-1}}{1-\mu} \quad (t = 1,2,3;\ \mathbf{M}_0 = \mathbf{0}) \tag{37}$$

and the *field gradient* $\mathbf{G}^{\pm} = \left(\begin{smallmatrix} 0 & \pm A \\ \pm A & -\gamma \end{smallmatrix}\right)$, where the prescribed buffer values $\mathbf{M}_{1,2,3}$ are given below. With $\bar{\mathbf{M}}^{\pm} = \left(\begin{smallmatrix} 0 & \pm 1 \\ \pm 1 & -\gamma \end{smallmatrix}\right)$, the factorization

$$\bar{\mathbf{M}}^{\pm} = \bar{\mathbf{D}}^{\pm} \left(\begin{smallmatrix} 24/25 & \mp 7/25 \\ \mp 7/25 & 337/300 \end{smallmatrix}\right) \qquad (\text{second factor} \succ 0,\ \det = 1) \tag{38}$$

shows $\mathrm{polar}(\bar{\mathbf{M}}^{\pm}) = \bar{\mathbf{D}}^{\pm}$, with singular values $\frac{3}{4}$ and $\frac{4}{3}$; this is the condition number $\frac{16}{9}$ cited in the proof idea.

Standard momentum ($\tilde{\mathbf{M}}_t = \mathbf{M}_t$). Take $\mathbf{M}_1 = \mathbf{S}_1$, $\mathbf{M}_2 = \mathbf{S}_2$, $\mathbf{M}_3 = \bar{\mathbf{M}}^-$; then $\hat{\mathbf{G}}_{1,2,3}$ are the explicit matrices (37). Feeding $\mathbf{G}^{(-1)^t}$ for $t \ge 4$ keeps $\mathbf{M}_t = \bar{\mathbf{M}}^{(-1)^t}$, because $(1-\mu)\mathbf{G}^{\pm} = \bar{\mathbf{M}}^{\pm} - \mu\bar{\mathbf{M}}^{\mp}$ with A as in (34). As $\mathrm{polar}(\mathbf{S}_1) = \mathbf{S}_1$ (orthogonal) and $\mathrm{polar}(\mathbf{S}_2) = \mathbf{S}_2$ (rank one), (38) yields the targets (30).

Nesterov momentum ($\tilde{\mathbf{M}}_t = (1+\mu)\mathbf{M}_t - \mu\mathbf{M}_{t-1}$). It is here that the orthogonality of $\mathbf{S}_1$ is needed. The Nesterov direction involves two consecutive buffers, so prescribing the tail $t \geq 4$ already fixes $\mathbf{M}_3$, and only the preamble remains to absorb the mismatch at $t = 3$; and the set of matrices with a given polar factor $\mathbf{D}$ is $\{\mathbf{DH} : \mathbf{H} \succ 0\}$, which is three-dimensional when $\mathbf{D}$ is orthogonal but only one-dimensional (a positive scalar) when $\mathbf{D}$ has rank one. A preamble of two rank-one targets leaves too little freedom: a symbolic check shows that it admits no realization once $\mu \gtrsim 0.3$. One orthogonal target supplies enough freedom, of which the construction uses a single scalar, the τ below. Take $\mathbf{M}_1 = \frac{1}{1+\mu}\mathbf{S}_1\mathbf{H}_1$, $\mathbf{M}_2 = \frac{1}{1+\mu}(\mathbf{S}_2 + \mu\mathbf{M}_1)$, $\mathbf{M}_3 = \bar{\mathbf{M}}'^{-}$ and $\mathbf{M}_t = \bar{\mathbf{M}}'^{(-1)^t}$ ($t \geq 4$), where $\bar{\mathbf{M}}'^{\pm} = \left(\begin{smallmatrix} 0 & \pm\nu \\ \pm\nu & -\gamma \end{smallmatrix}\right)$, $\mathbf{H}_1 = \mathrm{diag}(1, 1+\tau)$, and, for $\mu > 0$,

$$\tau = \left(\frac{140(1+\mu)^2}{1+2\mu} - 44\right)\frac{1+\mu}{117\mu} > 0.$$

(At $\mu = 0$ the Nesterov rule reads $\tilde{\mathbf{M}}_t = \mathbf{M}_t$ and is the standard case already treated, so nothing is left to prove there.) Then $\tilde{\mathbf{M}}_1 = \mathbf{S}_1\mathbf{H}_1$ and $\tilde{\mathbf{M}}_2 = \mathbf{S}_2$ have polar factors $\mathbf{S}_1, \mathbf{S}_2$, and $\tilde{\mathbf{M}}_t = \bar{\mathbf{M}}^{(-1)^t}$ for $t \geq 4$ (since $\nu(1+2\mu) = 1$). The one nontrivial step is $t = 3$. Set $\mathbf{H}_3 := \bar{\mathbf{D}}^{-}\tilde{\mathbf{M}}_3$; since $(\bar{\mathbf{D}}^{-})^2 = \mathbf{I}$, this is the same as $\tilde{\mathbf{M}}_3 = \bar{\mathbf{D}}^{-}\mathbf{H}_3$. The stated τ is exactly the value making $\mathbf{H}_3$ symmetric, and $\mathbf{H}_3 \succ 0$ for all $\mu \in (0,1)$: under $\mu = \frac{s}{1+s}$ ($s > 0$) the numerators of its two leading minors are polynomials in s with nonnegative coefficients. Hence $\mathrm{polar}(\tilde{\mathbf{M}}_3) = \bar{\mathbf{D}}^{-}$, completing (30).

The function delivers these gradients. It remains to verify $\nabla\tilde{f}(\tilde{\mathbf{X}}_{t-1}) = \hat{\mathbf{G}}_t$ for $t \leq 3$ and $= \mathbf{G}^{(-1)^t}$ for $t \geq 4$, by induction along the run: as long as the gradients match this prescription, the iterates are those computed in Part 2, and the prescription need only be checked at those points. By Lemma 3 the iterates $\tilde{\mathbf{X}}_0, \tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2$ are exactly the centers $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3$, where $\nabla\tilde{f} = \hat{\mathbf{G}}_{1,2,3}$ by construction. The three balls are disjoint ($\|\mathbf{Z}_j - \mathbf{Z}_k\|_F \geq \frac{7}{10} > 2r$), and every later iterate has $(1,2)$-entry $\geq \frac{131}{200}$ while the centers have it ≤ 0, so no ball is re-entered. For $t \geq 4$ the query lies in the field, where

$$\nabla g(\mathbf{W}) = \begin{pmatrix} 0 & A\psi_1(W_{12}) \\ A\psi_2(W_{21}) & -\gamma \end{pmatrix}.$$

The period shift $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t = -(\mathbf{d}_A + \mathbf{d}_B)$ advances W_{12} by exactly $+p_1$ and W_{21} by exactly $-p_2$ per period, so the two coordinates travel to $+\infty$ and $-\infty$ respectively while, mod p_i, they hold the residues ρ_i^{+} at odd indices and ρ_i^{-} at even ones; there $\psi_i = \pm 1$, giving $\nabla g = \mathbf{G}^{(-1)^t}$. ■

Proof of Theorem 4. By Lemma 4 the normalized run produces the targets (30), so by Lemma 3 its estimate locks onto the wrong-sign cycle and $\tilde{\mathbf{X}}_{t+2} - \tilde{\mathbf{X}}_t$ is a constant shift. Along it Φ_1, Φ_2 return to their values and the corrections b_k vanish, so only the linear term acts: $\tilde{f}(\tilde{\mathbf{X}}_{t+2}) - \tilde{f}(\tilde{\mathbf{X}}_t) = -\gamma(-\frac{7}{20}) = \frac{49}{240} > 0$. By Lemma 1, f then obeys (29) with $c = \frac{49}{240\bar{L}}$. As $(L, \eta, \mu, \text{variant})$ were arbitrary, no rule $\eta = \eta(L, \mu)$ can prevent divergence. ■

Remark 1 (Boundedness below) *Only the linear term $-\gamma W_{22}$ makes $\tilde{f}$ unbounded below, and only as $W_{22} \to +\infty$, a region the iterates never enter, since $(\tilde{\mathbf{X}}_t)_{22} \leq \frac{7}{10}$ throughout (it decreases after the transient). Replacing $-\gamma W_{22}$ by any C^∞ function that agrees with it on $\{W_{22} \leq 1\}$ and is constant on $\{W_{22} \geq 2\}$ therefore leaves the whole trajectory, and (29) with it, unchanged while rendering $\tilde{f}$ bounded below (Assumption 1); the theorem is stated with this modification in force.*

Remark 2 (The construction is not convex) *Theorems 1–3 run on a linear, hence convex, objective; $\tilde{f}$ is nonconvex, through the periodic terms $A\Phi_i$ and the corrections b_k. The nonconvexity is forced by the run rather than chosen by the realization: along the divergent trajectory,*

$$\left\langle \nabla\tilde{f}(\tilde{\mathbf{X}}_6) - \nabla\tilde{f}(\tilde{\mathbf{X}}_3),\ \tilde{\mathbf{X}}_6 - \tilde{\mathbf{X}}_3 \right\rangle = -\tfrac{9A}{50} < 0,$$

violating the gradient monotonicity that every convex function obeys, so no *convex function generates these iterates and gradients. The theorem is stated under Assumptions 1–2 because that is where it is needed: EF21-MuonUSign and EF21-MuonSign converge under these hypotheses together with Assumption 3 (including the deterministic case) and the prescribed tuning (Theorem 5), so the divergence and the guarantees concern one problem class. Whether some convex instance, necessarily through a different trajectory, also defeats EF21-SignMuon we leave open.*

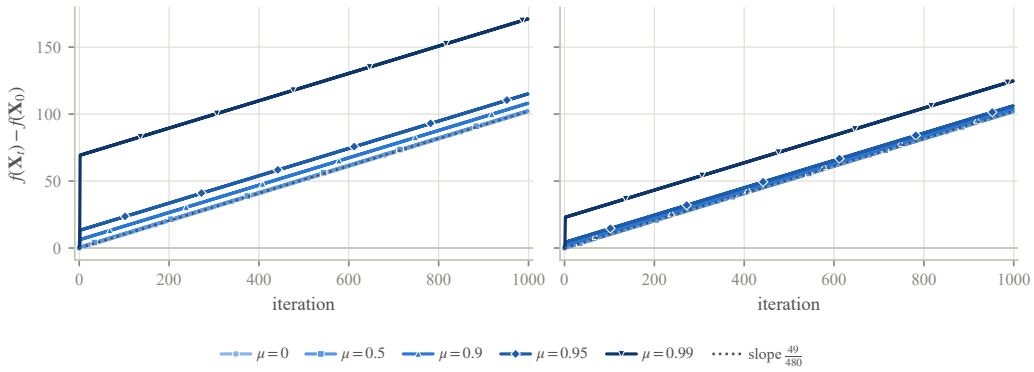


Figure 5: For each momentum coefficient μ and variant, EF21-SignMuon is run on the corresponding instance of Theorem 4 with standard (left) or Nesterov (right) momentum.

Remark 3 (Verification) *Theorem 4 and the plateau-ramp realization of Lemma 4 are formalized in Lean. The construction is also checked in exact rational arithmetic and by running the float64 reference implementation of Algorithm 3 on the assembled $\tilde{f}$. The right panel of Figure 1 confirms that at $\mu = 0$ EF21-SignMuon is the only one of the eight methods that diverges, the others (SignMuon, MuonUSign, MuonSign, EF21-MuonUSign, EF21-MuonSign, SignSGD, Muon) staying bounded; Figure 5 confirms that EF21-SignMuon diverges at the exact rate $\frac{49}{480}$ for every $\mu \in \{0, \frac{1}{2}, 0.9, 0.95, 0.99\}$ under both standard and Nesterov momentum, as the reduction predicts.*

I Convergence of EF21-MuonUSign and EF21-MuonSign

For EMA momentum, EF21-MuonUSign and EF21-MuonSign are exact instances of EF21-Muon [Gruntkowska et al., 2025, Algorithm 3]. The reduction below matches the LMO step and momentum of Gruntkowska et al. [2025], aligns the iteration indices (Proposition 2), and verifies compressor contractivity (Lemma 5). Table 4 gives the notation correspondence. The Nesterov momentum is analysed in Theorem 6.

Notation and constants. For the layer tuple $\mathbf{X} = [\mathbf{X}_1, \dots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, of the Problem Statement write $d_i := m_i n_i$, $r_i := \min(m_i, n_i)$, $d_{\max} := \max_i d_i$; a second subscript selects a layer ($\mathbf{X}_{t,i}$, $\mathbf{g}_{t,i}$). We use $\|\mathbf{Y}\|_{2\to 2} \le \|\mathbf{Y}\|_F \le \|\mathbf{Y}\|_* \le \sqrt{\operatorname{rank} \mathbf{Y}}\, \|\mathbf{Y}\|_F$. Smoothness constants: L_i for f and $L_{i,j}$ for f_j in Assumption 2, with $\tilde{L}_i^2 := \frac{1}{N} \sum_j L_{i,j}^2$ and $L := \max_i L_i$; in the (L^0, L^1) form the pairs are again per layer for f and per layer and client for f_j, with $L^1_{i,\max} := \max_j L^1_{i,j}$.

Assumption 3 (Stochastic gradient) *Samples are independent across workers and rounds. Each client's stochastic gradient is unbiased, $\mathbb{E}_\xi[\nabla f_j(\mathbf{X}; \xi)] = \nabla f_j(\mathbf{X})$, with bounded variance $\mathbb{E}_\xi \|\nabla f_j(\mathbf{X}; \xi) - \nabla f_j(\mathbf{X})\|_*^2 \le \sigma^2$ for all $\mathbf{X}$.*

Assumptions 1–3 are Assumptions 1–2 and 6–10 of Gruntkowska et al. [2025] with the layer norms taken spectral; our variance bound is stated in the nuclear norm and implies theirs via $\|\cdot\|_F \le \|\cdot\|_*$. The clause $f_j \ge f_j^*$ of Assumption 1 is needed only for the generalized-smooth results.

I.1 EMA convergence: main guarantees

Theorem 5 (Convergence of the EF21 methods) *Run the federated Algorithm 9 with the EF21 uplink and EMA momentum $\mu \in [0, 1)$. Then:*

(i) (smooth; EF21-MuonUSign and EF21-MuonSign) *under Assumptions 1–3, with the "sharp" learning rate $\eta_{t,i} = \gamma_i \|\mathbf{g}_{t,i}\|_*$ and tuned (γ_i, μ), both methods reach $\frac{1}{T} \sum_{t<T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$ (Corollary 1);*

(ii) (generalized smooth; EF21-MuonUSign only) *under Assumptions 1 and 3 and (L^0, L^1)-smoothness, EF21-MuonUSign with the horizon-dependent learning rate and momentum of*

Corollary 2 reaches $\min_{t\le T}\sum_i \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$. *Its learning rate is constant within each run.*

Both statements are written for a per-layer constant common to all layers ($\gamma_i \equiv \gamma$, $\eta_i \equiv \eta$); for unequal constants the left-hand sides carry the step-size weights of Corollaries 1–2.

Part (i) implies the rate announced in the main text, since the minimum over $t \le T$ is at most the average. The centralized Algorithms 6–7 are the federated method at $N = 1$, so both parts cover the centralized runs as well. Theorem 6 proves both rates for Nesterov momentum under its stated tunings.

I.2 EMA reduction to EF21-Muon

LMO step of Gruntkowska et al. [2025]. For $\mathbf{G} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ the framework's oracle over the spectral ball of radius τ is $\mathrm{lmo}(\mathbf{G}) = \mathbf{X} - \tau\mathbf{U}\mathbf{V}^\top$, which is our server step with $\tau = \eta_{t,i}$, since $\mathbf{D}_{t,i} = -A(\mathbf{g}_{t,i}) = \mathbf{U}_{t,i}\mathbf{V}_{t,i}^\top$:

$$\mathbf{X}_{t,i} = \mathbf{X}_{t-1,i} - \eta_{t,i}\,\mathbf{D}_{t,i} = \mathrm{lmo}_{\mathcal{B}(\mathbf{X}_{t-1,i},\eta_{t,i})}(\mathbf{g}_{t,i}). \tag{39}$$

Their "sharp" step $\mathbf{X} - \gamma\,\mathbf{G}^\sharp$ uses $\mathbf{G}^\sharp = \|\mathbf{G}\|_*\mathbf{U}\mathbf{V}^\top$, so a constant γ_i amounts to the schedule $\eta_{t,i} = \gamma_i\|\mathbf{g}_{t,i}\|_*$ and a constant $\eta_{t,i}$ to the plain Muon rate: the two differ as learning-rate choices, not as algorithms, and either run is an instance of the framework. The guarantees, however, are attached to specific choices. Part (i) of Theorem 5 assumes the nuclear-norm schedule, which our implementation does not use. Theorem 7 covers the update $\mathbf{X}_{t,i} = \mathbf{X}_{t-1,i} - \eta_{t,i}\,\mathrm{polar}(\mathbf{g}_{t,i})$ with deterministic learning rates $\eta_{t,i}$, including constant and cosine schedules. The spectral norm-equivalence constants are $\underline{\rho}_i = 1$, $\bar{\rho}_i = \sqrt{r_i}$, defined by $\underline{\rho}_i\|\mathbf{Y}\|_{2\to2} \le \|\mathbf{Y}\|_F \le \bar{\rho}_i\|\mathbf{Y}\|_{2\to2}$. At rank-deficient $\mathbf{G}$ the LMO is non-unique, but any selection serves: writing $\mathbf{\Delta} := \mathrm{lmo}_{\mathcal{B}(\mathbf{X},\tau)}(\mathbf{G}) - \mathbf{X}$ for the displacement it produces, the analysis uses only $\langle\mathbf{G},\mathbf{\Delta}\rangle = -\tau\|\mathbf{G}\|_*$ and $\|\mathbf{\Delta}\|_{2\to2} \le \tau$, with $\mathbf{G} = \mathbf{0}$ read as the zero step ($\mathbf{0}^\sharp = \mathbf{0}$).

Momentum of Gruntkowska et al. [2025]. Algorithm 9 uses $\mathbf{M}_t = \mu\mathbf{M}_{t-1} + (1-\mu)\mathbf{G}_t$, which is (6) and is the framework's momentum with $\beta = 1 - \mu$. Rescaling the momentum stream by a constant, as the heavy-ball convention does by the factor $1/(1-\mu)$, alters nothing: the EF21 recursion is positively homogeneous and the Muon LMO scale-invariant, so the factor leaves the iterates unchanged under a constant $\eta_{t,i}$ and is absorbed into γ_i under the "sharp" schedule. The Nesterov update is analysed in Theorem 6.

Proposition 2 (Exact instance) *Fix μ, a learning-rate schedule, and the LMO selection above. Started from $\mathbf{M}_0^{(j)} = \mathbf{g}_0^{(j)} = \mathbf{g}_0 = \mathbf{0}$, $\mathbf{W}_0 = \mathbf{X}_0$, Algorithm 9 with EMA momentum, the EF21 uplink and either downlink mode ($\mathcal{C}^\downarrow \in \{$exact, EF21-P$\}$) produces the same trajectory as Algorithm 3 of Gruntkowska et al. [2025] with spectral norms, scaled-sign worker compressors, identity/scaled-sign server compressor, $\beta = 1 - \mu$, and radii $t_i^k = \eta_{k,i}$, up to a one-step index shift $X^{t+1} = \mathbf{X}_t$.*

Proof. The loops differ only in where the round is cut: they order it *step → downlink → gradient → uplink*, we order it *gradient → uplink → step → downlink*. Their first step and downlink are vacuous under our initialization: $\mathbf{g}_0 = \mathbf{0}$ gives $X^1 = X^0 = \mathbf{X}_0$ and a zero downlink residual, so $W^1 = \mathbf{X}_0$. Their first gradient/uplink phase is our first gradient/uplink phase; each subsequent step/downlink phase completes that same paper round. Induction gives $X^{t+1} = \mathbf{X}_t$, $W^{t+1} = \mathbf{W}_t$, with matching momentum and worker estimators at the corresponding phases. Running their method for $K = T+1$ iterations therefore yields $\{\mathbf{X}_0, \mathbf{X}_0, \mathbf{X}_1, \ldots, \mathbf{X}_{T-1}\}$. Repeating $\mathbf{X}_0$ leaves the minimum stationarity value unchanged. For averages of nonnegative stationarity terms, the index shift contributes a factor at most $(T+1)/T$. □

Zero initialization also makes their initial-error constant Ψ^0 explicit: the $\|\mathbf{M}_0 - \mathbf{g}_0\|$ term vanishes and the gradient-deviation terms become $\|\nabla_i f(\mathbf{X}_0)\|$. The β^{-1} surviving in Ψ^0 enters only through the Ψ^0/T term, which $T^{-1/2}$ dominates.

	This paper	EF21-Muon
clients / rounds	N, T	$n, K = T+1$
iterate	$\mathbf{X}_t$	X^{t+1}
momentum	μ	$1-\beta$
learning rate	$\eta_{t,i}$	radius t_i^t
norm equivalence	$1,\ \sqrt{r_i}$	$\underline{\rho}_i,\ \bar{\rho}_i$
uplink compr.	scaled sign	$\mathbb{B}_2(\alpha_D)$
downlink compr.	exact / sign	$\mathcal{I}$ / $\mathbb{B}_2(\alpha_P)^\dagger$
compression α	$1/d_{\max}$	α_D

Table 4: Change of variables from our notation to that of Gruntkowska et al. [2025]. Layer norms $\|\cdot\|_{(i)}, \|\cdot\|_{(i)\star}$ are read as $\|\cdot\|_{2\to 2}, \|\cdot\|_*$ and smoothness constants $L_i^0, \tilde{L}_i^0$ as $L_i, \tilde{L}_i$; the index shift is Proposition 2. †Their Theorem 19 as printed asks for a server compressor in $\mathbb{B}(\alpha_P)$, contractive in the *layer* norm; the Euclidean class $\mathbb{B}_2(\alpha_P)$, to which the scaled sign does belong, is admitted by their Remark 23, which introduces the factor $\bar{\rho}_i^2$ appearing in Corollary 1 and Remark 5.

I.3 Contractivity of scaled-sign compression

The framework requires every transmitted message to originate in a *contractive compressor*: a map with $\mathbb{E}\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|^2 \le (1-\alpha)\|\mathbf{Y}\|^2$ for some $\alpha \in (0,1]$ [Gruntkowska et al., 2025, Def. 1]. This is precisely the property that fails for a bare sign and holds once it is scaled.

Lemma 5 (Contractivity of the scaled sign) *For every $\mathbf{Y} \in \mathbb{R}^{m\times n}$ ($d = mn$), the scaled sign $\mathcal{C}(\mathbf{Y}) = \mathrm{mean}(|\mathbf{Y}|)\,\mathrm{sign}(\mathbf{Y})$, with exact zeros resolved to ± 1 as in Section 3, satisfies*

$$\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - \tfrac{1}{d}\|\mathbf{Y}\|_1^2 \le \left(1 - \tfrac{1}{d}\right)\|\mathbf{Y}\|_F^2, \tag{40}$$

so $\mathcal{C}$ is Euclidean-contractive with $\alpha = 1/d$. The identity holds for every draw of the random signs and, for $\mathbf{Y} \neq \mathbf{0}$, gives the exact contraction $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2)$.

Proof. Write $c := \|\mathbf{Y}\|_1/d$ and $s_{kl} = \pm 1$ for the transmitted signs. A nonzero entry contributes $(|Y_{kl}| - c)^2$ and a zero entry contributes $(c\, s_{kl})^2 = c^2$ whichever sign was drawn, so $\|\mathcal{C}(\mathbf{Y}) - \mathbf{Y}\|_F^2 = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + c^2\|\mathbf{Y}\|_0 + c^2(d - \|\mathbf{Y}\|_0) = \|\mathbf{Y}\|_F^2 - 2c\|\mathbf{Y}\|_1 + dc^2$, which is (40) on substituting c. The bound follows from $\|\mathbf{Y}\|_1 \ge \|\mathbf{Y}\|_F$, with equality for a nonzero input exactly at one-sparse inputs. □

The convention $\mathrm{sign}(0) \in \{\pm 1\}$ ensures the exact identity in (40), including at sparse inputs.

The scaling is essential to Lemma 5: a bare sign contracts for no α at all, since as $\|\mathbf{Y}\|_F \to 0$ on a fixed support, $\|\mathrm{sign}(\mathbf{Y}) - \mathbf{Y}\|_F \to \sqrt{d}$. That is the dividing line between our divergent and convergent methods: the majority-vote methods transmit unscaled signs of full quantities, the EF21 variants the scaled sign of a *residual*, at the cost of one extra scalar per layer per round. Per layer the uplink scaled sign lies in $\mathbb{B}_2(1/d_i) \subseteq \mathbb{B}_2(\alpha)$ with $\alpha := 1/d_{\max}$; the downlink is exact for EF21-MuonUSign ($\alpha_P = 1$) and the same scaled sign for EF21-MuonSign.

I.4 EMA convergence bounds

The following guarantees specialize the proofs of Theorems 19 and 24 of Gruntkowska et al. [2025] through Table 4, using their Remark 23 for the scaled-sign downlink.

Corollary 1 (Smooth case; EF21-MuonUSign and EF21-MuonSign) *Let Assumptions 1–3 hold, with the zero initialization of Proposition 2. Run EF21-MuonUSign or EF21-MuonSign (Algorithm 9 with the EF21 uplink and the exact or the scaled-sign downlink, respectively) with EMA momentum and $\eta_{t,i} = \gamma_i\|\mathbf{g}_{t,i}\|_*$. The following explicit tuning suffices. Put $\tau_i = 0$ for EF21-MuonUSign and $\tau_i = r_i$ for EF21-MuonSign, and define the horizon-independent envelope*

$$Z_i = r_i\left(12L_i^2 + \frac{72\tau_i L_i^2}{\alpha_P^2} + \frac{180\tilde{L}_i^2}{\alpha_D^2} + \frac{1008\tau_i\tilde{L}_i^2}{\alpha_P^2\alpha_D^2}\right).$$

Choose fixed $g_i > 0$ with $g_i(2L_i + 2\sqrt{Z_i}) \le 1$, for example $g_i = (1 + 2L_i + 2\sqrt{Z_i})^{-1}$. For each positive horizon T, set $\beta = 1 - \mu = T^{-1/2}$ and $\gamma_i = g_i T^{-1/2}$. Then

$$\frac{1}{T}\sum_{t<T}\sum_{i=1}^{p} w_i\,\mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2}),$$

with $w_i := \gamma_i/(\frac{1}{p}\sum_l \gamma_l)$; for a common γ_i all $w_i = 1$ and the left side is $\frac{1}{T}\sum_{t<T}\mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2$.

Proof. By Proposition 2 and Lemma 5, the run is an instance of Algorithm 3 with $\alpha_D = \alpha$ and $\alpha_P \in \{1, \alpha\}$. We specialize the proof of Gruntkowska et al. [2025, Thm. 19], omitting the downlink-error terms when $W^k = X^k$ (exact downlink) and using their Remark 23 for the scaled-sign downlink. For $0 < \beta \le 1$, the resulting constant satisfies $\zeta_i \le Z_i/\beta^2$, so the displayed choice ensures $\gamma_i(2L_i + 2\sqrt{\zeta_i}) \le 1$. Under zero initialization the initial potential is bounded independently of T: the $\gamma_i/\beta = g_i$ terms are constant and the $\gamma_i\beta$ terms are bounded. The noise term is $O(\beta^2)$ before division by the mean sharp step $\bar\gamma = \bar g\beta$, where $\bar g = p^{-1}\sum_i g_i > 0$. The finite-time bound is therefore $O((\bar\gamma T)^{-1} + \beta) = O(T^{-1/2})$, with $w_i = g_i/\bar g$. A one-step shift changes the constant by at most $(T+1)/T$. □

Corollary 2 ((L^0, L^1) case; EF21-MuonUSign) *Let Assumptions 1 and 3 hold and let Assumption 2 hold in its (L^0, L^1) form. Use the zero initialization of Proposition 2. Put $q := \sqrt{1-\alpha_D} \in [0, 1)$. Choose fixed $0 < \eta_i \le 1$ satisfying the explicit sufficient bound*

$$24\eta_i^2\sqrt{r_i}\,(L^1_{i,\max})^2 \le 1.$$

Run EF21-MuonUSign (exact downlink) with $S := T + 2$, momentum $\mu = 1 - S^{-1/2}$, and per-layer rate $\eta_{t,i} \equiv \eta_i/S^{3/4}$, constant within each run. For all sufficiently large horizons the remaining sufficient conditions

$$6\eta_i^2(L_i^1)^2 \le \sqrt{S}, \qquad 24q\eta_i^2\sqrt{r_i}\,(L^1_{i,\max})^2 \le (1-q)\sqrt{S}$$

also hold.[1] *Then*

$$\min_{0\le t\le T}\ \sum_{i=1}^{p} v_i\,\mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4}), \quad v_i := \frac{\eta_i}{\frac{1}{p}\sum_l \eta_l}.$$

Proof. We follow the proof of Gruntkowska et al. [2025, Thm. 24] with $K = T + 1$, $\beta = S^{-1/2}$, Table 4, and the sufficient conditions above. □

I.5 Nesterov momentum

We prove convergence for Nesterov momentum in Algorithm 9. With $\beta = 1 - \mu$ and $a = 1 - \beta$:

$$\begin{aligned} M_{t,i}^{(j)} &= aM_{t-1,i}^{(j)} + \beta G_{t,i}^{(j)}, \\ V_{t,i}^{(j)} &= aM_{t,i}^{(j)} + \beta G_{t,i}^{(j)}, \\ g_{t,i}^{(j)} &= g_{t-1,i}^{(j)} + C_i^{\uparrow}(V_{t,i}^{(j)} - g_{t-1,i}^{(j)}). \end{aligned} \tag{41}$$

Initialize $M_{0,i}^{(j)} = g_{0,i}^{(j)} = 0$, $W_0 = X_0$, and set $M_{-1,i}^{(j)} = V_{0,i}^{(j)} = 0$ for every layer i and worker j. The oracle is the exact measurable polar selection, with zero direction at zero. Put $\rho_i = \sqrt{r_i}$ and $\alpha_D = 1/d_{\max}$, and take $\alpha_P = 1$ for the exact downlink and $\alpha_P = \alpha_D$ for the compressed downlink. Let σ_i^2 bound each worker's Frobenius noise variance in block i; Assumption 3 permits $\sigma_i = \sigma$.

Theorem 6 (Nesterov convergence) *With the initialization above, the Nesterov updates (41) have the following guarantees.*

(i) *Under Assumptions 1–3 with ordinary smoothness, EF21-MuonUSign and EF21-MuonSign satisfy*

$$\frac{1}{T}\sum_{t<T}\sum_i \mathbb{E}\|\nabla_i f(X_t)\|_*^2 = \mathcal{O}(T^{-1/2})$$

with $\beta = T^{-1/2}$ and sharp steps $\eta_{t,i} = \gamma\|g_{t,i}\|_*$, $\gamma = c\beta$. An explicit sufficient choice of the fixed $c > 0$ is given in the proof of Lemma 6.

[1] We use the sufficient inequalities displayed in the proof of Gruntkowska et al. [2025, Thm. 24]; the second restriction in its statement is stronger by a factor $K + 1$.

(ii) *Under the generalized form of Assumption 2 and Assumptions 1, 3, EF21-MuonUSign with exact downlink, $S = T + 2$, $\beta = S^{-1/2}$ and learning rates $\eta_{t,i} = \eta_i S^{-3/4}$ satisfies*

$$\min_{0\le t\le T} \sum_i \frac{\eta_i}{\bar\eta} \mathbb{E}\|\nabla_i f(X_t)\|_* = \mathcal{O}(T^{-1/4}), \qquad \bar\eta = \frac{1}{p}\sum_i \eta_i,$$

for any fixed $\eta_i > 0$.

All constants are independent of T.

The proof has two parts: a squared-error tracking estimate for ordinary smoothness and a first-moment estimate for generalized smoothness. For both, cut a round just before the server step. A state consists of $(X, W, M, M_{\mathrm{prev}}, g)$, and its successor first forms (X', W') by the step and downlink, then queries fresh gradients at W', and finally applies (41). Denote conditional expectation given this state by $\mathbb{E}_t$. This is the same rotation as in Proposition 2: the initial step is zero and $X_{t+1}^{\mathrm{rot}} = X_t^{\mathrm{paper}}$. The previous buffer is carried as an extra analysis state. In particular,

$$\begin{aligned} V_i^{(j)} &= (2-\beta)M_i^{(j)} - aM_{\mathrm{prev},i}^{(j)}, \\ V_i^{(j)\prime} - V_i^{(j)} &= (2-\beta)(M_i^{(j)\prime} - M_i^{(j)}) - a(M_i^{(j)} - M_{\mathrm{prev},i}^{(j)}). \end{aligned} \tag{42}$$

Thus tracking the Nesterov target also requires controlling momentum increments. Write $\langle z_j\rangle_j = N^{-1}\sum_j z_j$.

Lemma 6 (Smooth Nesterov tracking) *Under Assumptions 1–3 with ordinary smoothness and the initialization above, there is a constant $c_0 > 0$ such that, for $0 < c \le c_0$, $0 < \beta \le 1$ and $\gamma = c\beta$, a nonnegative potential Ψ satisfies*

$$\mathbb{E}_t\Psi' + \frac{\gamma}{4}\sum_i \|\nabla_i f(X)\|_*^2 \le \Psi + \gamma\beta\mathcal{S}, \qquad \Psi_0 \le \mathcal{I}.$$

The constants $c_0, \mathcal{I}, \mathcal{S}$ are independent of β, T; explicit values are given in the proof.

Proof. We track the averaged momentum error, the local momentum errors, the uplink errors, the momentum increments, and the downlink gap. For each layer, put $\ell_i^2 = \tilde L_i^2$ and define

$$\begin{aligned} E_i &= \|\langle M_i^{(j)}\rangle_j - \nabla_i f(X)\|_F^2, \\ F_i &= \langle \|M_i^{(j)} - \nabla_i f_j(X)\|_F^2\rangle_j, \\ H_i &= \langle \|V_i^{(j)} - g_i^{(j)}\|_F^2\rangle_j, \\ K_i &= \|\langle M_i^{(j)} - M_{\mathrm{prev},i}^{(j)}\rangle_j\|_F^2, \\ J_i &= \langle \|M_i^{(j)} - M_{\mathrm{prev},i}^{(j)}\|_F^2\rangle_j, \qquad P_i = \|W_i - X_i\|_F^2. \end{aligned}$$

Let $A_i = \langle\|\nabla_i f_j(X') - \nabla_i f_j(X)\|_F^2\rangle_j$ and $B_i = \langle\|\nabla_i f_j(W') - \nabla_i f_j(X')\|_F^2\rangle_j$. Conditional centering, Young's inequality, and compressor contractivity give

$$\begin{aligned} \mathbb{E}_t F_i' &\le (1-\beta/2)F_i + 2A_i/\beta \\ &\quad + \beta^2(1+2/\beta)B_i + \beta^2\sigma_i^2, \\ \mathbb{E}_t J_i' &\le 3\beta^2(F_i + A_i + B_i) + \beta^2\sigma_i^2, \\ \mathbb{E}_t H_i' &\le (1-\alpha_D/2)H_i + \frac{16}{\alpha_D}\mathbb{E}_t J_i' + \frac{4}{\alpha_D}J_i. \end{aligned} \tag{43}$$

The same bounds hold for E_i', K_i' with F_i, J_i replaced by E_i, K_i and σ_i^2 replaced by σ_i^2/N. Indeed, average the centered worker noises before squaring; their conditional cross terms vanish. Jensen bounds the averaged gradient movements by A_i, B_i. Thus worker averaging is retained in the direction error, while the local recurrences still control compression. For the last inequality, (42) yields, for each worker, $\|V_i^{(j)\prime} - V_i^{(j)}\|_F^2 \le 8\|M_i^{(j)\prime} - M_i^{(j)}\|_F^2 + 2\|M_i^{(j)} - M_{\mathrm{prev},i}^{(j)}\|_F^2$; apply the contracted Young inequality to $(V_i^{(j)} - g_i^{(j)}) + (V_i^{(j)\prime} - V_i^{(j)})$ before averaging. Smoothness gives $A_i \le \ell_i^2\gamma^2\|g_i\|_*^2$ and $B_i \le \ell_i^2 P_i'$, where $g_i = \langle g_i^{(j)}\rangle_j$. The downlink satisfies

$$P_i' \le (1-\alpha_P/2)P_i + \frac{2\tau_i}{\alpha_P}\gamma^2\|g_i\|_*^2,$$

with $\tau_i = 0$ for the exact downlink and $\tau_i = r_i$ for the scaled-sign downlink.

To combine these recurrences with objective descent, choose the following nonnegative weights:

$$\begin{aligned}
b_i &= 8\rho_i^2, & h_i &= 2b_i/\alpha_D,\\
j_i &= b_i + 4h_i/\alpha_D, & R_i &= 16h_i/\alpha_D + j_i,\\
\widehat{R}_i &= R_i - b_i = 40b_i/\alpha_D^2, & u_i &= 2b_i/\beta + 6\beta R_i,\\
e_i &= 2b_i/\beta + 6\beta b_i, & v_i &= 6\beta\widehat{R}_i,\\
I_i &= [u_i\beta^2(1+2/\beta) + 3\beta^2 R_i]\ell_i^2, & p_i &= 2I_i/\alpha_P,\\
\zeta_i^{\mathrm{N}} &= 2u_i\ell_i^2/\beta + 3\beta^2 R_i\ell_i^2 + (I_i + p_i)2\tau_i/\alpha_P.
\end{aligned}$$

The identities $e_i + v_i = u_i$ and $b_i + \widehat{R}_i = R_i$ preserve the movement and broadcast coefficients. Moreover, $e_i\beta/2 - 3\beta^2 b_i = b_i$, $v_i\beta/2 - 3\beta^2\widehat{R}_i = 0$, $h_i\alpha_D/2 = b_i$, and $j_i - b_i = 4h_i/\alpha_D$ absorb the global and local tracking errors. Indeed, Jensen's inequality and (42) imply $\|g_i - \nabla_i f(X)\|_F^2 \le 3(E_i + H_i + K_i)$, and the polar identities imply

$$\begin{aligned}
\langle \nabla_i f(X), X_i' - X_i\rangle \le \gamma b_i(E_i + H_i + K_i)\\
- \tfrac{\gamma}{4}\|\nabla_i f(X)\|_*^2 - \tfrac{\gamma}{4}\|g_i\|_*^2.
\end{aligned}$$

Consequently the nonnegative potential

$$\Psi = f(X) - f_* + \sum_i \gamma\big[e_i E_i + v_i F_i + h_i H_i + b_i K_i + (j_i - b_i)J_i + p_i P_i\big]$$

satisfies, whenever $2L_i\gamma + 4\zeta_i^{\mathrm{N}}\gamma^2 \le 1$,

$$\mathbb{E}_t\Psi' + \frac{\gamma}{4}\sum_i \|\nabla_i f(X)\|_*^2 \le \Psi + \gamma\sum_i \nu_i\sigma_i^2, \tag{44}$$

where $\nu_i = \beta^2[(e_i + b_i)/N + v_i + \widehat{R}_i]$. Here smooth descent contributes $L_i\gamma^2\|g_i\|_*^2/2$; all tracking movement terms contribute at most $\zeta_i^{\mathrm{N}}\gamma^3\|g_i\|_*^2$, which the stated condition absorbs.

It remains to choose a sharp-step scale that satisfies this condition uniformly over β. Define

$$\begin{aligned}
Z_i^{\mathrm{N}} = (4b_i + 15R_i)\ell_i^2\\
+ (6b_i + 21R_i)\ell_i^2(1 + 2/\alpha_P)\,2\tau_i/\alpha_P.
\end{aligned}$$

Direct substitution gives $\beta^2\zeta_i^{\mathrm{N}} \le Z_i^{\mathrm{N}}$, $\beta u_i \le 2b_i + 6R_i$, and $\nu_i \le \beta(9b_i/N + 7\widehat{R}_i)$. Take $c_0 = [1 + \sum_i(2L_i + 2\sqrt{Z_i^{\mathrm{N}}})]^{-1}$. Every $0 < c \le c_0$ satisfies $c(2L_i + 2\sqrt{Z_i^{\mathrm{N}}}) \le 1$ for every layer. Then $\gamma = c\beta$ ensures the condition in (44). Put

$$\begin{aligned}
\mathcal{I} &= f(X_0) - f_* + c\sum_i\big[8b_i\|\nabla_i f(X_0)\|_F^2 + 6\widehat{R}_i\langle\|\nabla_i f_j(X_0)\|_F^2\rangle_j\big],\\
\mathcal{S} &= \sum_i (9b_i/N + 7\widehat{R}_i)\sigma_i^2.
\end{aligned}$$

Zero initialization gives $\Psi_0 \le \mathcal{I}$, and $\sum_i \nu_i\sigma_i^2 \le \beta\mathcal{S}$ proves the drift claimed in the lemma. □

Lemma 7 (Generalized-smooth Nesterov tracking) *Under Assumptions 1, 3, generalized smoothness, and the initialization above, use the exact downlink. Fix $\eta_i > 0$ and set $s_i = \eta_i S^{-3/4}$, $\beta = S^{-1/2}$ for $S \ge 2$. There are a nonnegative potential Φ and finite nonnegative constants K, B, I, independent of S, such that*

$$\mathbb{E}_t\Phi' + \sum_i s_i\|\nabla_i f(X)\|_* \le (1 + K/S)\Phi + B/S, \qquad \Phi_0 \le I.$$

Proof. The exact downlink gives $W = X$. We first separate the averaged gradient noise from the tracking bias, and then bound the gradient-dependent smoothness terms by the objective gap. Let $q = \sqrt{1 - \alpha_D} < 1$ and let $Z_i' = aZ_i + \beta\langle\varepsilon_i^{(j)}\rangle_j$, $Z_{0,i} = 0$, be an analysis-only noise EMA.

Replace F_i, H_i, J_i by their first-moment versions $F_i^{(1)}, H_i^{(1)}, J_i^{(1)}$ (remove the squares), and put $D_i = \|\langle M_i^{(j)}\rangle_j - \nabla_i f(X) - Z_i\|_F$. For $d_i^g = \|\nabla_i f(X') - \nabla_i f(X)\|_F$ and $d_i^\ell = \langle \|\nabla_i f_j(X') - \nabla_i f_j(X)\|_F\rangle_j$, the same updates give

$$\begin{aligned}
\mathbb{E}_t D_i' &\le a(D_i + d_i^g),\\
\mathbb{E}_t F_i^{(1)\prime} &\le a(F_i^{(1)} + d_i^\ell) + \beta\sigma_i,\\
\mathbb{E}_t J_i^{(1)\prime} &\le \beta(F_i^{(1)} + d_i^\ell + \sigma_i),\\
\mathbb{E}_t H_i^{(1)\prime} &\le qH_i^{(1)} + 2q\mathbb{E}_t J_i^{(1)\prime} + qJ_i^{(1)},\\
\mathbb{E}_t\|Z_i'\|_F^2 &\le a\|Z_i\|_F^2 + \beta^2\sigma_i^2/N.
\end{aligned}$$

The last line uses independent centered worker noise. The first line is noise-free after debiasing. Define

$$\begin{aligned}
\delta_i &= 2s_i\rho_i/\beta, & h_i^g &= 4s_i\rho_i/(1-q),\\
j_i^g &= 2s_i\rho_i + qh_i^g, & R_i^g &= 2qh_i^g + j_i^g,\\
z_i^g &= s_i\rho_i/(\beta\sqrt{\beta}).
\end{aligned}$$

Since $\|g_i - \nabla_i f(X)\|_F \le D_i + H_i^{(1)} + J_i^{(1)} + \|Z_i\|_F$, these recursions and $2\|Z_i\|_F \le \|Z_i\|_F^2/\sqrt{\beta} + \sqrt{\beta}$ show that

$$\Phi = f(X) - \bar{f}_* + \sum_i \left[\delta_i D_i + R_i^g F_i^{(1)} + h_i^g H_i^{(1)} + j_i^g J_i^{(1)} + z_i^g\|Z_i\|_F^2\right]$$

obeys

$$\mathbb{E}_t\Phi' + \sum_i s_i\|\nabla_i f(X)\|_* \le \Phi + \sum_i \left[A_i^g C_i^g(X) + A_i^\ell C_i^\ell(X) + n_i\right], \tag{45}$$

$$\begin{aligned}
A_i^g &= s_i^2/2 + \delta_i a s_i, \quad A_i^\ell = R_i^g s_i,\\
n_i &= 2R_i^g\beta\sigma_i + z_i^g\beta^2\sigma_i^2/N + s_i\rho_i\sqrt{\beta},
\end{aligned}$$

where $\bar{f}_* = N^{-1}\sum_j f_j^*$, $C_i^g(X) = L_i^0 + L_i^1\|\nabla_i f(X)\|_*$, and $C_i^\ell(X) = \langle L_{i,j}^0 + L_{i,j}^1\|\nabla_i f_j(X)\|_*\rangle_j$. Indeed, generalized smoothness bounds the gradient movements by $d_i^g \le s_i C_i^g(X)$, $d_i^\ell \le s_i C_i^\ell(X)$. Its descent inequality contributes $s_i^2 C_i^g(X)/2$ and $2s_i\rho_i\|g_i - \nabla_i f(X)\|_F$; the weighted recursions absorb the latter term, giving (45).

To bound the right-hand side of (45), we use the lower bounds on the local objectives. They give finite nonnegative constants $a_i^g, b_i^g, a_i^\ell, b^\ell$ such that

$$C_i^g(X) \le a_i^g + b_i^g(f(X) - \bar{f}_*), \quad C_i^\ell(X) \le a_i^\ell + b^\ell(f(X) - \bar{f}_*).$$

For completeness, set $M = 1 + \sum_i L_i^1 + \sum_{j,i} L_{i,j}^1$, $o = \sum_i L_i^0/M$ and $\tilde{o} = \langle\sum_i L_{i,j}^0/M\rangle_j$. A generalized-smooth descent step of radius $1/M$ in each block gives $\sum_i \|\nabla_i f(X)\|_* \le 2M(f(X) - \bar{f}_*) + o$, and the analogous bound holds for each $f_j - f_j^*$. Thus one can take $a_i^g = L_i^0 + L_i^1 o$, $b_i^g = 2ML_i^1$, $a_i^\ell = \langle L_{i,j}^0\rangle_j + M\tilde{o}$ and $b^\ell = 2M^2$. For the chosen s_i and β, each remaining coefficient is of order $1/S$. More explicitly, define

$$\begin{aligned}
c_i^g &= \tfrac{1}{2} + 2\rho_i, & c_i^\ell &= 2\rho_i + 12\rho_i q/(1-q),\\
c_i^n &= 2c_i^\ell\sigma_i + \rho_i\sigma_i^2/N + \rho_i.
\end{aligned}$$

The coefficients above satisfy $A_i^g \le (s_i^2/\beta)c_i^g$, $A_i^\ell \le (s_i^2/\beta)c_i^\ell$ and $n_i \le s_i\sqrt{\beta}\,c_i^n$. Both $s_i^2/\beta = \eta_i^2/S$ and $s_i\sqrt{\beta} = \eta_i/S$. Consequently (45) implies

$$\mathbb{E}_t\Phi' + \sum_i s_i\|\nabla_i f(X)\|_* \le (1 + K/S)\Phi + B/S,$$

where

$$K = \sum_i \eta_i^2 (c_i^g b_i^g + c_i^\ell b^\ell),$$
$$B = \sum_i [\eta_i^2 (c_i^g a_i^g + c_i^\ell a_i^\ell) + \eta_i c_i^n],$$
$$I = f(X_0) - \bar{f}_* + \sum_i \big[2\eta_i \rho_i \|\nabla_i f(X_0)\|_F + \eta_i c_i^\ell \langle \|\nabla_i f_j(X_0)\|_F \rangle_j\big].$$

Initialization gives $\Phi_0 \le I$. This proves the lemma. □

Proof of Theorem 6. For part (i), sum the inequality of Lemma 6 over $T+1$ rotated rounds. Discard the nonnegative final potential and the duplicated initial gradient term. With $\beta = T^{-1/2}$ and $(T+1)/T \le 2$, this gives

$$\frac{1}{T} \sum_{t<T} \sum_i \mathbb{E}\|\nabla_i f(X_t)\|_*^2 \le \frac{4\mathcal{I}/c + 8\mathcal{S}}{\sqrt{T}}. \tag{46}$$

For part (ii), apply Lemma 7 with $S = T+2$. Divide its inequality at round t by $(1+K/S)^{t+1}$ and sum. For any $k \le S$, the total expected gradient cost is at most $e^K(I+B)$, since $(1+K/S)^k \le e^K$. Taking $k = T+1$, dividing by $\bar{\eta}(T+1)S^{-3/4}$, and using $S/(T+1) \le 2$ gives $2e^K(I+B)/(\bar{\eta}S^{1/4})$ for the rotated average. Its minimum of expectations is no larger. The round rotation transfers this minimum to the paper's iterates $0, \dots, T$: it removes a duplicated initial model and adds the final paper model. This proves the second claim. All conditional recursions are integrable: the smooth proof propagates second moments, and the generalized proof propagates the displayed first moments and the squared noise EMA. Finite-product integration justifies these iterations. □

I.6 Accuracy, momentum, and worker averaging

The next bound separates initialization, averaged noise, and compression costs. It gives a momentum choice and a gradient-query budget for any target accuracy. Use a common sharp-step scale $\gamma = c\beta$, where $c > 0$ satisfies the envelope condition of Corollary 1 for EMA or Theorem 6 for Nesterov. Write $Q_T = T^{-1} \sum_{t<T} \sum_i \mathbb{E}\|\nabla_i f(X_t)\|_*^2$.

Proposition 3 (Finite-time smooth bound) *Under Assumptions 1–3 with ordinary smoothness and zero initialization, either momentum rule and either downlink satisfy*

$$Q_T \le 4\left[\frac{A}{\beta T} + \frac{H\beta}{T} + S_1\beta + S_2\beta^2 + S_3\beta^3\right] \tag{47}$$

for $T \ge 1$ and $0 < \beta \le 1$. The nonnegative coefficients A, H depend on initialization, and S_1, S_2, S_3 are noise coefficients. All five are independent of β, T; their explicit values are given in the proof.

Proof. Put $\Delta = f(X_0) - f_*$, $G_i^0 = \|\nabla_i f(X_0)\|_F^2$ and $H_i^0 = N^{-1} \sum_j \|\nabla_i f_j(X_0)\|_F^2$. Let σ_i^2 bound the worker noise variance in Frobenius norm; Assumption 3 permits $\sigma_i = \sigma$. Define $A = \Delta/c + \sum_i a_i$, $H = \sum_i h_i$, and $S_k = 2\sum_i s_{ki}\sigma_i^2$, using the following coefficients:

	EMA	Nesterov
a_i	$6\rho_i^2 G_i^0$	$2b_i G_i^0$
h_i	$72\rho_i^2 H_i^0/\alpha_D^2$	$6b_i G_i^0 + 6\widehat{R}_i H_i^0$
s_{1i}	$6\rho_i^2/N$	$2b_i/N$
s_{2i}	$6\rho_i^2(1-\alpha_D)/\alpha_D$	$b_i/N + \widehat{R}_i$
s_{3i}	$72\rho_i^2/\alpha_D^2$	$6b_i/N + 6\widehat{R}_i$

Here $\rho_i = \sqrt{r_i}$, $b_i = 8\rho_i^2$ and $\widehat{R}_i = 40b_i/\alpha_D^2$ as above. The factor two in S_k accounts for the round rotation in Proposition 2.

For Nesterov, substitution into (44) gives $\Psi_0 = c(A+H\beta^2)$ and $\sum_i \nu_i \sigma_i^2 = (S_1\beta+S_2\beta^2+S_3\beta^3)/2$. For EMA, the global and local squared momentum errors receive weights $6\rho_i^2/\beta$ and $72\beta\rho_i^2/\alpha_D^2$; the compression error receives weight $6\rho_i^2(1-\alpha_D)/\alpha_D$. The usual conditional tracking inequalities give the same initial-potential identity and noise polynomial with the EMA column. Their movement terms obey the envelope in Corollary 1. In both cases telescope the Lyapunov drift over $T+1$ rotated rounds. The extra initial gradient cost is nonnegative and $(T+1)/T \le 2$; the initial-potential term is $4\Psi_0/(c\beta T)$ and the noise term is at most $4(S_1\beta + S_2\beta^2 + S_3\beta^3)$. □

Corollary 3 (Tuning for a target accuracy) *For any $\varepsilon > 0$, set*

$$D_\varepsilon = 1 + \frac{24S_1}{\varepsilon^2} + \sqrt{\frac{24S_2}{\varepsilon^2}} + \left(\frac{24S_3}{\varepsilon^2}\right)^{1/3},$$
$$\beta = D_\varepsilon^{-1}, \qquad \gamma = c\beta,$$
$$T_\varepsilon = \max\left\{1, \left\lceil \frac{8(A+H)D_\varepsilon}{\varepsilon^2} \right\rceil\right\}.$$

Then $Q_{T_\varepsilon} \le \varepsilon^2$, using exactly T_ε synchronous gradient-query rounds and NT_ε worker-gradient evaluations. In particular, the sufficient budget has the dependence

$$T_\varepsilon = \mathcal{O}\Big(1 + (A+H)[\varepsilon^{-2} + S_1\varepsilon^{-4} + \sqrt{S_2}\varepsilon^{-3} + S_3^{1/3}\varepsilon^{-8/3}]\Big).$$

With zero noise, $\beta = 1$ (no momentum) and $T_\varepsilon = \mathcal{O}(\varepsilon^{-2})$. For fixed $S_1 > 0$, $T_\varepsilon = \mathcal{O}(\varepsilon^{-4})$. For both momentum rules, the leading variance coefficient S_1 is proportional to $1/N$; S_2, S_3 also contain local compression costs.

Proof. Each of $S_1\beta$, $S_2\beta^2$, $S_3\beta^3$ is at most $\varepsilon^2/24$. The two transient terms in (47) total at most $4(A+H)/(\beta T_\varepsilon) \le \varepsilon^2/2$. The noise terms contribute at most the other half. Substitution of D_ε gives the stated budget and its noiseless and stochastic specializations. The query count follows from one fresh gradient per worker per paper round. □

I.7 Fixed momentum and learning-rate schedules

We now bound the gradient norm for deterministic learning rates, including the constant and cosine schedules used in experiments. The argument has two steps: bound the tracking error, then sum the resulting descent inequality. Throughout this section the polar directions are exact and momentum is fixed within a run. Use $0 < \beta \le 1$, $\mu = 1-\beta$, and $\theta = 0$ for EMA or $\theta = \beta$ for Nesterov, with the Frobenius noise bounds σ_i defined above. A round queries $a_{t,i}^{(j)} = \nabla_i f_j(W_t) + \xi_{t,i}^{(j)}$ and performs

$$\begin{aligned}
M_{t+1,i}^{(j)} &= \mu M_{t,i}^{(j)} + \beta a_{t,i}^{(j)},\\
V_{t,i}^{(j)} &= (1-\theta)M_{t+1,i}^{(j)} + \theta a_{t,i}^{(j)},\\
g_{t+1,i}^{(j)} &= g_{t,i}^{(j)} + C_i^{\uparrow}(V_{t,i}^{(j)} - g_{t,i}^{(j)}),\\
h_{t,i} &= N^{-1}\sum_j g_{t+1,i}^{(j)}, \qquad X_{t+1,i} = X_{t,i} - \eta_{t,i}\,\mathrm{polar}(h_{t,i}),\\
W_{t+1,i} &= W_{t,i} + C_i^{\downarrow}(X_{t+1,i} - W_{t,i}).
\end{aligned}$$

Initialize $X_0 = W_0$ and all momentum and gradient memories at zero. The indexing starts immediately before the gradient query. Put $\rho_i = \sqrt{r_i}$ and $\Delta = f(X_0) - f_*$. For $0 \le \eta_{t,i} \le \bar{\eta}_i$, write $S_i = \sum_{t<T}\eta_{t,i}$ and $Q_i = \sum_{t<T}\eta_{t,i}^2$. Let $q_{Ui}, q_{Pi} \in [0,1)$ satisfy, for every compressor input,

$$\|R - C_i^{\uparrow}(R)\|_F \le q_{Ui}\|R\|_F, \qquad \|R - C_i^{\downarrow}(R)\|_F \le q_{Pi}\|R\|_F.$$

These hold for scaled sign with $q = \sqrt{1-\alpha}$ and $\alpha = 1/d_i$, and for the exact downlink with $q_{Pi} = 0$.

Theorem 7 (Convergence with deterministic learning rates) *Under Assumptions 1–3 with ordinary smoothness, the updates above, for either momentum rule and either downlink, satisfy*

$$\sum_{t<T}\sum_i \eta_{t,i}\mathbb{E}\|\nabla_i f(X_t)\|_* \le \Delta + \sum_i \left(2\rho_i\mathcal{B}_{i,T} + \frac{L_i Q_i}{2}\right), \tag{48}$$

where the tracking budget separates into initialization, movement, and noise:

$$\mathcal{B}_{i,T} = \bar{\eta}_i I_{i,\beta} + S_i(\bar{\eta}_i D_{i,\beta} + R_{i,\beta}).$$

The nonnegative coefficients $I_{i,\beta}, D_{i,\beta}, R_{i,\beta}$ are independent of T and the learning-rate schedule. Their explicit values are given in the proof of Lemma 8.

Lemma 8 (Tracking with fixed momentum) *Under the assumptions of Theorem 7,*

$$\sum_{t<T} \sum_i \eta_{t,i} \mathbb{E}\|h_{t,i} - \nabla_i f(X_t)\|_F \le \sum_i \mathcal{B}_{i,T}. \tag{49}$$

For fixed problem and compressor constants, as $\beta \downarrow 0$, $I_{i,\beta}, D_{i,\beta} = \mathcal{O}(\beta^{-1})$, and

$$R_{i,\beta} \le 4\sigma_i \left[\sqrt{\beta/N} + \kappa_{Ui}\beta\right], \qquad \kappa_{Ui} = q_{Ui}/(1-q_{Ui}).$$

Proof. We bound the downlink gap, momentum bias, and momentum noise separately. Put $\widehat{L}_i = \max_j L_{i,j}$, $B_i = \rho_i$, $\kappa_{Pi} = q_{Pi}/(1-q_{Pi})$, $g_i^0 = \|\nabla_i f(X_0)\|_F$ and $h_i^0 = N^{-1}\sum_j \|\nabla_i f_j(X_0)\|_F$. The needed coefficients are

$$\begin{aligned}
k &= \beta + \mu\theta, \qquad v = \frac{\beta}{2-\beta}, \qquad v_\theta = k^2 + (1-\theta)^2\mu^2 v, \\
I_{i,\beta} &= (1-\theta)\mu g_i^0/\beta + \kappa_{Ui} h_i^0, \\
D_{i,\beta} &= \widehat{L}_i B_i[\kappa_{Pi} + (1-\theta)\mu/\beta + \kappa_{Ui}(1+2k\kappa_{Pi})], \\
R_{i,\beta} &= \sigma_i[\sqrt{v_\theta/N} + \kappa_{Ui}((1-\theta)\beta(1+\sqrt{v}) + 2\theta)].
\end{aligned}$$

For either momentum rule, $v \le \beta$, $v_\theta \le 5\beta$ and $(1-\theta)\beta(1+\sqrt{v}) + 2\theta \le 4\beta$. These inequalities give the stated bounds on the coefficients.

To verify the tracking budget, first note that

$$\|X_{t+1,i} - W_{t+1,i}\|_F \le q_{Pi}(\|X_{t,i} - W_{t,i}\|_F + B_i\eta_{t,i}).$$

Thus the downlink gap is at most $P_i = \kappa_{Pi} B_i \bar{\eta}_i$. Separate the noise EMA $Z_{t+1,i}^{(j)} = \mu Z_{t,i}^{(j)} + \beta\xi_{t,i}^{(j)}$ from the momentum bias. Conditional centering gives $\mathbb{E}\|Z_{t,i}^{(j)}\|_F^2 \le \sigma_i^2 v$; averaging workers first gives $\sigma_i^2 v/N$. The remaining bias, measured against $\nabla_i f_j(X_t)$, is at most $\mu^t\|\nabla_i f_j(X_0)\|_F + \widehat{L}_i P_i + \widehat{L}_i B_i \bar{\eta}_i/\beta$. The momentum target's centered noise is $(1-\theta)\mu Z_{t,i}^{(j)} + k\xi_{t,i}^{(j)}$. Its two terms are orthogonal in expectation, giving the exact variance factor v_θ. Consequently the averaged target bias and noise obey

$$\mathbb{E}\left\| N^{-1}\sum_j V_{t,i}^{(j)} - \nabla_i f(X_t)\right\|_F \le (1-\theta)\mu^{t+1} g_i^0 + A_i,$$

where $A_i = \widehat{L}_i P_i + (1-\theta)\mu\widehat{L}_i B_i \bar{\eta}_i/\beta + \sigma_i\sqrt{v_\theta/N}$.

For $t \ge 1$, the target increment is

$$V_{t,i}^{(j)} - V_{t-1,i}^{(j)} = (1-\theta)\beta(a_{t,i}^{(j)} - M_{t,i}^{(j)}) + \theta(a_{t,i}^{(j)} - a_{t-1,i}^{(j)}).$$

The preceding bias and noise bounds show its expected norm is at most $(1-\theta)\beta\mu^t\|\nabla_i f_j(X_0)\|_F + J_i$, where

$$J_i = \widehat{L}_i B_i \bar{\eta}_i + 2k\widehat{L}_i P_i + \sigma_i[(1-\theta)\beta(1+\sqrt{v}) + 2\theta].$$

Contractivity propagates the uplink error with multiplier q_{Ui}. Sum the geometric initialization terms and use $\eta_{t,i} \le \bar{\eta}_i$ to obtain $\sum_t \eta_{t,i}\mathbb{E}\|h_{t,i} - \nabla_i f(X_t)\|_F \le \bar{\eta}_i I_{i,\beta} + S_i(A_i + \kappa_{Ui} J_i)$ for each block. Here $A_i + \kappa_{Ui} J_i = \bar{\eta}_i D_{i,\beta} + R_{i,\beta}$, which gives $\mathcal{B}_{i,T}$. For simultaneous blocks, Assumption 2 implies each local gradient block depends only on its own parameter block, as explained in Appendix F; therefore these estimates apply to the common full-model query. □

Proof of Theorem 7. Combine Lemma 8 with the polar pairing inequality

$$\langle \nabla_i f(X_t), \mathrm{polar}(h_{t,i})\rangle \ge \|\nabla_i f(X_t)\|_* - 2\rho_i\|h_{t,i} - \nabla_i f(X_t)\|_F.$$

Apply smooth descent with remainder $\sum_i L_i \eta_{t,i}^2/2$, take expectations, and telescope the single global objective gap Δ. This gives (48). □

Corollary 4 (Constant and cosine schedules) *For common learning rates $\eta_{t,i} = \eta_t$, put $S = \sum_{t<T} \eta_t > 0$, $\bar{\eta} = \max_{t<T} \eta_t$ and*

$$\mathcal{G}_T = S^{-1} \sum_{t<T} \eta_t \mathbb{E} \sum_i \|\nabla_i f(X_t)\|_*.$$

With $I_\beta = 2\sum_i \rho_i I_{i,\beta}$, $D_\beta = \frac{1}{2}\sum_i L_i + 2\sum_i \rho_i D_{i,\beta}$ and $R_\beta = 2\sum_i \rho_i R_{i,\beta}$, one has

$$\mathcal{G}_T \le \frac{\Delta + \bar{\eta} I_\beta}{S} + \bar{\eta} D_\beta + R_\beta. \tag{50}$$

For constant $\eta_t = \bar{\eta}$ and cosine $\eta_t = \bar{\eta}[1 + \cos(\pi t/T)]/2$, this gives:

(i) *With fixed momentum and fixed peak, an $\mathcal{O}(T^{-1})$ transient plus the explicit upper-bound residual $\bar{\eta} D_\beta + R_\beta$.*

(ii) *With fixed momentum and peak $\bar{\eta} \asymp T^{-1/2}$, $\mathcal{G}_T \le \mathcal{O}(T^{-1/2}) + R_\beta$.*

(iii) *With $\beta \asymp T^{-1/2}$ and $\bar{\eta} \asymp T^{-3/4}$, $\mathcal{G}_T = \mathcal{O}(T^{-1/4})$ for either momentum and downlink.*

For both momentum rules, the noise residual satisfies

$$R_\beta \le 8 \sum_i \rho_i \sigma_i [\sqrt{\beta/N} + \kappa_{Ui}\beta].$$

Fixed positive layer multipliers can be absorbed into $\eta_{t,i}$; the same conclusions then concern the corresponding weighted block norm.

Proof. Substitute the common schedule into (48) and use $\sum_t \eta_t^2 \le \bar{\eta} S$. For the constant schedule $S = T\bar{\eta}$, so the transient is $\Delta/(T\bar{\eta}) + I_\beta/T$. For cosine, $S = \bar{\eta}(T+1)/2$, so it is at most twice this quantity. For fixed problem and compressor constants, as $\beta \downarrow 0$ both momentum rules have $I_\beta, D_\beta = \mathcal{O}(\beta^{-1})$, $v_\theta = \mathcal{O}(\beta)$ and $R_\beta = \mathcal{O}(\sqrt{\beta})$. These estimates give the stated rates. The noise bound follows from Lemma 8. □

Smaller β reduces this noise budget but increases the drift bound $\bar{\eta} D_\beta = \mathcal{O}(\bar{\eta}/\beta)$. The averaged noise retains $1/\sqrt{N}$; the local compression term remains. For $q_{Pi} = \sqrt{1-\alpha_P}$ and small α_P, $\kappa_{Pi} \sim 2/\alpha_P$; thus poor downlink contraction increases D_β and the gap bound $P_i = \kappa_{Pi} B_i \bar{\eta}_i$. The exact downlink sets these terms to zero. The guarantee controls the learning-rate-weighted first moment $\mathcal{G}_T$ under the layerwise smoothness of Assumption 2 and fixed momentum.

I.8 Compression constants and other layer norms

Remark 4 (Worst-case and realized contraction) *The uplink parameter $\alpha = 1/d_{\max}$ enters the threshold of Corollary 1 through the $1/\alpha^2$-terms of ζ_i, so the admissible γ_i shrinks linearly in $d_{\max}$, the dimension of the largest layer; the same dimension factor arises for Top-1 compressors in Euclidean EF21 [Richtárik et al., 2021]. The measured uplink residuals in Table 10 have contraction between 0.32 and 0.64. A uniform residual bound $\|\mathbf{\Delta}_t\|_1^2 \ge \kappa d \|\mathbf{\Delta}_t\|_F^2$ permits the choice $\alpha = \kappa$ in these bounds. For iteration-dependent contraction bounds, see Gruntkowska et al. [2025, Rem. 12].*

The downlink residual of EF21-MuonSign can become concentrated in magnitude even when every entry is nonzero. Unlike the uplink residual, which every round's gradient refreshes, it is produced by the compressor's own recursion

$$\mathbf{\Delta}_{t+1}^{\downarrow} = \mathbf{\Delta}_t^{\downarrow} - \eta_{t+1}\mathbf{D}_{t+1} - \operatorname{mean}|\mathbf{\Delta}_t^{\downarrow}| \operatorname{sign}(\mathbf{\Delta}_t^{\downarrow}),$$

which corrects every coordinate by the same scalar $\operatorname{mean}|\mathbf{\Delta}_t^{\downarrow}|$. A coordinate whose per-step drive $\eta_{t+1}(\mathbf{D}_{t+1})_{kl}$ exceeds that scalar receives a correction smaller than its drive at every step, while the remaining coordinates keep the scalar small, so the residual concentrates on few coordinates; by the exact expression $\alpha(\mathbf{Y}) = \|\mathbf{Y}\|_1^2/(d\|\mathbf{Y}\|_F^2)$ of Lemma 5, concentration is precisely what lowers α. Section 4.2 measures the effect: on the layers built from a zero initialization, $\alpha(\mathbf{\Delta}^{\downarrow})$ falls to 1.2×10^{-4}, about four orders of magnitude below the uplink value on the same layers, yet still far above the floor $1/d = 4.2 \times 10^{-7}$.

Remark 5 (The scaled sign is not layer-norm contractive) *For EF21-MuonSign we use the Frobenius-contractivity extension of Gruntkowska et al. [2025, Rem. 23], which introduces $\bar{\rho}_i^2 = r_i$ in the downlink terms. Scaled sign has no uniform spectral-norm contraction guarantee in general: for $\mathbf{Y}_\delta = (1-\delta)\mathbf{I}_n + \delta\mathbf{J}_n$ ($\mathbf{J}_n$ all ones, $0 < \delta < 1$, $n \geq 2$), every entry is positive, so $\operatorname{sign}(\mathbf{Y}_\delta) = \mathbf{J}_n$, $\operatorname{mean}|\mathbf{Y}_\delta| = (1+(n-1)\delta)/n$, and*

$$\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta = (1-\delta)\big(\tfrac{1}{n}\mathbf{J}_n - \mathbf{I}_n\big),$$

a matrix with eigenvalues 0 and $-(1-\delta)$; hence $\|\mathcal{C}(\mathbf{Y}_\delta) - \mathbf{Y}_\delta\|_{2\to 2} = 1-\delta$ while $\|\mathbf{Y}_\delta\|_{2\to 2} = 1+(n-1)\delta$, and the ratio tends to 1 as $\delta \downarrow 0$.

Remark 6 (From Muon to Gluon) *Only $(\underline{\rho}_i, \bar{\rho}_i) = (1, \sqrt{r_i})$ is spectral-norm-specific: Lemma 5 is Euclidean and the framework's theorems hold for arbitrary layer norms. A new geometry has to supply only its LMO and its norm-equivalence pair; both corollaries then hold with those constants in place of $(1, \sqrt{r_i})$, giving* EF21-GluonUSign *and* EF21-GluonSign *for the Gluon setting [Riabinin et al., 2025]. Admissible geometries abound: $\ell_1 \to \ell_\infty$ for embeddings [Pethick et al., 2025a], the Schatten-p norms [Cesista, 2025], the Ky Fan duals of the Fanion family [Kravatskiy et al., 2025], and, by the closure property of the last work, the norms whose LMO is a conic combination of these LMOs; their unit ball is the corresponding Minkowski sum, whence $\bar{\rho}_i \leq \sum_j \alpha_j \bar{\rho}_i^{(j)}$. Convergence is thus a property of error feedback together with scaled-sign compression, not of the spectral geometry.*

J Reproducibility Details

Computing infrastructure. The synthetic study ran on one NVIDIA RTX A4000 (Python 3.12, PyTorch 2.7.0, CUDA 12.8). Centralized and federated CIFAR-10 runs used an NVIDIA RTX A4500 (19.6 GB) with Python 3.12.11, PyTorch 2.5.1+cu124, CUDA 12.4. Language-modelling hardware is specified in Appendix O.

Randomness and seeds. The network experiments seed Python's `random` module, NumPy and PyTorch on all CUDA devices; the federated runs additionally pin cuDNN to deterministic kernels, while the centralized sweep leaves cuDNN autotuning enabled. CIFAR-10 uses seeds 0–2 for centralized learning and 0–4 for federated learning, reporting the mean $\pm$ sample standard deviation. The weight-decay ablations use seed 0. NanoGPT uses one unseeded run per method. Synthetic seeds and averaging are specified in Appendix K.

CIFAR-10 settings. For the primary CIFAR-10 comparisons, only η_0 is tuned. Every method with a norm-fixed step is tuned and reported under the unit-gain rule (5); SGD and Adam use one global rate. Momentum is fixed at 0.9 and weight decay at 0 in the primary tables. Biases, normalization parameters and the classifier head use AdamW at 10^{-3}. Both learning rates follow cosine schedules.

Learning-rate selection. Both CIFAR-10 studies select η_0 at seed 0 on a fixed 45k/5k train/validation partition; the federated validation split is held out before client partitioning. Each method starts from a five-point 1–2–5 lattice, widened and rerun when the optimum lies at an endpoint. Selection uses the full reporting horizon: 75 epochs for centralized learning or 2000 rounds for federated learning, ranked by validation accuracy averaged over the final five evaluations. The selected rate is reused for all final runs on the full 50k training set. Test accuracy is averaged over the final five evaluations of each run.

Numerical conventions. The Muon LMO is computed in `bfloat16` (five Newton–Schulz steps) unless stated otherwise. Federated BatchNorm statistics remain at their initialization in training and evaluation; affine parameters are learned. For EF21-MuonSign, training metrics are logged at the broadcast model $\mathbf{W}$ while validation and test metrics are evaluated at the server model $\mathbf{X}$ of (8), except where a table states otherwise.

K The smooth convex problem

We compare optimization trajectories, gradient–step alignment, and sensitivity to step size and conditioning on a deterministic matrix quadratic.

Algorithm	**iters**	**best** F	$\min \|\nabla F\|_F$	**tuned** (η, μ)
SignMuon	364	$6.9 \cdot 10^{-4}$	$2.4 \cdot 10^{-2}$	$(1.0 \cdot 10^{-3}, 0)$
MuonUSign	556	$7.1 \cdot 10^{-4}$	$2.3 \cdot 10^{-2}$	$(1.0 \cdot 10^{-2}, 0.5)$
MuonSign	595	$4.8 \cdot 10^{-4}$	$1.9 \cdot 10^{-2}$	$(6.8 \cdot 10^{-4}, 0.5)$
EF21-SignMuon	401	$5.4 \cdot 10^{-4}$	$2.1 \cdot 10^{-2}$	$(6.8 \cdot 10^{-3}, 0)$
EF21-MuonUSign	521	$5.3 \cdot 10^{-4}$	$1.8 \cdot 10^{-2}$	$(6.8 \cdot 10^{-3}, 0)$
EF21-MuonSign	695	$3.8 \cdot 10^{-4}$	$1.7 \cdot 10^{-2}$	$(4.6 \cdot 10^{-3}, 0)$
Muon	267	$4.7 \cdot 10^{-4}$	$2.0 \cdot 10^{-2}$	$(1.0 \cdot 10^{-2}, 0.5)$
SignSGD	486	$4.3 \cdot 10^{-4}$	$2.2 \cdot 10^{-2}$	$(6.8 \cdot 10^{-4}, 0.5)$
SGD	66	$1.9 \cdot 10^{-8}$	$1.6 \cdot 10^{-6}$	$(2.15, 0.5)$
Adam	85	$9.2 \cdot 10^{-6}$	$6.5 \cdot 10^{-4}$	$(6.8 \cdot 10^{-2}, —)$

Table 5: Fewest iterations to $F \leq 10^{-3}$ within 5000 iterations, with $(\eta, \mu, \text{schedule})$ tuned per method. The selected schedules are constant. The remaining columns give the minimum objective and gradient norm over the full run.

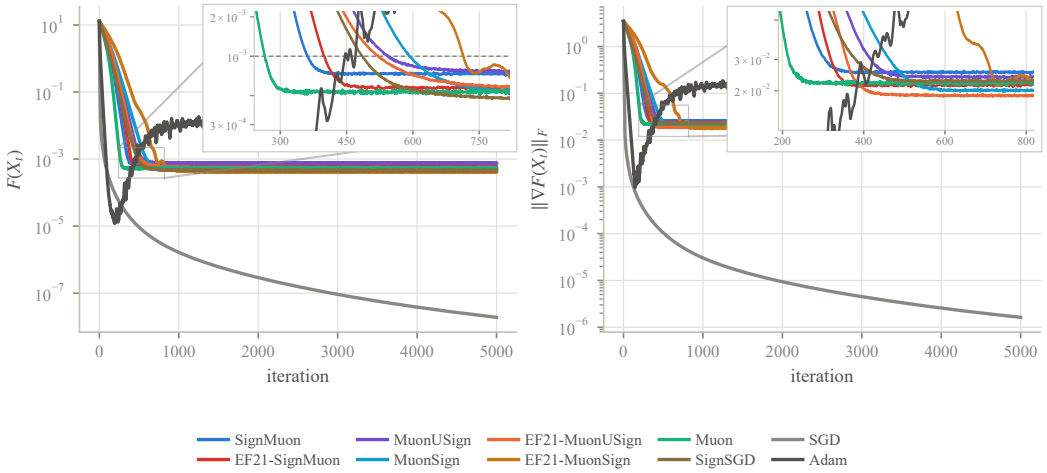


Figure 6: Objective and gradient-norm trajectories for the smooth convex problem (51) at the settings of Table 5. Curves show geometric means.

Quadratic objective and initialization. We use

$$F(\mathbf{X}) = \frac{1}{2}\|\mathbf{A}^{1/2}\mathbf{X}\mathbf{B}^{1/2}\|_F^2 = \frac{1}{2}\langle \mathbf{X}, \mathbf{A}\mathbf{X}\mathbf{B}\rangle, \qquad \mathbf{X} \in \mathbb{R}^{m\times n}, \tag{51}$$

with $m = n = 100$, positive-definite $\mathbf{A}, \mathbf{B}$, and initialization entries drawn from $\mathcal{N}(0, 0.01)$. The eigenvalues of $\mathbf{A}$ and $\mathbf{B}$ are uniform on $(0, 1)$ in independent Haar-random eigenbases. The minimizer is $\mathbf{X}^\star = \mathbf{0}$, and $\nabla F(\mathbf{X}) = \mathbf{A}\mathbf{X}\mathbf{B}$. The Hessian $\mathbf{B} \otimes \mathbf{A}$ gives the Frobenius smoothness and strong-convexity constants

$$L = \lambda_{\max}(\mathbf{A})\lambda_{\max}(\mathbf{B}), \qquad \sigma = \lambda_{\min}(\mathbf{A})\lambda_{\min}(\mathbf{B}).$$

For the conditioning sweep, log-spaced spectra set $L = 1$ and $\kappa = L/\sigma$ explicitly. EF21-MuonSign evaluates gradients at the broadcast model $\mathbf{W}$ and reports metrics at the server model $\mathbf{X}$.

Tuning for a fixed objective threshold. We tune learning rate, momentum and schedule to minimize the iterations required for $F \leq 10^{-3}$. Table 5 reports the crossing times and full-run minima; Figure 6 shows the trajectories. The separate sweeps below measure constant-step plateaus and dependence on the iteration budget.

Gradient–update alignment. We record the alignment between the gradient and the direction actually used in the update:

$$\rho_t = \frac{\langle \nabla F(\mathbf{X}_t), \mathbf{D}_t\rangle}{\|\nabla F(\mathbf{X}_t)\|_F \|\mathbf{D}_t\|_F}. \tag{52}$$

Algorithm	$\min_t \rho_t$	**median** ρ_t	**mean** ρ_t	**% of steps with** $\rho_t < 0$	**closed form**	**tuned** (η, μ)
SignMuon	0.208	0.381	0.399	0.00%	—	$(2.2 \cdot 10^{-4}, 0)$
MuonUSign	0.161	0.357	0.378	0.00%	—	$(3.2 \cdot 10^{-3}, 0)$
MuonSign	0.142	0.360	0.364	0.00%	—	$(2.2 \cdot 10^{-4}, 0)$
EF21-SignMuon	0.333	0.659	0.573	0.00%	—	$(1.5 \cdot 10^{-3}, 0)$
EF21-MuonUSign	0.346	0.558	0.553	0.00%	—	$(1.5 \cdot 10^{-3}, 0)$
EF21-MuonSign	−0.026	0.489	0.419	0.95%	—	$(1.5 \cdot 10^{-3}, 0)$
Muon	0.454	0.666	0.633	0.00%	0.695	$(1.5 \cdot 10^{-3}, 0)$
SignSGD	0.174	0.503	0.499	0.00%	0.794	$(2.2 \cdot 10^{-4}, 0.5)$
SGD	—	—	—	—	1	$(3.2, 0.95)$

Table 6: Alignment along the tuned trajectories. Analytic references are evaluated at $\mathbf{X}_0$ without momentum using exact directions; trajectory statistics use the listed momentum. SGD is shown by its analytic reference only.

	floor		**budget**				**stability**
Algorithm	$\mathrm{d}\log g_\infty / \mathrm{d}\log \eta$	R^2	p	R^2	q	R^2	$\eta_{\max} s$
SignMuon	1.000	1.000	1.88	0.996	0.39	0.94	13.4
MuonUSign	1.000	1.000	1.86	0.995	0.39	0.94	14.9
MuonSign	1.000	1.000	1.93	0.993	0.44	0.94	11.8
EF21-SignMuon	1.000	1.000	2.07	1.000	0.55	1.00	18.9
EF21-MuonUSign	1.000	1.000	2.04	1.000	0.55	1.00	14.9
EF21-MuonSign	1.000	1.000	2.14	1.000	0.55	1.00	14.9
Muon	1.000	1.000	2.03	0.999	0.55	1.00	17.6
SignSGD	0.989	1.000	1.76	0.990	0.39	0.94	11.1
SGD	*no floor*		8.97	0.985	−0.11	0.50	$2.06^{\dagger}$
Adam	1.000	1.000	2.65	0.915	−0.66	0.62	$> 100^{\ddagger}$

Table 7: Empirical step-size and budget exponents, with log–log fit R^2, and stability edges. For the scaled edge, $s = \sqrt{mn}$ for SignMuon, MuonSign and SignSGD, and $s = \sqrt{r}$ for polar-based methods, where $r = \min(m, n)$. † SGD reports $\eta_{\max}$. ‡ Adam reports the upper search limit for η, reached without divergence.

Table 6 summarizes the trajectories and includes no-momentum references at $\mathbf{G} = \nabla F(\mathbf{X}_0)$: 1 for SGD, $\|\mathbf{G}\|_1/(\|\mathbf{G}\|_F\sqrt{mn})$ for SignSGD, and $\|\mathbf{G}\|_*/(\|\mathbf{G}\|_F\sqrt{\operatorname{rank}\mathbf{G}})$ for exact Muon. The three sign placements retain positive alignment in these runs.

Step-size, budget and conditioning sweeps. For constant-step runs, we fit the settled Frobenius gradient norm g_∞ against η. At each horizon T, we re-tune $(\eta, \mu, \text{schedule})$ and fit

$$\mathrm{err}(T) \propto T^{-p}, \qquad \eta^\star(T) \propto T^{-q}.$$

Here $\mathrm{err}(T) = \min_{t \le T} \|\nabla F(\mathbf{X}_t)\|^2$. The norm is the entrywise ℓ_1 norm for SignMuon, MuonSign and SignSGD, the nuclear norm for polar-based methods, and the Frobenius norm for SGD and Adam. Table 7 reports these empirical fits and the largest stable step sizes. Figure 7 also sweeps $\kappa \in [10, 10^6]$ at fixed $L = 1$, re-tuning at each value.

Random seeds and tuning grids. We use three problem draws (seeds 1337–1339) and initialization seed 42, with the LMO computed in `bfloat16`. Error metrics are geometric means across draws; iteration counts are arithmetic means. Learning-rate grids span five decades per step family, with boundary optima flagged. At the largest condition numbers, SGD and SignSGD select the upper momentum-grid value 0.99. The benchmark records the configuration and run metadata for each measurement.

L Centralized training results

Table 8 reports the centralized CIFAR-10 results on ResNet-18 in full: the selected η_0, the final train and test accuracies, the epoch count to the 90% threshold, and the per-epoch cost. Figure 2 starts at

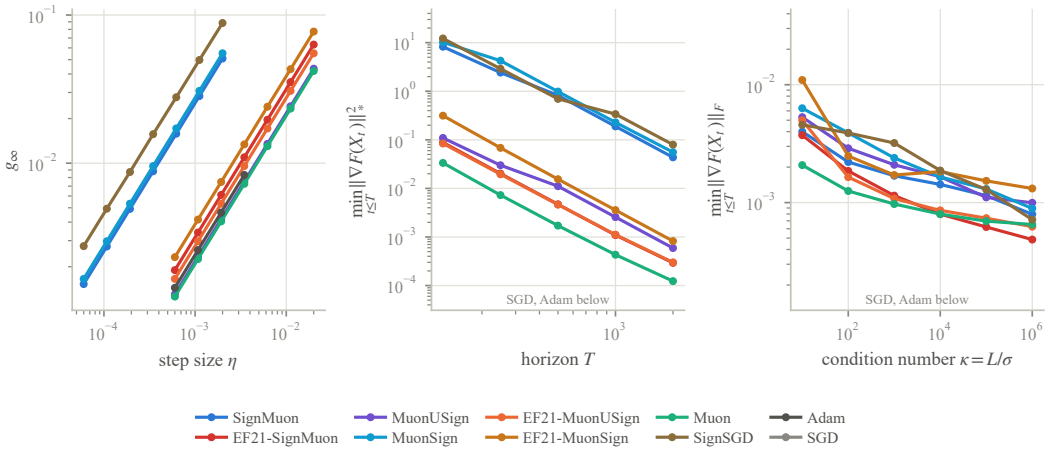


Figure 7: Step-size, horizon and conditioning sweeps for the quadratic benchmark. Hyperparameters are re-tuned at each horizon and condition number.

Dataset	Optimizer	Epochs	η_0	Train Acc	Test Acc	Ep. to 90%	s/epoch
CIFAR-10	SignMuon	75	0.02	99.99	94.60 ± 0.15	7.7	16.6
	Muon		0.1	99.99	94.35 ± 0.27	7.7	14.4
	EF21-SignMuon		0.02	100.00	94.31 ± 0.11	9.0	16.6
	EF21-MuonUSign		0.05	99.99	94.14 ± 0.07	10.3	16.7
	EF21-MuonSign		0.005	99.99	94.04 ± 0.10	11.0	18.1
	MuonUSign		0.02	99.99	93.98 ± 0.12	10.3	16.2
	Adam		0.001	99.97	93.37 ± 0.27	20.0	12.7
	SignSGD		0.002	99.97	93.37 ± 0.26	19.0	12.5
	MuonSign		0.1	99.98	93.31 ± 0.21	17.7	17.2
	SGD		0.02	99.96	93.04 ± 0.14	21.3	12.2

Table 8: CIFAR-10: centralized learning on ResNet-18, 75 epochs and three seeds. Test accuracy is averaged over the last five epochs and reported as mean $\pm$ s.d. across seeds; the other performance metrics are seed means. Learning-rate selection: Appendix J.

epoch 25, where the methods are a few points apart rather than twenty. Every run uses batch 128, momentum 0.9, auxiliary rate 10^{-3}, and zero weight decay, so that the matrix rule and η_0 are the only quantities that vary. Figure 8 gives the accuracy curves from epoch 3, Figure 9 the training loss, and Figure 10 the learning-rate sweep behind the selection.

Selection horizon. Selection and reporting share the 75-epoch horizon, so each η_0 above is the argmax of a validation sweep conducted at the length the table reports. An earlier protocol that selected at 15 epochs and reported at 75 was abandoned because the two horizons selected different rates for two of the methods. The selection is not sharply peaked for the sign-after methods: over the rates within a factor of five of its own optimum, SignMuon's validation accuracy varies by 0.13 points, against 0.65 for Muon and 3.00 for SignSGD (Figure 10).

Weight decay. The primary table is unregularized, the setting the theorems analyse and the one Mishra et al. [2026]'s sweep selects. Repeating the top three at the same η_0 with decoupled decay 5×10^{-4} (seed 0, decay not re-tuned) displaces each by at most 0.34 points, of the order of the three-seed standard deviation of the undecayed runs, and leaves the three within 0.23 of one another: Muon 94.46% against 94.12% undecayed at that seed, EF21-SignMuon 94.45% against 94.42%, SignMuon 94.23% against 94.43%. The ordering within that interval does change, but at a single seed and over so narrow a range it is not a measured effect; what the ablation establishes is that decay introduces no separation where the primary table shows none. Decay is applied decoupled, $\mathbf{X} \mathrel{*}= 1 - \eta\lambda$, so the LMO sees the true gradient; the coupled convention would only rotate the direction, since every step here is scale-invariant.

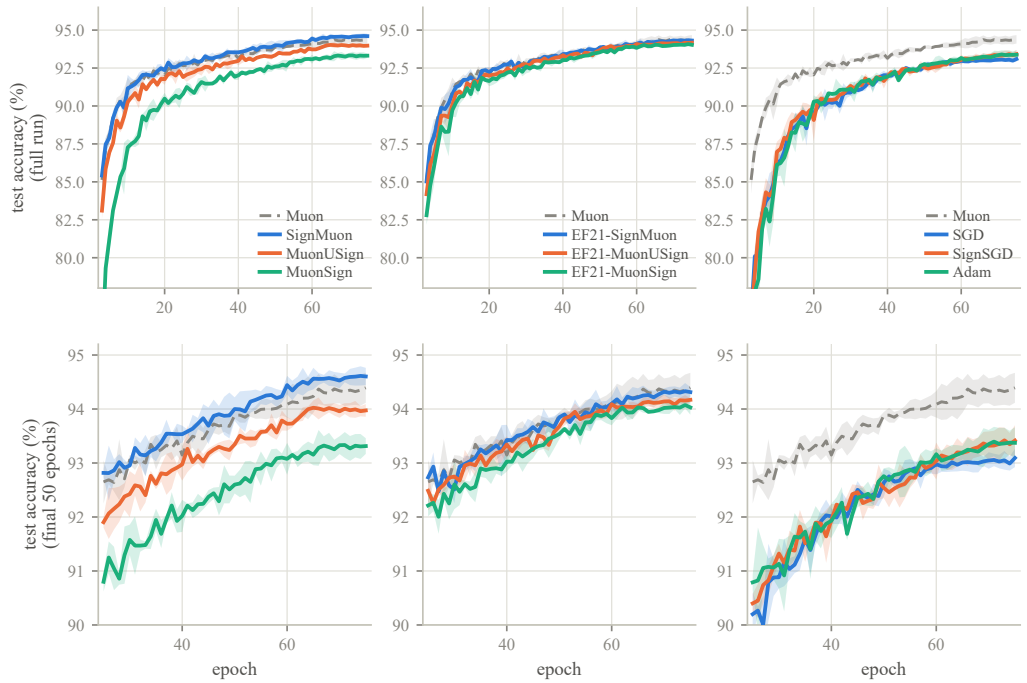


Figure 8: Test accuracy for centralized training of ResNet-18 on CIFAR-10. Bands are ± 1 standard deviation over three seeds.

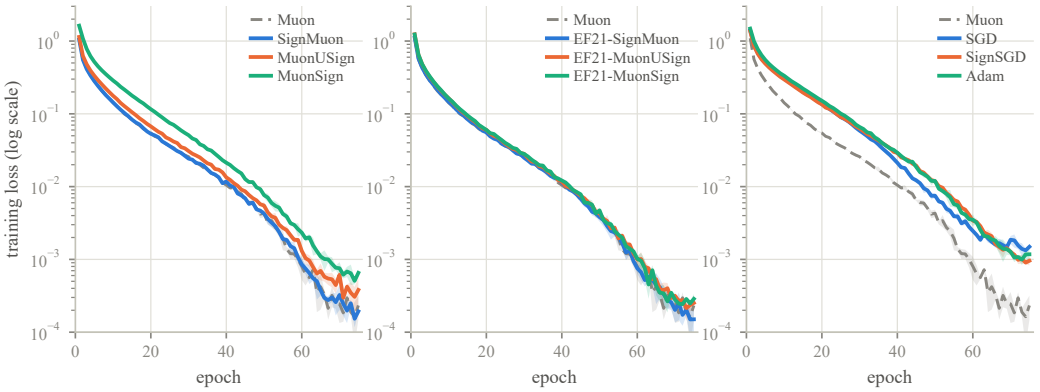


Figure 9: Training loss for centralized training of ResNet-18 on CIFAR-10 over 75 epochs.

M Communication accounting

We count communication per client per round in bits per model parameter.

Write P_{mat} for the number of matrix parameters, P_{aux} for the auxiliary group (biases, BatchNorm affine parameters, the classifier head), L for the number of matrix layers, and $P = P_{\text{mat}} + P_{\text{aux}}$. Auxiliary parameters are transmitted at 32 bits each in both directions. The sign convention of Section 3 gives binary messages. Each EF21 channel transmits the pair $(\mathbf{s}_t, \alpha_t)$ with one full-precision α_t per matrix layer.

For a full-precision channel, a binary sign channel, and a scaled-sign EF21 channel, respectively, the costs are

$$c_{\text{full}} = 32, \qquad c_{\text{sign}} = \frac{P_{\text{mat}} + 32P_{\text{aux}}}{P}, \qquad c_{\text{EF21}} = c_{\text{sign}} + \frac{32L}{P}.$$

A channel with cost c has a communication reduction of $32/c$ relative to full precision.

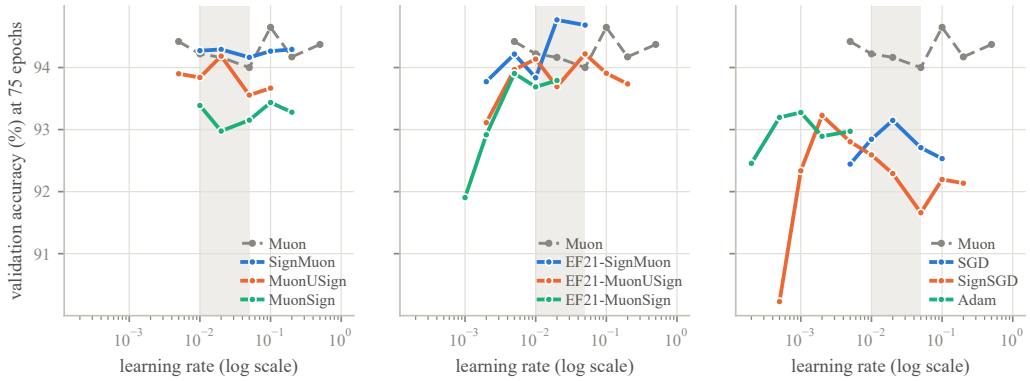


Figure 10: Learning-rate sweep for centralized training of ResNet-18 on CIFAR-10. Validation accuracy is measured on a held-out split after 75 epochs.

Method	Up (bits)	Down (bits)	Up reduction	Down reduction
Muon, MuonServer	32	32	1.0×	1.0×
SGD, Adam	32	32	1.0×	1.0×
SignMuon	1.0870	1.0870	29.4×	29.4×
SignSGD	1.0870	1.0870	29.4×	29.4×
MuonSign	1.0870	1.0870	29.4×	29.4×
EF21-MuonSign	1.0871	1.0871	29.4×	29.4×
MuonUSign	1.0870	32	29.4×	1.0×
EF21-SignMuon	1.0871	32	29.4×	1.0×
EF21-MuonUSign	1.0871	32	29.4×	1.0×

Table 9: Communication per client per round on CNN2, under the randomized-zero convention. Bits are per model parameter; reductions are against a 32-bit baseline.

SignMuon and SignSGD broadcast the majority vote, MuonSign broadcasts a signed LMO direction, and EF21-MuonSign broadcasts a scaled-sign model residual. MuonUSign, EF21-SignMuon and EF21-MuonUSign broadcast a full-precision model.

On CNN2, $P_{\text{mat}} = 762{,}560$, $P_{\text{aux}} = 2{,}146$ and $L = 3$. Table 9 gives the resulting costs, used in the Up and Down columns of Table 1.

N Federated training results

Figure 11 shows learning curves for the comparison in Table 1. The figure additionally shows Muon with a server-side LMO, whose final accuracy is close to client-side Muon's.

EF21-SignMuon finishes below SignMuon, with a larger accuracy gap than in centralized training (Table 2).

MuonUSign reaches the target accuracy earlier than SignMuon and EF21-SignMuon, but finishes below both. EF21-SignMuon achieves the lowest test cross-entropy, while Muon has the highest final accuracy.

O Language-modelling details

Setup. We use modded-nanoGPT record #40 (2025-10-04), replacing Muon on the hidden matrices and gates. Embeddings, scalar parameters, and the output head retain distributed Adam. Model: 12 layers, model dimension 768, 6 heads of dimension 128, vocabulary 50,257; hidden matrices are 768×3072. The merged attention weight `qkvo_w` is used as four 768×768 blocks, and both the LMO and the compressor scale are applied per block. The two MLP matrices are the up-projection `c_fc` and the zero-initialized output projection `c_proj`, which maps the 3072-dimensional hidden

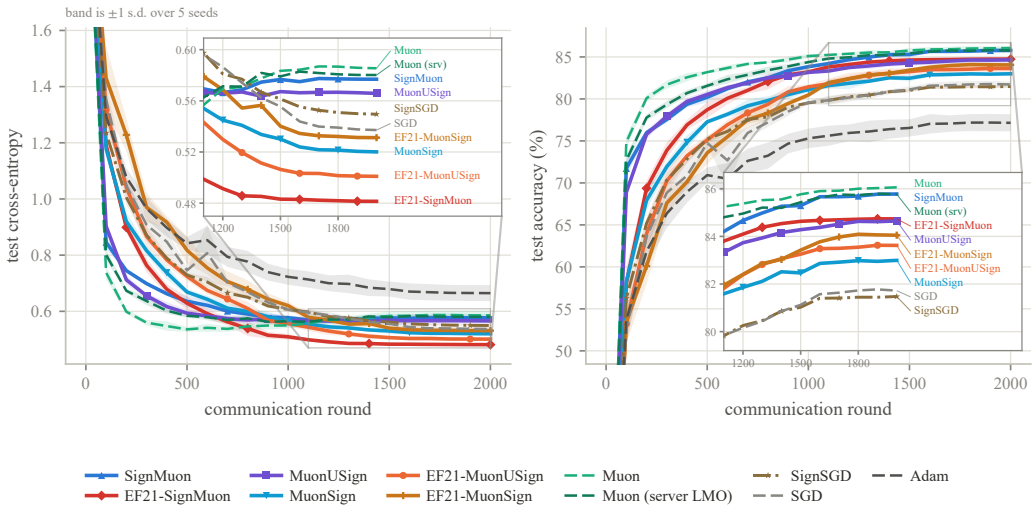


Figure 11: CIFAR-10 federated learning on CNN2 at $N = 11$ clients and batch 192 per client, every method evaluated on the exact server model $\mathbf{X}_t$.

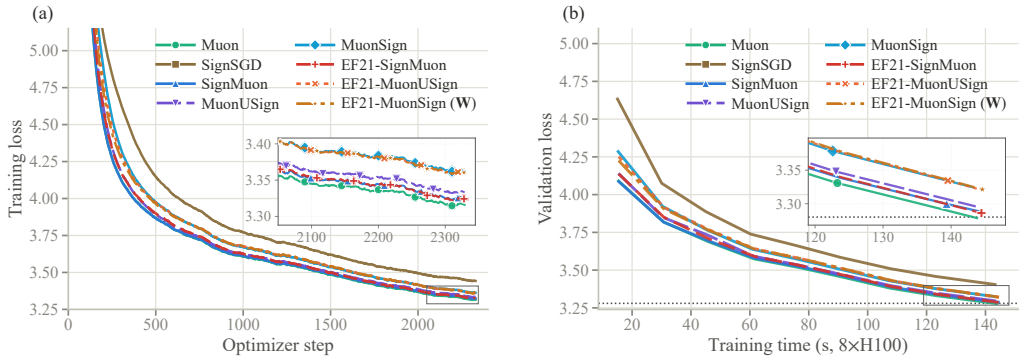


Figure 12: NanoGPT, supporting curves. Training loss is EMA-smoothed and normalized per token; the speedrun clock excludes validation and compilation. EF21-MuonSign training loss is evaluated at $\mathbf{W}$.

activation back to the model dimension. Data: FineWeb10B, the 10B-token sample of FineWeb [Penedo et al., 2024]; 262,144 tokens per step, 2330 steps (= 611M tokens), validation on the fixed 10,485,760-token split.

Hyperparameters. Matrix/gate optimizer: $\eta_0 = 0.06$ (LMO family) or 0.03 (SIGN family), per-layer scaled by the unit-gain rule (Appendix P); Nesterov momentum $\mu = 0.95$, warmed up linearly from 0.85 over the first 300 steps and cooled back to 0.85 over the last 50; weight decay 0; η constant then linearly cooled to $0.1\eta_0$ over the final 45% of the 2290 scheduled iterations, with the 40-step extension held at that floor; LMO by 5 Polar-Express iterations. Distributed Adam: $\eta = 0.008$, $\beta = (0.65, 0.95)$, $\varepsilon = 10^{-8}$, no weight decay, per-parameter multipliers 75 on embeddings and 5 on scalars, stepped every other iteration.

Hardware and timing. Runs use one 8×H100 SXM node (80 GB per GPU), driver 595.71.05, running PyTorch 2.10.0+cu128 under Python 3.12.3, with one process per GPU. Each update uses gradients averaged across the eight GPUs.

Compressor diagnostics. Table 10 reports, per layer type at the final step, the contraction each scaled sign achieves, $\alpha(\mathbf{\Delta}) = \|\mathbf{\Delta}\|_1^2/(d\|\mathbf{\Delta}\|_F^2)$, and the relative estimator lag. By Lemma 5, $1 -$

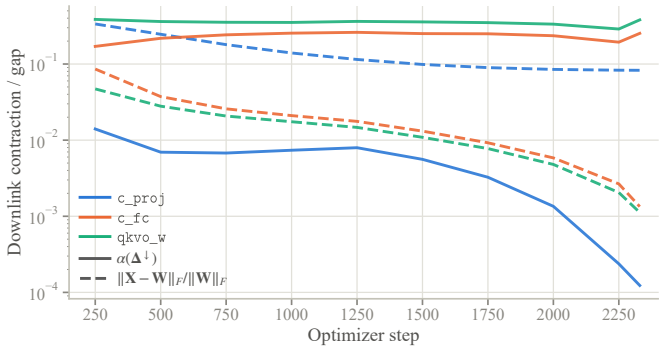


Figure 13: The downlink measurement for EF21-MuonSign, per layer type: the contraction $\alpha(\mathbf{\Delta}^{\downarrow})$ and relative server/broadcast gap $\|\mathbf{X}_t - \mathbf{W}_t\|_F / \|\mathbf{W}_t\|_F$.

	uplink		downlink	
Layer type	α	lag	α	gap
EF21-SignMuon				
`qkvo_w` ($\times 10$)	0.640	0.66	–	–
`c_fc` ($\times 11$)	0.633	0.62	–	–
`c_proj` ($\times 11$)	0.634	0.62	–	–
EF21-MuonUSign				
`qkvo_w`	0.374	0.81	–	–
`c_fc`	0.323	0.82	–	–
`c_proj`	0.597	0.62	–	–
EF21-MuonSign				
`qkvo_w`	0.406	0.79	0.380	0.0011
`c_fc`	0.355	0.78	0.253	0.0013
`c_proj`	0.596	0.61	$\mathbf{1.2\times 10^{-4}}$	**0.083**

Table 10: Compressor diagnostics at step 2330, medians over the identical layers of each type. α is the contraction the scaled sign achieves on that round's residual; "lag" is $\|\text{target} - \text{estimator}\|_F / \|\text{target}\|_F$; "gap" is $\|\mathbf{X}_t - \mathbf{W}_t\|_F / \|\mathbf{W}_t\|_F$.

$\alpha(\mathbf{\Delta})$ is the fraction of the squared Frobenius residual remaining after compression. The small downlink contraction on `c_proj` therefore gives weak residual correction.

Server and broadcast models. Figure 13 tracks EF21-MuonSign's relative model gap by layer type. The persistent gap is concentrated in the zero-initialized MLP output projections (`c_proj`). The server model has higher final validation loss than the broadcast model (Table 3).

P Per-layer step sizes: unit-gain normalization

We use $\eta_\ell = \eta_0 \lambda_\ell$, with shape-dependent multipliers λ_ℓ fixed before tuning η_0. SGD and Adam use a global learning rate.

P.1 Derivation of the shape multipliers

For a layer $\mathbf{X} \in \mathbb{R}^{m \times n}$, let $r = \min(m, n)$. The step matrices fall into two families:

$$\mathbf{P} = \mathbf{U}\mathbf{V}^\top \ \text{(LMO)}, \qquad \mathbf{P} \in \{\pm 1\}^{m \times n} \ \text{(SIGN)}. \tag{53}$$

The LMO family comprises Muon, MuonUSign, EF21-MuonUSign and EF21-MuonSign; the SIGN family comprises SignMuon, MuonSign and SignSGD. A polar input of rank q gives $\|\mathbf{P}\|_F^2 = q$; at full rank, $\|\mathbf{P}\|_F = \sqrt{r}$. A binary sign matrix has $\|\mathbf{P}\|_F = \sqrt{mn}$.

Rule	LMO	SIGN	Equalizes
global ($a=0$)	1	1	nothing
RMS→RMS ball	$\sqrt{\frac{m}{n}}$	—	LMO trust region
Muon default	$\sqrt{\max(1, \frac{m}{n})}$	1	LMO gain
unit gain	$\sqrt{\max(1, \frac{m}{n})}$	$n^{-1/2}$	per-step gain, both
μP ($a=1$)	$\sqrt{\max(1, \frac{m}{n})}$	n^{-1}	aligned accumulation
Mishra et al. [2026]	$r^{-1/2}$	$(mn)^{-1/2}$	$\|\mathbf{P}\|_F$

Table 11: Layerwise multipliers for full-rank polar and binary sign steps, with $r = \min(m, n)$. Only the global rate η_0 is tuned. The RMS→RMS row gives the multiplier induced by that operator-norm ball; the final row equalizes the Frobenius step norm.

Method	Muon	SignMuon	SignSGD	MuonSign
h	0.513	0.515	0.490	0.561

Table 12: Accumulation exponents for ResNet-18 trained on CIFAR-10 over 20 epochs at a constant learning rate; all fits have $R^2 \geq 0.999$.

For an independent isotropic input, define the RMS gain by

$$\gamma(\mathbf{A})^2 := \frac{\mathbb{E}_{\mathbf{u}}[\mathrm{rms}(\mathbf{A}\mathbf{u})^2]}{\mathbb{E}_{\mathbf{u}}[\mathrm{rms}(\mathbf{u})^2]}, \qquad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \tag{54}$$

where $\mathrm{rms}(\mathbf{v}) = \|\mathbf{v}\|_2/\sqrt{\dim \mathbf{v}}$. Since $\mathbb{E}\|\mathbf{A}\mathbf{u}\|_2^2 = \|\mathbf{A}\|_F^2$,

$$\gamma(\mathbf{A}) = \frac{\|\mathbf{A}\|_F}{\sqrt{m}}. \tag{55}$$

This is the isotropic-input counterpart of operator-norm scaling [Yang et al., 2023, Large et al., 2024]. Requiring equal per-step gain, $\gamma(\eta_\ell \mathbf{P}) = \eta_0$, gives

$$\boxed{\eta_\ell = \eta_0 \lambda_\ell, \qquad \lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{P}\|_F}}. \tag{56}$$

Substituting the full-rank polar and binary sign norms yields the shape multipliers used in our experiments:

$$\lambda_\ell^{\mathrm{LMO}} = \sqrt{\max\left(1, \tfrac{m}{n}\right)}, \qquad \lambda_\ell^{\mathrm{SIGN}} = \frac{1}{\sqrt{n}}. \tag{57}$$

For EF21-SignMuon, we apply the LMO multiplier to the error-feedback estimator $\mathbf{d}_t^{\mathrm{est}}$ of the polar update. The multipliers are known from the layer shapes and require no communication.

Relation to other normalizations. The polar multiplier matches the reference Muon implementation [Jordan et al., 2024b]. The RMS→RMS oracle instead prescribes $\sqrt{m/n}$ (Appendix F); the two coincide for $m \geq n$. Table 11 compares these choices with a global rate, μP scaling [Yang et al., 2021], and the Frobenius normalization of Mishra et al. [2026].

P.2 Experimental validation of sign scaling

We fit $\|\mathbf{X}_t - \mathbf{X}_0\|_F/\sqrt{m} \propto t^h$ at a constant learning rate to measure update accumulation. Incoherent and aligned accumulation have reference exponents $h = \frac{1}{2}$ and $h = 1$, respectively. The fitted exponents are consistent with incoherent accumulation (Table 12).

Normalization ablation. For federated learning on CNN2, re-tuning the learning rates under each normalization preserves the method ranking (Table 13).

Method	Rule	λ_ℓ	η_0	Test acc (%)
SignMuon	global	1	0.002	85.68 ± 0.17
SignMuon	**unit gain**	$n^{-1/2}$	0.1	85.72 ± 0.24
SignMuon	μP	n^{-1}	2	85.52 ± 0.01
MuonSign	global	1	0.001	82.26 ± 0.40
MuonSign	**unit gain**	$n^{-1/2}$	0.02	82.94 ± 0.19
MuonSign	μP	n^{-1}	0.5	82.37 ± 0.27
SignSGD	global	1	0.0005	81.37 ± 0.21
SignSGD	**unit gain**	$n^{-1/2}$	0.01	81.44 ± 0.15
SignSGD	μP	n^{-1}	0.5	81.47 ± 0.25

Table 13: Layerwise normalization on federated CNN2 (Table 1). Each (method, rule) pair is retuned on the five-point lattice of Appendix J and evaluated at 2000 rounds: three seeds for global and μP, five for unit gain.

P.3 Weight decay

Coupled decay inserts $\widetilde{\mathbf{G}}_t = \mathbf{G}_t + \lambda_{\mathrm{wd}} \mathbf{X}_t$ before sign or polar normalization, changing the step direction. We use decoupled decay,

$$\mathbf{X}_{t+1} = (1 - \eta_t \lambda_{\mathrm{wd}}) \mathbf{X}_t - \eta_t \lambda_\ell \mathbf{P}_t,$$

which applies multiplicative shrinkage directly; coupled decay is used only in an ablation.

Convergence and bounded iterates. For coupled decay, the ordinary-smooth convergence bounds apply to $f_\lambda = f + \frac{\lambda_{\mathrm{wd}}}{2} \|\mathbf{X}\|_F^2$, with each L_i and $L_{i,j}$ increased by $\lambda_{\mathrm{wd}} r_i$. For decoupled decay, suppose $\lambda_{\mathrm{wd}} > 0$, $0 \le \eta_t \lambda_{\mathrm{wd}} \le 1$, and $\lambda_\ell \|\mathbf{P}_t\|_F \le \sqrt{m}$. The triangle inequality gives $\|\mathbf{X}_{t+1}\|_F \le (1 - \eta_t \lambda_{\mathrm{wd}}) \|\mathbf{X}_t\|_F + \eta_t \sqrt{m}$, hence

$$\gamma(\mathbf{X}_t) \le \max\{\gamma(\mathbf{X}_0), \lambda_{\mathrm{wd}}^{-1}\} \quad \text{for all } t. \tag{58}$$

Q Algorithms

All optimization algorithms take an initial model $\mathbf{X}_0$, momentum coefficient μ, learning rates η_t, and horizon T, and return $\mathbf{X}_T$. They act per matrix block with the shape multipliers of Appendix P; auxiliary parameters use AdamW (Appendix J). The routine $\mathrm{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$ returns $(\mathbf{M}_t, \tilde{\mathbf{M}}_t)$ defined by

$$\mathbf{M}_t = \mu \mathbf{M}_{t-1} + (1 - \mu) \mathbf{G}_t, \qquad \tilde{\mathbf{M}}_t = \begin{cases} \mathbf{M}_t, & \text{EMA}, \\ (1 - \mu) \mathbf{G}_t + \mu \mathbf{M}_t, & \text{Nesterov}. \end{cases} \tag{59}$$

Algorithm 1 MuonLMO

Input: tensor $\mathbf{Y}$; Newton-Schulz coefficients $a = 3.4445$, $b = 4.7750$, $c = 2.0315$; iteration count ns_steps
Output: polar factor $\mathbf{D} \approx \mathbf{U}\mathbf{V}^\top$ of $\mathbf{Y}$

$\mathbf{Y} \leftarrow \mathrm{ReshapeTo2D}(\mathbf{Y})$
$\mathbf{Y} \leftarrow \mathbf{Y} / \|\mathbf{Y}\|_F$
for $k = 1$ to ns_steps **do**
  $\mathbf{A} \leftarrow \mathbf{Y}\mathbf{Y}^\top$
  $\mathbf{Y} \leftarrow a\,\mathbf{Y} - b\,\mathbf{A}\mathbf{Y} + c\,\mathbf{A}^2\,\mathbf{Y}$
end for
$\mathbf{D} \leftarrow \mathrm{ReshapeToOriginal}(\mathbf{Y})$
return $\mathbf{D}$

Algorithm 2 SignMuon

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$
  $(\mathbf{M}_t, \tilde{\mathbf{M}}_t) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\mathbf{s}_t^{\uparrow} \leftarrow \text{sign}(\mathbf{D}_t)$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\uparrow}$
end for

Algorithm 3 EF21-SignMuon

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{d}_0^{\text{est}} \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$
  $(\mathbf{M}_t, \tilde{\mathbf{M}}_t) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\tilde{\mathbf{M}}_t)$
  $\Delta_t^{\uparrow} \leftarrow \mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}$
  $\alpha_t^{\uparrow} \leftarrow \text{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{d}_t^{\text{est}} \leftarrow \mathbf{d}_{t-1}^{\text{est}} + \alpha_t^{\uparrow} \text{sign}(\Delta_t^{\uparrow})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\text{est}}$
end for

Algorithm 4 MuonUSign

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$
  $(\mathbf{M}_t, \tilde{\mathbf{M}}_t) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$
  $\mathbf{s}_t^{\uparrow} \leftarrow \text{sign}(\tilde{\mathbf{M}}_t)$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$
end for

Algorithm 5 MuonSign

$\mathbf{M}_0 \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$
  $(\mathbf{M}_t, \tilde{\mathbf{M}}_t) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$
  $\mathbf{s}_t^{\uparrow} \leftarrow \text{sign}(\tilde{\mathbf{M}}_t)$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\mathbf{s}_t^{\uparrow})$
  $\mathbf{s}_t^{\downarrow} \leftarrow \text{sign}(\mathbf{D}_t)$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\downarrow}$
end for

Algorithm 6 EF21-MuonUSign

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\text{est}} \leftarrow 0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{X}_{t-1}; \xi_t)$
  $(\mathbf{M}_t, \tilde{\mathbf{M}}_t) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$
  $\Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$
  $\alpha_t^{\uparrow} \leftarrow \text{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_t^{\text{est}} \leftarrow \mathbf{g}_{t-1}^{\text{est}} + \alpha_t^{\uparrow} \text{sign}(\Delta_t^{\uparrow})$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\mathbf{g}_t^{\text{est}})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$
end for

Algorithm 7 EF21-MuonSign

$\mathbf{M}_0 \leftarrow 0, \quad \mathbf{g}_0^{\text{est}} \leftarrow 0, \mathbf{W}_0 \leftarrow \mathbf{X}_0$
for $t = 1$ to T **do**
  $\mathbf{G}_t \leftarrow \nabla f(\mathbf{W}_{t-1}; \xi_t)$
  $(\mathbf{M}_t, \tilde{\mathbf{M}}_t) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}, \mathbf{G}_t)$
  $\Delta_t^{\uparrow} \leftarrow \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$
  $\alpha_t^{\uparrow} \leftarrow \text{mean}(|\Delta_t^{\uparrow}|)$
  $\mathbf{g}_t^{\text{est}} \leftarrow \mathbf{g}_{t-1}^{\text{est}} + \alpha_t^{\uparrow} \text{sign}(\Delta_t^{\uparrow})$
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\mathbf{g}_t^{\text{est}})$
  $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$
  $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$
  $\alpha_t^{\downarrow} \leftarrow \text{mean}(|\Delta_t^{\downarrow}|)$
  $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \text{sign}(\Delta_t^{\downarrow})$
end for

Federated methods. Algorithms 8–9 use the compressor choices in Table 14. Clients evaluate gradients at the broadcast model and take no local parameter steps. CIFAR-10 gradients accumulate three mini-batches of 64 at fixed weights with frozen BatchNorm statistics (Appendix J). Plain signs are aggregated by majority vote, with ties resolved by a fair coin [Bernstein et al., 2019]; EF21 scaled residuals are averaged [Gruntkowska et al., 2025]. Communication costs are given in Appendix M.

Method	LMO	Uplink $\mathcal{C}^{\uparrow}$	Downlink $\mathcal{C}^{\downarrow}$
SignMuon	worker	sign / MV	exact
EF21-SignMuon	worker	EF21	exact
MuonUSign	server	sign / MV	exact
MuonSign	server	sign / MV	sign
EF21-MuonUSign	server	EF21	exact
EF21-MuonSign	server	EF21	EF21-P

Table 14: The six federated methods from Algorithms 8–9. "MV": majority vote; "EF21-P": primal (model-side) error feedback; "exact": the server applies no downlink compressor. Communication costs: Appendix M.

Algorithm 8 Federated SignMuon / EF21-SignMuon (worker-side LMO)

Input: clients N; uplink compressor $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$ (Table 14)

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{d}_0^{(j)} \leftarrow 0$ for all j; $\quad \mathbf{d}_0 \leftarrow 0$
broadcast $\mathbf{X}_0$
for $t = 1$ to T **do**
  for $j = 1$ to N **in parallel do**
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{X}_{t-1}; \xi_t^{(j)})$
    $(\mathbf{M}_t^{(j)}, \tilde{\mathbf{M}}_t^{(j)}) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}^{(j)}, \mathbf{G}_t^{(j)})$
    $\mathbf{D}_t^{(j)} \leftarrow \text{MuonLMO}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$
    if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then**
      $\Delta_t^{(j)} \leftarrow \mathbf{D}_t^{(j)} - \mathbf{d}_{t-1}^{(j)}; \quad \alpha_t^{(j)} \leftarrow \text{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{d}_t^{(j)} \leftarrow \mathbf{d}_{t-1}^{(j)} + \alpha_t^{(j)} \,\text{sign}(\Delta_t^{(j)})$
      send $\big(\text{sign}(\Delta_t^{(j)}),\, \alpha_t^{(j)}\big)$
    else
      send $\mathbf{s}_t^{(j)} \leftarrow \text{sign}(\mathbf{D}_t^{(j)})$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then**
    $\mathbf{d}_t \leftarrow \mathbf{d}_{t-1} + \frac{1}{N} \sum_j \alpha_t^{(j)} \,\text{sign}(\Delta_t^{(j)})$
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \,\mathbf{d}_t$; **broadcast** $\mathbf{X}_t$
  else
    $\hat{\mathbf{s}}_t \leftarrow \text{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \,\hat{\mathbf{s}}_t$; **broadcast** $\hat{\mathbf{s}}_t$
  end if
end for

Algorithm 9 Federated MuonUSign / MuonSign / EF21-MuonUSign / EF21-MuonSign (server-side LMO)

Input: clients N; compressors $\mathcal{C}^{\uparrow} \in \{\text{sign}, \text{EF21}\}$, $\mathcal{C}^{\downarrow} \in \{\text{exact}, \text{sign}, \text{EF21-P}\}$ (Table 14)

$\mathbf{M}_0^{(j)} \leftarrow 0$, $\mathbf{g}_0^{(j)} \leftarrow 0$ for all j; $\mathbf{g}_0 \leftarrow 0$
broadcast $\mathbf{X}_0$; $\mathbf{W}_0 \leftarrow \mathbf{X}_0$
for $t = 1$ to T **do**
  for $j = 1$ to N **in parallel do**
    $\mathbf{G}_t^{(j)} \leftarrow \nabla f_j(\mathbf{W}_{t-1}; \xi_t^{(j)})$
    $(\mathbf{M}_t^{(j)}, \tilde{\mathbf{M}}_t^{(j)}) \leftarrow \text{Momentum}(\mathbf{M}_{t-1}^{(j)}, \mathbf{G}_t^{(j)})$
    if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then**
      $\Delta_t^{(j)} \leftarrow \tilde{\mathbf{M}}_t^{(j)} - \mathbf{g}_{t-1}^{(j)}$; $\alpha_t^{(j)} \leftarrow \text{mean}(|\Delta_t^{(j)}|)$
      $\mathbf{g}_t^{(j)} \leftarrow \mathbf{g}_{t-1}^{(j)} + \alpha_t^{(j)} \text{sign}(\Delta_t^{(j)})$
      send $\big(\text{sign}(\Delta_t^{(j)}), \alpha_t^{(j)}\big)$
    else
      send $\mathbf{s}_t^{(j)} \leftarrow \text{sign}\big(\tilde{\mathbf{M}}_t^{(j)}\big)$
    end if
  end for
  on the server:
  if $\mathcal{C}^{\uparrow} = \text{EF21}$ **then**
    $\mathbf{g}_t \leftarrow \mathbf{g}_{t-1} + \frac{1}{N} \sum_j \alpha_t^{(j)} \text{sign}(\Delta_t^{(j)})$
  else
    $\mathbf{g}_t \leftarrow \text{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$
  end if
  $\mathbf{D}_t \leftarrow \text{MuonLMO}(\mathbf{g}_t)$
  if $\mathcal{C}^{\downarrow} = \text{sign}$ **then**
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \text{sign}(\mathbf{D}_t)$; **broadcast** $\text{sign}(\mathbf{D}_t)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} - \eta_t \text{sign}(\mathbf{D}_t)$
  else if $\mathcal{C}^{\downarrow} = \text{EF21-P}$ **then**
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$
    $\Delta_t^{\downarrow} \leftarrow \mathbf{X}_t - \mathbf{W}_{t-1}$; $\alpha_t^{\downarrow} \leftarrow \text{mean}(|\Delta_t^{\downarrow}|)$
    $\mathbf{W}_t \leftarrow \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \text{sign}(\Delta_t^{\downarrow})$; **broadcast** $\big(\text{sign}(\Delta_t^{\downarrow}), \alpha_t^{\downarrow}\big)$
  else
    $\mathbf{X}_t \leftarrow \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$; **broadcast** $\mathbf{X}_t$; $\mathbf{W}_t \leftarrow \mathbf{X}_t$
  end if
end for